Online Appendix 1

Appendix OA.A. Preliminaries, Notation and Assumptions Recap

Before starting we recap the main notation, definitions and assumptions.

Notation for Random Variables We denote the covariate domain space by $\Omega_{\mathcal{X}} \subset \mathbb{R}^{d_{\mathcal{X}}}$ and a single covariate (random variable) by calligraphic $\mathcal{X}$. Similarly, single response variable is denoted by calligraphic $\mathcal{Y} \in \mathbb{R}$. The covariate/response distributions are denoted by $P_{\mathcal{X}}$ and $P_{\mathcal{Y}}$ and their joint distribution is $P_{\mathcal{X},\mathcal{Y}}$. The random variables defined as i.i.d. samples of size n of covariate-response pairs are denoted by uppercase boldface letters $(\boldsymbol{X}_n, \boldsymbol{Y}_n)$, where $\boldsymbol{X}_n = (\mathcal{X}_1, \ldots, \mathcal{X}_n)$ and $\boldsymbol{Y}_n = (\mathcal{Y}_1, \ldots, \mathcal{Y}_n)$. Single data realisations are denoted by lowercase letters. A realisation of $\mathcal{X}$ is $\boldsymbol{x} \in \mathbb{R}^{d_{\mathcal{X}}}$ and a relisation of $\mathcal{Y}$ is y. An observed covariate sample is $X_n = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ (a matrix of size $n \times d$) and an observed response sample is the vector $\boldsymbol{y}_n = (y_1, \ldots, y_n)$. Then, the regression function can be written as $f(\boldsymbol{x}) = \mathbb{E}[\mathcal{Y}|\mathcal{X} = \boldsymbol{x}]$. Similarly, we denote the noise random variable as Ξ, it's single realisation as ξ and a sample vector of length n is $\boldsymbol{\xi}_n$. Any lowercase boldface characters will always denote vectors.

$GPnn$ **Response Model.** In $GPnn$ (Allison et al., 2023) We assume that the response variables are generated as

$$\mathcal{Y}_i = f(\mathcal{X}_i) + \Xi_i, \quad i = 1, \ldots, n. \tag{1}$$

$NNGP$ **Response Model.** The responses $\mathcal{Y}$ are assumed to be generated according to

$$\mathcal{Y}_i = \boldsymbol{t}(\mathcal{X}_i)^T . \boldsymbol{b} + w(\mathcal{X}_i) + \Xi_i, \tag{2}$$

where $\boldsymbol{b} \in \mathbb{R}^{d_T}$ is the vector of regression coefficients, Ξ_i is the independent and identically distributed noise and $w(\boldsymbol{x})$ is a sample path drawn from a GP with mean-zero and covariance function $\tilde{k} : \mathbb{R}^{d_{\mathcal{X}}} \times \mathbb{R}^{d_{\mathcal{X}}} \to \mathbb{R}$. The role of $w(\boldsymbol{x})$ is to model the effect of unknown/unobserved spatially-dependent covariates.

$GPnn/NNGP$ **Estimators.** We fix a continuous symmetric and positive definite kernel function $c : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ normalised so that $c(\boldsymbol{x}, \boldsymbol{x}) = 1$ and which determines the exact form of the $GPnn$ estimator. Consider a sequence of n training points $X_n = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$ together with their response values $\boldsymbol{y}_n = (y_1, \ldots, y_n)$, and a test point $\boldsymbol{x}_*$. Let $\mathcal{N}_m(\boldsymbol{x}_*, X_n)$ be the set of m-nearest neighbours of $\boldsymbol{x}_*$ in X_n. Let $X_{\mathcal{N}}(\boldsymbol{x}_*) = (\boldsymbol{x}_{n,1}(\boldsymbol{x}_*), \ldots, \boldsymbol{x}_{n,m}(\boldsymbol{x}_*))$ be the sequence of the m-nearest neighbours of $\boldsymbol{x}_*$ ordered increasingly according to their distance from $\boldsymbol{x}_*$ (we assume that ties occur with probability zero) and let $\boldsymbol{y}_{\mathcal{N}}$ be their corresponding responses. Given the hyperparameters: $\hat{\sigma}_f^2 > 0$ (the kernelscale), $\hat{\sigma}_\xi^2 \geq 0$ (the noise variance) and $\hat{\ell} > 0$ (the lengthscale) we define the (shifted) Gram matrix for m-nearest neighbours of $\boldsymbol{x}_*$ as

$$[K_{\mathcal{N}}]_{ij} := \hat{\sigma}_f^2\, c(\boldsymbol{x}_{n,i}/\hat{\ell}, \boldsymbol{x}_{n,j}/\hat{\ell}) + \hat{\sigma}_\xi^2\, \delta_{ij}, \quad [\boldsymbol{k}_{\mathcal{N}}^*]_j := \hat{\sigma}_f^2\, c(\boldsymbol{x}_*/\hat{\ell}, \boldsymbol{x}_{n,j}/\hat{\ell}), \quad 1 \leq i, j \leq m, \tag{3}$$

where δ_{ij} is the Kronecker delta. In $GPnn$ the predicted mean and variance of the distribution of the response $\hat{y}_*$ at $\boldsymbol{x}_*$ is normally are given by the standard GP regression formulae

	Response Model	Predictive Mean	Predictive Variance
$GPnn$	$f(\boldsymbol{x}) + \xi(\boldsymbol{x})$	$\Gamma\, \boldsymbol{k}_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1}\, \boldsymbol{y}_{\mathcal{N}}$	$\hat{\sigma}_{\xi}^{2} + \hat{\sigma}_{f}^{2} - \boldsymbol{k}_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1} \boldsymbol{k}_{\mathcal{N}}^{*}$
$NNGP$	$\boldsymbol{t}(\boldsymbol{x})^{T}.\boldsymbol{b} + w(\boldsymbol{x}) + \xi(\boldsymbol{x})$	$\boldsymbol{t}_{*}^{T}.\hat{\boldsymbol{b}} + \Gamma\, k_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1} \left(\boldsymbol{y}_{\mathcal{N}} - T_{\mathcal{N}}.\hat{\boldsymbol{b}}\right)$	

Table 1: Summary of response models, predictive mean and variance in $GPnn$ and $NNGP$. See the main text for the explanation of the symbols. The predictive means are corrected by the coefficient $\Gamma = \frac{\hat{\sigma}_{\xi}^{2} + m\hat{\sigma}_{f}^{2}}{m\hat{\sigma}_{f}^{2}}$ making them unbiased when m is fixed and in the limit $n \to \infty$, as opposed to standard formulas used in the literature.

(Rasmussen and Williams, 2005).

$$\mu_{GPnn} = \boldsymbol{k}_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1}\, \boldsymbol{y}_{n,m}, \quad \sigma_{\mathcal{N}}^{*}{}^{2} = \hat{\sigma}_{\xi}^{2} + \hat{\sigma}_{f}^{2} - \boldsymbol{k}_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1} \boldsymbol{k}_{\mathcal{N}}^{*}. \tag{4}$$

The asymptotically unbiased counterparts of the $GPnn$ estimator reads

$$\tilde{\mu}_{GPnn}(\boldsymbol{x}_{*}) = \Gamma\, \boldsymbol{k}_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1}\, \boldsymbol{y}_{\mathcal{N}}, \quad \Gamma = \frac{\hat{\sigma}_{\xi}^{2} + m\hat{\sigma}_{f}^{2}}{m\hat{\sigma}_{f}^{2}}. \tag{5}$$

The variance of the hyperparameter-conditional predictive distribution in $NNGP$ is the same as the predictive variance in $GPnn$, while the predictive mean is given by the following formula

$$\tilde{\mu}_{NNGP}(\boldsymbol{x}_{*}) = \boldsymbol{t}_{*}^{T}.\hat{\boldsymbol{b}} + \Gamma\, k_{\mathcal{N}}^{*}{}^{T}\, K_{\mathcal{N}}^{-1} \left(\boldsymbol{y}_{\mathcal{N}} - T_{\mathcal{N}}.\hat{\boldsymbol{b}}\right), \tag{6}$$

where we have adjusted the version given in Finley et al. (2019) by incorporating the factor Γ thereby ensuring asymptotic unbiasedness when m is fixed. $T_{\mathcal{N}}$ is the $m \times d_{T}$-matrix of regressors at the nearest-neighbours $(\boldsymbol{t}\left(\boldsymbol{x}_{n,1}(\boldsymbol{x}_{*})\right), \ldots, \boldsymbol{t}\left(\boldsymbol{x}_{n,m}(\boldsymbol{x}_{*})\right))$ and $\boldsymbol{t}_{*} := \boldsymbol{t}(\boldsymbol{x}_{*})$. Table 1 summarises the $GPnn$ and $NNGP$ setups described above.

OA.A.1 L_2-risk, Universal Consistency and Stone's Optimal Convergence Rates

In the task of estimating (noiseless) $f(\boldsymbol{x}_{*})$ in the generative model (1) given noisy data $(X_n, \boldsymbol{y}_n)$ we denote the estimated value of f at test point $\boldsymbol{x}_{*}$ as $\hat{f}_n(\boldsymbol{x}_{*})$, where the subscript n refers to the size of the training dataset. Assume that the training data are i.i.d. samples from the distribution $P_{\mathcal{X},\mathcal{Y}}$.

The $L_2(P_{\mathcal{X}})$-risk (which we simply call *risk* throughout the paper) is defined as

$$\mathcal{R}\left(\hat{f}_n\right) := \mathbb{E}\left[\int \left(\hat{f}_n(\boldsymbol{x}) - f(\boldsymbol{x})\right)^{2} dP_{\mathcal{X}}(\boldsymbol{x})\right], \tag{7}$$

where the inner integral is taken over the test data given a training sample $X_n, \boldsymbol{y}_n$ and it can be viewed as the squared $L_2(P_{\mathcal{X}})$-distance between $\hat{f}_n$ and f. The outer expectation is taken over all the training samples of size n coming from $P_{\mathcal{X},\mathcal{Y}}^{n}$. Similarly, we can define an $L_2(P_{\mathcal{X}})$-risk directly using the observed noisy responses (rather than the exact values of f) which is more applicable to the $GPnn$ and $NNGP$ response models (1) and (2) as follows.

$$\mathcal{R}_{Y}\left(\hat{f}_n\right) := \mathbb{E}\left[\int \left(\hat{f}_n(\boldsymbol{x}) - y\right)^{2} dP_{\mathcal{X},\mathcal{Y}}(\boldsymbol{x}, \boldsymbol{y})\right].$$

In our noise model specified in the assumption (AC.4) the above two $L_2(P_{\mathcal{X}})$-risk measures differ by an additive constant, i.e.,

$$\mathcal{R}_Y\left(\hat{f}_n\right) = \mathcal{R}\left(\hat{f}_n\right) + \int \sigma_\xi^2\left(\mathcal{X}\right) dP_{\mathcal{X}}(\boldsymbol{x}),$$

where $\sigma_\xi^2(\mathcal{X})$ is the variance of the noise variable Ξ at $\mathcal{X}$.

We say that the estimator $\hat{f}_n(\boldsymbol{x}_*)$ is *universally consistent* with respect to a family of training data distributions $\mathcal{D}$ if it satisfies the following conditions.

Definition OA.A.1 (Universal Consistency) *A sequence of regression function estimates $(\hat{f}_n)$ is universally consistent with respect to $\mathcal{D}$ if for all distributions $P_{\mathcal{X},\mathcal{Y}} \in \mathcal{D}$ we have*

$$\mathcal{R}\left(\hat{f}_n\right) \xrightarrow{n\to\infty} 0.$$

In this work, we study nearest-neighbour-based estimators which are indexed by n (the training data size) and m (the number of nearest-neighbours). There, we also distinguish the notion of approximate universal consistency.

Definition OA.A.2 (Approximate Universal Consistency) *A sequence of nearest - neighbour regression function estimates $(\hat{f}_{n,m})$ is approximately universally consistent with respect to $\mathcal{D}$ if for all distributions $P_{\mathcal{X},\mathcal{Y}} \in \mathcal{D}$ we have*

$$\inf_{m\in\mathbb{N}} \lim_{n\to\infty} \mathcal{R}\left(\hat{f}_{n,m}\right) = 0.$$

Stone (1982) found the best possible minimax rate at which the risk of a universally consistent estimator $\hat{f}_n$ can tend to zero with n. More precisely, denote $\mathcal{D}_q$ the class of distributions of $(\mathcal{X},\mathcal{Y})$ where $\mathcal{X}$ is uniformly distributed on the unit hypercube $[0,1]^d$ and $\mathcal{Y} = f(\mathcal{X}) + \Xi$ with some q-smooth function $f : \mathbb{R}^d \to \mathbb{R}$ and the noise variable Ξ is drawn from the standard normal distribution independently of $\mathcal{X}$. Function f is q-smooth if all its partial derivatives of the order $\lfloor q \rfloor$ exist and are β-Hölder continuous with $\beta = q - \lfloor q \rfloor$ with respect to the Euclidean metric on $\mathbb{R}^d$. Stone showed that there exists a positive constant $\mathcal{C} > 0$ such that

$$\lim_{n\to\infty} \inf_{\hat{f}_n} \sup_{P\in\mathcal{D}_q} \mathcal{P}_P\left[\int \left(\hat{f}_n(\boldsymbol{x}) - f(\boldsymbol{x})\right)^2 dP_{\mathcal{X}} > \mathcal{C}\, n^{-\frac{2q}{2q+d}}\right] = 1,$$

where the outer probability is taken with respect to the training data samples coming from the product distribution P^n. This means that the best universally achievable risk cannot decay faster than $\mathcal{O}\left(n^{-\frac{2q}{2q+d}}\right)$. In this work, we prove that $GPnn$ and $NNGP$ achieve the optimal minimax convergence rates when $0 < q \leq 1$ and provide experimental evidence that $GPnn$ can achieve these rates also when $q > 1$.

OA.A.2 Performance Metrics

Let $\hat{f}_n$ be equal to $\tilde{\mu}_{GPnn}$ or $\tilde{\mu}_{NNGP}$ defined in (5) and (6). Define the following metrics.

$$\begin{aligned} se(y_*, \boldsymbol{y}_n) &:= \left(y_* - \hat{f}_n(\boldsymbol{x}_*)\right)^2, \quad cal(y_*, \boldsymbol{y}_n) := \frac{\left(y_* - \hat{f}_n(\boldsymbol{x}_*)\right)^2}{{\sigma^*_{\mathcal{N}}}^2}, \\ nll(y_*, \boldsymbol{y}_n) &:= \frac{1}{2}\left(\log\left({\sigma^*_{\mathcal{N}}}^2\right) + \frac{\left(\boldsymbol{y}_* - \hat{f}_n(\boldsymbol{x}_*)\right)^2}{{\sigma^*_{\mathcal{N}}}^2} + \log 2\pi\right). \end{aligned} \tag{8}$$

We focus on the above performance metrics averaged over the noise component i.e. we treat the training set $\boldsymbol{X}_n$ and the test point $\mathcal{X}_*$ as given and define the respective conditional expectations over the test response $\mathcal{Y}_*$ and the training responses $\boldsymbol{Y}_n$ as follows.

$$MSE := \mathbb{E}\left[se(\mathcal{Y}_*, \boldsymbol{Y}_n) \mid \mathcal{X}_*, \boldsymbol{X}_n\right], \tag{9}$$

$$CAL := \mathbb{E}\left[cal(\mathcal{Y}_*, \boldsymbol{Y}_n) \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \frac{MSE}{{\sigma^*_{\mathcal{N}}}^2}, \tag{10}$$

$$NLL := \mathbb{E}\left[nll(\mathcal{Y}_*, \boldsymbol{Y}_n) \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \frac{1}{2}\left(\log\left({\sigma^*_{\mathcal{N}}}^2\right) + CAL + \log 2\pi\right). \tag{11}$$

We use the following shorthand notation for the MSE derivatives. For each $\phi \in \{\hat{\sigma}^2_\xi, \hat{\sigma}^2_f, \hat{\ell}, \hat{\boldsymbol{b}}\}$ we define

$$D_\phi(\mathcal{X}_*, \boldsymbol{X}_n) := \begin{cases} \left|\partial_\phi MSE(\mathcal{X}_*, \boldsymbol{X}_n)\right|, & \phi \in \{\hat{\sigma}^2_\xi, \hat{\sigma}^2_f, \hat{\ell}\}, \\ \left\|\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP}(\mathcal{X}_*, \boldsymbol{X}_n)\right\|_2, & \phi = \hat{\boldsymbol{b}}. \end{cases} \tag{12}$$

OA.A.3 Assumptions Recap

Below, we list all the assumptions that are used throughout the proofs. Note that the assumption differ between the theorems/proofs, so use the list below as a lookup-list when reading the proofs.

Assumptions Related to Consistency.

(AC.1) The training covariates $\{\mathcal{X}_i\}_{i=1}^n$ and the test covariate $\mathcal{X}_*$ are i.i.d. distributed according to the probability measure $P_{\mathcal{X}}$ on $\mathbb{R}^{d_{\mathcal{X}}}$.

(AC.2) The nearest neighbours are chosen according to the kernel-induced metric ρ_c.

(AC.3) The function f in the $GPnn$ response model (1) and the functions t_i, $i = 1, \ldots, d_T$ in the $NNGP$ response model (2) are continuous almost everywhere with respect to the kernel-induced (pseudo)metric ρ_c and are integrable i.e., they are measurable and satisfy

$$\int f(\boldsymbol{x}) dP_{\mathcal{X}}(\boldsymbol{x}) < \infty, \quad \int t_i(\boldsymbol{x}) dP_{\mathcal{X}}(\boldsymbol{x}) < \infty.$$

(AC.4) The noise Ξ is heteroscedastic with mean zero and

$$\mathbb{E}[\Xi_i \mid \mathcal{X}_i] = 0, \qquad \mathbb{E}[\Xi_i^2 \mid \mathcal{X}_i] = \sigma_\xi^2(\mathcal{X}_i),$$

for some function $\sigma_\xi^2 : \Omega_{\mathcal{X}} \to \mathbb{R}_{>0}$ and the noise random variables are uncorrelated given the covariates, i.e.,

$$\mathrm{Cov}\left[\Xi_i, \Xi_j \mid \mathcal{X}_i, \mathcal{X}_j\right] = 0 \quad \text{for} \quad i \neq j, \quad \mathrm{Cov}\left[\Xi_*, \Xi_i \mid \mathcal{X}_*, \mathcal{X}_i\right] = 0.$$

In the $NNGP$ response model (2) we also assume that $\{\Xi_i\} \cup \{\Xi_*\}$ are independent of the sample path $w(\cdot)$. We further assume that the variance function $\sigma_\xi^2(\cdot)$ is almost continuous with respect to the kernel metric ρ_c and is an integrable function of $\boldsymbol{x}$ i.e.,

$$\int \sigma_\xi^2(\boldsymbol{x}) dP_{\mathcal{X}}(\boldsymbol{x}) \leq \infty.$$

(AC.5) The covariance function of the GP sample paths generating the $NNGP$ responses (2) satisfies $\tilde{k}(\boldsymbol{x}, \boldsymbol{x}) = \sigma_w^2$ for all $\boldsymbol{x} \in \mathbb{R}^{d_{\mathcal{X}}}$. Define $\tilde{c}(\cdot,\cdot) := \tilde{k}(\cdot,\cdot)/\sigma_w^2$. The (pseudo)metrics ρ_c and $\rho_{\tilde{c}}$ are equivalent.

Assumptions Related to Convergence Rates.

(AR.1) The (normalised) GP kernel function is an isotropic and a strictly decreasing function of the Euclidean distance, i.e.,

$$c(\boldsymbol{x}, \boldsymbol{x}') \equiv c\left(r\right), \quad r = \left\|\boldsymbol{x} - \boldsymbol{x}'\right\|_2, \quad c(r_1) < c(r_2) \quad \text{if} \quad r_1 > r_2.$$

(AR.2) There exist constants $L_c > 0$ and $0 < p \leq 1$ such that the (isotropic and normalised) GP kernel function $c : \mathbb{R}^{d_{\mathcal{X}}} \times \mathbb{R}^{d_{\mathcal{X}}} \to \mathbb{R}_{\geq 0}$ used in the $GPnn/NNGP$ estimators (5) and (6) is lower bounded as

$$c(r) \geq 1 - L_c\, r^{2p}.$$

(AR.3) The normalised covariance function of the GP sample paths that generate the $NNGP$ responses (2) satisfies

$$\tilde{c}\left(\boldsymbol{x}, \boldsymbol{x}'\right) \geq 1 - L_{\tilde{c}} \left\|\boldsymbol{x} - \boldsymbol{x}'\right\|_2^{2q_0}, \quad L_{\tilde{c}} > 0. \tag{13}$$

(AR.4) The function f in the $GPnn$ response model (1) is bounded in absolute value by some constant $\infty > B_f \geq 1$ and is q-Hölder-continuous, i.e., there exist constants $1 \leq L_f < \infty$ and $0 < q \leq 1$ such that for every $\boldsymbol{x}, \boldsymbol{x}'$

$$|f(\boldsymbol{x}) - f(\boldsymbol{x}')| \leq L_f \left\|\boldsymbol{x} - \boldsymbol{x}'\right\|_2^q.$$

Each function t_i, $i = 1, \ldots, d_T$ in the $NNGP$ response model (2) is bounded and q_i-Hölder continuous for, i.e.,

$$|t_i(\boldsymbol{x})| \leq B_T < \infty, \quad \left|t_i(\boldsymbol{x}) - t_i(\boldsymbol{x}')\right| \leq L_i \|\boldsymbol{x} - \boldsymbol{x}'\|_2^{q_i}, \quad i \in \{1, \ldots, d_T\}$$

with $0 < q_i \leq 1$ and $1 \leq L_i < \infty$.

(AR.5) There exists $\beta > \frac{4d\,(p+\alpha)}{d-4(p+\alpha)}$ for which $\mathbb{E}\left[\|\mathcal{X}\|_2^{\beta}\right] < \infty$ under the probability distribution $P_{\mathcal{X}}$ on $\mathbb{R}^{d_{\mathcal{X}}}$ with $d > 4(p+\alpha)$ where $\alpha = \min\{q, p\}$ for $GPnn$ and $\alpha = \min\{q_0, q_1, \ldots, q_{d_T}, p\}$ for $NNGP$ with p defined in (AR.2).

(AR.6) The noise is homoscedastic, i.e., the noise Ξ_i in $GPnn$ responses (1) and $NNGP$ responses (2) is i.i.d. from the probability distribution P_ξ with mean zero and fixed variance $\sigma_\xi^2 < \infty$.

Assumptions Related to MSE Derivatives.

(AD.1) The normalised kernel function $c(\cdot\,)$ is isotropic and such that $c(u)$ is differentiable for $u > 0$, the limit $\lim_{u \to 0^+} c'(u)$ exists (but may not be finite), and $0 \leq c(u) \leq 1$ for all $u \geq 0$, and $c(0) = 1$.

(AD.2) The normalised kernel function $c(u)$ is differentiable and satisfies for all $u \geq 0$

$$\left|u \frac{dc(u)}{du}\right| \leq B_c, \quad \left|u \frac{dc(u)}{du}\right| \leq L_c'\, u^{2p'}$$

for some $B_c, L_c' \geq 1$, and $0 < p' \leq 1$.

Appendix OA.B. Some Key Matrix Inequalities

In this section, we review and provide several generalisations of matrix inequalities relating to the sensitivity of linear equation systems under perturbations which can be found in (Golub and Van Loan, 2013). The proofs provided below are almost taken verbatim from the proofs of Lemma 2.6.1 and Theorem 2.6.2 in (Golub and Van Loan, 2013). We subsequently apply these results to derive some key inequalities that involve Gram matrices used to prove the main results of this paper.

Below, $||\cdot||$ denotes arbitrary matrix norm as well as its compatible column vector norm. The condition number $\kappa(A)$ for $A \in \mathbb{R}^{n \times n}$ pertaining to the norm $||\cdot||$ is defined as

$$\kappa(A) := ||A||\,||A^{-1}||.$$

Lemma OA.B.1 *Suppose*

$$A\boldsymbol{x} = \boldsymbol{b}, \quad A \in \mathbb{R}^{n \times n},\ \boldsymbol{0} \neq \boldsymbol{b} \in \mathbb{R}^{n \times 1},$$

$$(A + \Delta A)\boldsymbol{y} = \boldsymbol{b} + \Delta\boldsymbol{b}, \quad \Delta A \in \mathbb{R}^{n \times n},\ \Delta\boldsymbol{b} \in \mathbb{R}^{n \times 1},$$

with $||\Delta A|| \leq \epsilon_A ||A||$ *and* $||\Delta\boldsymbol{b}|| \leq \epsilon_b ||\boldsymbol{b}||$ *for some* $\epsilon_A, \epsilon_b > 0$ *such that* $\epsilon_A\, \kappa(A) < 1$. *Define*

$$r_A := \epsilon_A\, \kappa(A), \quad r_b := \epsilon_b\, \kappa(A).$$

Then, $A + \Delta A$ is nonsingular and

$$\frac{||\boldsymbol{y}||}{||\boldsymbol{x}||} \leq \frac{1 + r_b}{1 - r_A}, \tag{14}$$

$$\frac{||\boldsymbol{y} - \boldsymbol{x}||}{||\boldsymbol{x}||} \leq \frac{r_A + r_b}{1 - r_A}. \tag{15}$$

Proof The matrix $A + \Delta A$ is nonsingular due to Theorem 2.3.4 in Golub and Van Loan (2013) and the fact that $||A^{-1}\Delta A|| \leq \epsilon_A ||A^{-1}||\,||A|| = r_A < 1$.

In order to prove the second part of this Lemma, we first note the equality

$$(\mathbb{1} + A^{-1}\Delta A)\boldsymbol{y} = \boldsymbol{x} + A^{-1}\Delta \boldsymbol{b}.$$

Using the above equality and the fact that $||(\mathbb{1} - F)^{-1}|| \leq (1 - ||F||)^{-1}$ (Golub and Van Loan, 2013, Lemma 2.3.3), we find

$$||\boldsymbol{y}|| \leq \left(1 - ||A^{-1}\Delta A||\right)^{-1} \left(||\boldsymbol{x}|| + \epsilon_b ||A^{-1}||\,||\boldsymbol{b}||\right) \leq \frac{||\boldsymbol{x}|| + \epsilon_b ||A^{-1}||\,||\boldsymbol{b}||}{1 - r_A}.$$

Finally, using the fact that $\epsilon_b ||A^{-1}|| = r_b ||A||^{-1}$ and $||\boldsymbol{b}|| \leq ||A||||\boldsymbol{x}||$ we get the inequality (14).

In order to prove inequality (15), we first note that

$$\boldsymbol{y} - \boldsymbol{x} = A^{-1}\Delta \boldsymbol{b} - A^{-1}\Delta A \boldsymbol{y}.$$

Thus,

$$||\boldsymbol{y} - \boldsymbol{x}|| \leq \epsilon_b ||A^{-1}||\,||\boldsymbol{b}|| + \epsilon_A ||A^{-1}||\,||A||\,||\boldsymbol{y}|| = r_b \frac{||\boldsymbol{b}||}{||A||} + r_A ||\boldsymbol{y}|| \leq r_b ||\boldsymbol{x}|| + r_A ||\boldsymbol{y}||.$$

It follows that

$$\frac{||\boldsymbol{y} - \boldsymbol{x}||}{||\boldsymbol{x}||} \leq r_b + r_A \frac{||\boldsymbol{y}||}{||\boldsymbol{x}||} \leq r_b + r_A \frac{1 + r_b}{1 - r_A},$$

where in the last step we applied inequality (14). The result is equivalent to (15). ■

Corollary OA.B.2 *By taking the transpose of all the equations from Lemma OA.B.1, we obtain the following result. Suppose*

$$\boldsymbol{x}^T \tilde{A} = \boldsymbol{b}^T, \quad \tilde{A} \in \mathbb{R}^{n \times n},\, \boldsymbol{0} \neq \boldsymbol{b} \in \mathbb{R}^{n \times 1},$$

$$\boldsymbol{y}^T (\tilde{A} + \Delta \tilde{A}) = \boldsymbol{b}^T + \Delta \boldsymbol{b}^T, \quad \Delta \tilde{A} \in \mathbb{R}^{n \times n},\, \Delta \boldsymbol{b} \in \mathbb{R}^{n \times 1},$$

with $||\Delta \tilde{A}|| \leq \tilde{\epsilon}_A ||\tilde{A}||$ and $||\Delta \boldsymbol{b}^T|| \leq \tilde{\epsilon}_b ||\boldsymbol{b}^T||$ for some $\tilde{\epsilon}_A, \tilde{\epsilon}_b > 0$ such that $\tilde{\epsilon}_A\, \kappa(\tilde{A}) < 1$. Define

$$\tilde{r}_A := \tilde{\epsilon}_A\, \kappa(\tilde{A}), \quad \tilde{r}_b := \tilde{\epsilon}_b\, \kappa(\tilde{A}).$$

Then, $\tilde{A} + \Delta \tilde{A}$ is nonsingular and

$$\frac{||\boldsymbol{y}^T||}{||\boldsymbol{x}^T||} \leq \frac{1 + \tilde{r}_b}{1 - \tilde{r}_A}, \tag{16}$$

$$\frac{||\boldsymbol{y}^T - \boldsymbol{x}^T||}{||\boldsymbol{x}^T||} \leq \frac{\tilde{r}_A + \tilde{r}_b}{1 - \tilde{r}_A}. \tag{17}$$

Let us next move to applying Lemma OA.B.1 and Corollary OA.B.2 to derive useful inequalities involving Gram matrices.

Lemma OA.B.3 *Assume $\hat{\sigma}_f > 0$ and fix $\boldsymbol{x}_* \in \mathbb{R}^d$, X_n - the training dataset, the kernel function $c(\cdot\,,\cdot)$ and m - the number of nearest neighbours of $\boldsymbol{x}_*$ in X_n selected according to the kerne-induced metric ρ_c. Denote the nearest-neighbour (shifted) Gram matrix as $K_{\mathcal{N}}$. Assume that $K_{\mathcal{N}}$ is invertible and define $K_{\mathcal{N}}^{\infty}$ as in Lemma OA.C.5. Then, we have*

$$\frac{\left\| K_{\mathcal{N}}^{-1}\,\mathbf{k}_{\mathcal{N}}^{*} - \hat{\sigma}_f^2\,(K_{\mathcal{N}}^{\infty})^{-1}\,\mathbf{1} \right\|_1}{\left\| \hat{\sigma}_f^2\,\left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1} \right\|_1} \leq \frac{\epsilon_m + \epsilon_E}{1-\epsilon_E}, \tag{18}$$

$$\frac{\left\| (\mathbf{k}_{\mathcal{N}}^{*})^T\;K_{\mathcal{N}}^{-1} - \hat{\sigma}_f^2\,\mathbf{1}^T\;(K_{\mathcal{N}}^{\infty})^{-1} \right\|_1}{\left\| \hat{\sigma}_f^2\,\mathbf{1}^T\;\left(K_{\mathcal{N}}^{\infty}\right)^{-1} \right\|_1} \leq \frac{\epsilon_m + \epsilon_E}{1-\epsilon_E}, \tag{19}$$

where

$$\epsilon_E := \frac{1}{m}\,\left\| E(\boldsymbol{x}_*, X_n) \right\|_1, \quad \epsilon_m := \max_{1\leq i\leq m} \rho_c^2\left(\boldsymbol{x}_*, \boldsymbol{x}_{n,i}(\boldsymbol{x}*)\right). \tag{20}$$

with $E_{i,j} = \epsilon_{i,j} := \rho_c^2\left(\boldsymbol{x}_{n,i}(\boldsymbol{x}*), \boldsymbol{x}_{n,j}(\boldsymbol{x}*)\right)$, $1 \leq i,j \leq m$. For any function $f : \mathbb{R}^d \to \mathbb{R}$ that satisfies the q-Hölder condition (AR.4) we also have

$$\frac{\left\| K_{\mathcal{N}}^{-1}\,f(X) - f\left(\boldsymbol{x}_*\right)(K_{\mathcal{N}}^{\infty})^{-1}\,\mathbf{1} \right\|_1}{\left\| \left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1} \right\|_1} \leq B_f\,\frac{2L_f\,\min\{d_m^q, 1\} + \epsilon_E}{1-\epsilon_E}. \tag{21}$$

What is more,

$$\frac{\left\| (\mathbf{k}_{\mathcal{N}}^{*})^T\;K_{\mathcal{N}}^{-2} - \hat{\sigma}_f^2\,\mathbf{1}^T\;(K_{\mathcal{N}}^{\infty})^{-2} \right\|_1}{\left\| \hat{\sigma}_f^2\,\mathbf{1}^T\;\left(K_{\mathcal{N}}^{\infty}\right)^{-2} \right\|_1} \leq \frac{\epsilon_m + \epsilon_{E,2}}{1-\epsilon_{E,2}}, \tag{22}$$

where

$$\epsilon_{E,2} = \left(\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\right)^2 \left\| 2\frac{\hat{\sigma}_\xi^2}{\hat{\sigma}_f^2}\,E + E\,\mathbf{1}.\mathbf{1}^T + \mathbf{1}.\mathbf{1}^T\,E - E^2 \right\|_1. \tag{23}$$

Proof The proof of the inequality (18) follows from the application of Lemma OA.B.1 with $A = K_{\mathcal{N}}^{\infty}$, $\boldsymbol{b} = \lim_{n\to\infty}\mathbf{k}_{\mathcal{N}}^{*} = \hat{\sigma}_f^2\,\mathbf{1}$, $A + \Delta A = K_{\mathcal{N}}$ and $\boldsymbol{b} + \Delta\boldsymbol{b} = \mathbf{k}_{\mathcal{N}}^{*}$.

We first calculate the relevant condition number

$$\kappa(A) = \|K_{\mathcal{N}}^{\infty}\|_1\,\left\|(K_{\mathcal{N}}^{\infty})^{-1}\right\|_1.$$

By a direct calculation using the exact forms of the above matrices from Lemma OA.C.5, we find that

$$\|K_{\mathcal{N}}^{\infty}\|_1 = \hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2 = \left(\left\|(K_{\mathcal{N}}^{\infty})^{-1}\right\|_1\right)^{-1}. \tag{24}$$

Thus, $\kappa(A) = 1$. To satisfy the conditions of Lemma OA.B.1, we set $\epsilon_A = ||\Delta A||_1/||A||_1$ and $\epsilon_b = ||\Delta \boldsymbol{b}||_1/||\boldsymbol{b}||_1$. Let us set $\epsilon_A = r_A$ and $\epsilon_b = r_b$. We have

$$[\Delta A]_{ij} = [K_{\mathcal{N}}]_{ij} - [K_{\mathcal{N}}^{\infty}]_{ij} = \hat{\sigma}_f^2 \left(c(\boldsymbol{x}_i/\hat{\ell}, \boldsymbol{x}_j/\hat{\ell}) - 1 \right) = -\hat{\sigma}_f^2 \epsilon_{i,j},$$

thus we get

$$\epsilon_A = \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2} \, \|E\|_1 \leq \frac{1}{m} \, \|E\|_1 = \frac{1}{m} \max_j \sum_{i=1}^m \epsilon_{i,j} < \frac{1}{m} \max_j \sum_{i=1}^m 1 = 1.$$

This proves the first part of this Lemma. For the second part, we note that $||\boldsymbol{b}||_1 = \hat{\sigma}_f^2 \, ||\mathbf{1}||_1 = m\hat{\sigma}_f^2$ and $||\Delta \boldsymbol{b}||_1 = ||\mathbf{k}_{\mathcal{N}}^* - \hat{\sigma}_f^2 \, \mathbf{1}||_1 = \hat{\sigma}_f^2 \sum_{i=0}^m \epsilon_i$, where $\epsilon_i := \rho_c^2 \left(\boldsymbol{x}_{n,i}(\boldsymbol{x}*), \boldsymbol{x}* \right)$. Finally,

$$\epsilon_b = \frac{1}{m} \sum_{i=0}^m \epsilon_i = \frac{1}{m} \sum_{i=1}^m (1 - c(\boldsymbol{x}_i/\hat{\ell}, \boldsymbol{x}_*/\hat{\ell})) < 1$$

and we note that $\epsilon_b \leq \epsilon_m$.

The proof of the inequality (19) is fully analogous to the proof of the inequality (18). It follows from the application of Corollary OA.B.2 with $A = K_{\mathcal{N}}^{\infty}$, $\boldsymbol{b}^T = \lim_{n\to\infty} \left(\mathbf{k}_{\mathcal{N}}^*\right)^T = \hat{\sigma}_f^2 \, \mathbf{1}^T$, $A + \Delta A = K_{\mathcal{N}}$ and $\boldsymbol{b}^T + \Delta \boldsymbol{b}^T = \left(\mathbf{k}_{\mathcal{N}}^*\right)^T$.

The proof of (21) is fully analogous to the proof of (18) with $A = K_{\mathcal{N}}^{\infty}$, $\boldsymbol{b} = \lim_{n\to\infty} f(X) = f\left(\boldsymbol{x}_*\right) \, \mathbf{1}$, $A + \Delta A = K_{\mathcal{N}}$ and $\boldsymbol{b} + \Delta \boldsymbol{b} = f(X)$. Lemma OA.B.1 asserts that

$$\frac{\left\| K_{\mathcal{N}}^{-1} \, f(X) - f\left(\boldsymbol{x}_*\right) \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} \right\|_1}{|f\left(\boldsymbol{x}_*\right)| \, \left\| \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} \right\|_1} \leq \frac{\epsilon_b + \epsilon_E}{1 - \epsilon_E},$$

where $\epsilon_b = ||\Delta \boldsymbol{b}||_1/||\boldsymbol{b}||_1$. Using the Hölder property and the boundedness of the function f, we get

$$\begin{aligned} \epsilon_b = \frac{1}{m|f\left(\boldsymbol{x}_*\right)|} \sum_{i=1}^m |f(\boldsymbol{x}_i) - f(\boldsymbol{x}_*)| &\leq \frac{1}{m|f\left(\boldsymbol{x}_*\right)|} \sum_{i=1}^m \min\{L_f \|\boldsymbol{x}_i - \boldsymbol{x}_*\|_2^q, 2B_f\} \\ &\leq \frac{1}{|f\left(\boldsymbol{x}_*\right)|} \, \min\{L_f d_m^q, 2B_f\} \leq \frac{2L_f B_f}{|f\left(\boldsymbol{x}_*\right)|} \min\{d_m^q, 1\}. \end{aligned}$$

In order to prove the inequality (22) we use Corollary OA.B.2 with $\tilde{A} = \left(K_{\mathcal{N}}^{\infty}\right)^2$, $\boldsymbol{b} = \lim_{n\to\infty} \mathbf{k}_{\mathcal{N}}^* = \hat{\sigma}_f^2 \, \mathbf{1}$, $\tilde{A} + \Delta\tilde{A} = K_{\mathcal{N}}^2$ and $\boldsymbol{b} + \Delta \boldsymbol{b} = \mathbf{k}_{\mathcal{N}}^*$. We first calculate the relevant condition number

$$\kappa(\tilde{A}) = \left\| \left(K_{\mathcal{N}}^{\infty}\right)^2 \right\|_1 \, \left\| \left(K_{\mathcal{N}}^{\infty}\right)^{-2} \right\|_1 .$$

By a direct calculation using the exact forms of the above matrices using Lemma OA.C.5, we find that

$$\left\| \left(K_{\mathcal{N}}^{\infty}\right)^2 \right\|_1 = \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)^2 = \left(\left\| \left(K_{\mathcal{N}}^{\infty}\right)^{-2} \right\|_1 \right)^{-1}.$$

Thus, $\kappa(\tilde{A}) = 1$. To satisfy the conditions of Corollary OA.B.2, we set $\tilde{\epsilon}_A = ||\Delta\tilde{A}||_1/||\tilde{A}||_1$ and $\tilde{\epsilon}_b = \left\|\Delta \boldsymbol{b}^T\right\|_1 / \left\|\boldsymbol{b}^T\right\|_1$. Let us set $\tilde{\epsilon}_A = \tilde{r}_A =: \epsilon_{E,2}$ and $\tilde{\epsilon}_b = \tilde{r}_b$. We have

$$\left|\left[\Delta\tilde{A}\right]_{ij}\right| = \left|\left[K_{\mathcal{N}}^2\right]_{ij} - \left[(K_{\mathcal{N}}^{\infty})^2\right]_{ij}\right| = 2\hat{\sigma}_\xi^2\left(\hat{\sigma}_f^2 - k_\theta(\boldsymbol{x}_i, \boldsymbol{x}_j)\right)$$
$$+ \sum_{k=1}^{m}\left(\hat{\sigma}_f^4 - k_\theta(\boldsymbol{x}_i, \boldsymbol{x}_k)k_\theta(\boldsymbol{x}_j, \boldsymbol{x}_k)\right) = 2\hat{\sigma}_\xi^2\hat{\sigma}_f^2\,\epsilon_{ij} + \hat{\sigma}_f^4\sum_{k=1}^{m}\left(\epsilon_{ik} + \epsilon_{jk} - \epsilon_{ik}\epsilon_{jk}\right).$$

Thus,

$$\left|\left[\Delta\tilde{A}\right]_{ij}\right| < 2\hat{\sigma}_\xi^2\hat{\sigma}_f^2 + m\hat{\sigma}_f^4,$$

which implies that

$$\epsilon_{E,2} = \frac{1}{\left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)^2}\max_{1\le i\le m}\sum_{j=1}^{m}\left|\left[\Delta\tilde{A}\right]_{ij}\right| < m\frac{2\hat{\sigma}_\xi^2\hat{\sigma}_f^2 + m\hat{\sigma}_f^4}{\left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)^2} \le 1.$$

Finally, note that $\left\|\boldsymbol{b}^T\right\|_1 = \hat{\sigma}_f^2\,\left\|\mathbf{1}^T\right\|_1 = \hat{\sigma}_f^2$ and

$$||\Delta\boldsymbol{b}^T||_1 = \left\|(\mathbf{k}_{\mathcal{N}}^*)^T - \hat{\sigma}_f^2\,\mathbf{1}^T\right\|_1 = \hat{\sigma}_f^2\max_{1\le i\le m}\rho_c^2(\boldsymbol{x}_*, \boldsymbol{x}_{n,i}) \le \hat{\sigma}_f^2\,\epsilon_m.$$

Thus, $\tilde{\epsilon}_b \le \epsilon_m < 1$. ■

Lemma OA.B.4 (Relations between the epsilons.) *Let ϵ_m, ϵ_E and $\epsilon_{E,2}$ be as defined in Equations* (20) *and* (23) *in Lemma OA.B.3. The following bounds hold*

$$\epsilon_E \le 4\,\epsilon_m, \qquad \epsilon_{E,2} \le 2\,\epsilon_E.$$

Proof Recall the definition of ϵ_E, i.e., $\epsilon_E = \max_{1\le j\le m}\sum_{i=1}^{m}\epsilon_{i,j}$. The kernel-induced metric satisfies the triangle inequality, i.e.,

$$\sqrt{\epsilon_{i,j}} = \rho_c(\boldsymbol{x}_{n,i}(\boldsymbol{x}_*), \boldsymbol{x}_{n,j}(\boldsymbol{x}_*)) \le \rho_c(\boldsymbol{x}_*, \boldsymbol{x}_{n,i}(\boldsymbol{x}_*)) + \rho_c(\boldsymbol{x}_*, \boldsymbol{x}_{n,j}(\boldsymbol{x}_*)) \le 2\sqrt{\epsilon_m}.$$

By squaring both sides of this inequality we get that $\epsilon_E \le 4\,\epsilon_m$.

In order to derive the bound for $\epsilon_{E,2}$, we first note that

$$\epsilon_{E,2} \le \frac{1}{m}\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\left(2\frac{\hat{\sigma}_\xi^2}{\hat{\sigma}_f^2}\,\|E\|_1 + \left\|E\,\mathbf{1}.\mathbf{1}^T + \mathbf{1}.\mathbf{1}^T\,E - E^2\right\|_1\right).$$

Next,

$$\left\|E\,\mathbf{1}.\mathbf{1}^T + \mathbf{1}.\mathbf{1}^T\,E - E^2\right\|_1 = \max_{1\le j\le m}\sum_{i=1}^{m}\sum_{k=1}^{m}\left(\epsilon_{ik} + \epsilon_{jk} - \epsilon_{ik}\epsilon_{jk}\right) \le m\,\|E\|_1 + \sum_{i=1}^{m}\sum_{k=1}^{m}\epsilon_{ik},$$

where we have used the facts that

$$\left[E^2\right]_{i,j} = \sum_{k=1}^{m}\epsilon_{ik}\epsilon_{jk} \ge 0, \quad \max_{1\le j\le m}\sum_{i=1}^{m}\sum_{k=1}^{m}\epsilon_{jk} = m\max_{1\le j\le m}\sum_{k=1}^{m}\epsilon_{jk} = m\,\|E\|_1.$$

Finally, note that $\sum_i \sum_k \epsilon_{ik} \leq m \|E\|_1$. Thus, we obtain

$$\epsilon_{E,2} \leq \frac{1}{m} \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2} \left(2\frac{\hat{\sigma}_\xi^2}{\hat{\sigma}_f^2} \|E\|_1 + 2m \|E\|_1 \right) = \frac{2}{m} \|E\|_1 = 2\,\epsilon_E.$$

■

Appendix OA.C. Proving Consistency of $GPnn$ and $NNGP$ Regression

We first derive the limits of the performance measures defined in Equations (9) - (11) when the training data size n tends to infinity. Firstly, we prove the bias-variance decomposition of the MSE.

Lemma OA.C.1 (Bias-variance decomposition of MSE.) *Let the test and training covariates $\mathcal{X}_*, \mathcal{X}_1, \ldots, \mathcal{X}_n$ be i.i.d. from $P_\mathcal{X}$ and let $\boldsymbol{X}_n = (\mathcal{X}_1, \ldots, \mathcal{X}_n)$. Assume that the training and test responses are generated as $\mathcal{Y}_i = g(\mathcal{X}_i) + \Xi_i$, where g is a (possibly random) measurable function that is sampled independently of the covariates and Ξ is the random noise which is mean-zero at every location and are uncorrelated given the covariates, i.e.,*

$$\mathrm{Cov}\left[\Xi_i, \Xi_j \mid \mathcal{X}_i, \mathcal{X}_j\right] = 0 \quad \text{for} \quad i \neq j, \quad \mathrm{Cov}\left[\Xi_*, \Xi_i \mid \mathcal{X}_*, \mathcal{X}_i\right] = 0,$$

and are independent of g. Let $\hat{\mu} = \hat{\mu}\left(\mathcal{X}_*, \boldsymbol{X}_n, \boldsymbol{y}_n\right)$ be an estimator of the test response $\mathcal{Y}_*$ that depends linearly on the training responses $\boldsymbol{y}_n$. We have the following bias-variance decomposition of the corresponding MSE

$$MSE := \mathbb{E}\left[(\hat{\mu} - \mathcal{Y}_*)^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \sigma_\xi^2\left(\mathcal{X}_*\right) + \mathrm{Bias}^2\left(\mathcal{X}_*, \boldsymbol{X}_n\right) + \mathrm{Var}\left[\hat{\mu} - g(\mathcal{X}_*) \mid \mathcal{X}_*, \boldsymbol{X}_n\right], \tag{25}$$

where

$$\sigma_\xi^2(\mathcal{X}) := \mathrm{Var}\left[\Xi \mid \mathcal{X}\right], \quad \mathrm{Bias}\left(\mathcal{X}_*, \boldsymbol{X}_n\right) := \mathbb{E}\left[\hat{\mu} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - \mathbb{E}\left[g(\mathcal{X}_*) \mid \mathcal{X}_*\right].$$

When applied to GPnn response model (1) we have $g \equiv f$ deterministic, thus

$$\begin{aligned} \mathrm{Bias}_{GPnn}\left(\mathcal{X}_*, \boldsymbol{X}_n\right) &= \Gamma\, \mathbf{k}_\mathcal{N}^{*T} K_\mathcal{N}^{-1} f(\boldsymbol{X}_\mathcal{N}) - f(\mathcal{X}_*) \\ \mathrm{Var}\left[\tilde{\mu}_{GPnn} - g\left(\mathcal{X}_*\right) \mid \mathcal{X}_*, \boldsymbol{X}_n\right] &= \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma^2\, \mathbf{k}_\mathcal{N}^{*T} K_\mathcal{N}^{-1} \Sigma_{\boldsymbol{\xi}} K_\mathcal{N}^{-1} \mathbf{k}_\mathcal{N}^*, \end{aligned} \tag{26}$$

where $\boldsymbol{X}_\mathcal{N} = \boldsymbol{X}_\mathcal{N}\left(\mathcal{X}_*, \boldsymbol{X}_n\right)$ is the set of nearest neighbours of $\mathcal{X}_*$ in $\boldsymbol{X}_n$ and

$$\Sigma_{\boldsymbol{\xi}} = \mathrm{diag}\left\{\sigma_\xi^2\left(\mathcal{X}_{n,1}\right), \ldots, \sigma_\xi^2\left(\mathcal{X}_{n,m}\right)\right\}.$$

In the $NNGP$ response model (2) we have $g\left(\mathcal{X}\right) = \boldsymbol{t}\left(\mathcal{X}\right)^T.\boldsymbol{b} + w\left(\mathcal{X}\right)$, thus

$$\begin{aligned} \mathrm{Bias}_{NNGP}\left(\mathcal{X}_*, \boldsymbol{X}_n\right) &= \Gamma\, k_\mathcal{N}^{*T} K_\mathcal{N}^{-1} T_\mathcal{N}(\boldsymbol{b} - \hat{\boldsymbol{b}}) - \boldsymbol{t}_*^T.(\boldsymbol{b} - \hat{\boldsymbol{b}}). \\ \mathrm{Var}\left[\tilde{\mu}_{NNGP} - g(\mathcal{X}_*) \mid \mathcal{X}_*, \boldsymbol{X}_n\right] &= \sigma_w^2 + \Gamma^2\, k_\mathcal{N}^{*T} K_\mathcal{N}^{-1} \left(\tilde{K}_\mathcal{N} + \Sigma_\xi\right) K_\mathcal{N}^{-1} k_\mathcal{N}^* \\ &\quad - 2\Gamma\, k_\mathcal{N}^{*T} K_\mathcal{N}^{-1} \tilde{k}_\mathcal{N}^*, \end{aligned} \tag{27}$$

where $\left[\tilde{k}_\mathcal{N}^*\right]_i := \tilde{k}\left(\boldsymbol{x}_{n,i}\left(\boldsymbol{x}_*\right), \boldsymbol{x}_*\right)$ and $\left[\tilde{K}_\mathcal{N}\right]_{i,j} := \tilde{k}\left(\boldsymbol{x}_{n,i}\left(\boldsymbol{x}_*\right), \boldsymbol{x}_{n,j}\left(\boldsymbol{x}_*\right)\right)$, $1 \leq i, j \leq m$.

Proof In the assumed response model, we have

$$\mathbb{E}\left[(\hat{\mu}-\mathcal{Y}_*)^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \sigma_\xi^2(\mathcal{X}_*) + \mathbb{E}\left[\delta^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right], \quad \delta := \hat{\mu} - g(\mathcal{X}_*).$$

By the definition of variance, we have

$$\mathbb{E}\left[\delta^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \mathrm{Bias}^2(\mathcal{X}_*, \boldsymbol{X}_n) + \mathrm{Var}\left[\delta \mid \mathcal{X}_*, \boldsymbol{X}_n\right]$$

where $\mathrm{Bias}(\mathcal{X}_*, \boldsymbol{X}_n) := \mathbb{E}\left[\delta \mid \mathcal{X}_*, \boldsymbol{X}_n\right]$. Using the standard identity $\mathrm{Var}[A-B] = \mathrm{Var}[A] + \mathrm{Var}[B] - 2\mathrm{Cov}[A,B]$, we get

$$\mathrm{Var}\left[\delta \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \mathrm{Var}\left[\hat{\mu} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] + \mathrm{Var}\left[g(\mathcal{X}_*) \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - 2\mathrm{Cov}\left[\hat{\mu}, g(\mathcal{X}_*) \mid \mathcal{X}_*, \boldsymbol{X}_n\right].$$

Finally, $\mathrm{Var}\left[g(\mathcal{X}_*) \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \mathrm{Var}\left[g(\mathcal{X}_*) \mid \mathcal{X}_*\right]$, since g is drawn independently of the covariates. This proves Equation (25).

For $GPnn$ we have

$$\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\, \mathbb{E}\left[\boldsymbol{Y}_{\mathcal{N}} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}),$$
$$\mathbb{E}\left[(\tilde{\mu}_{GPnn})^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma^2\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\, \mathbb{E}\left[\boldsymbol{Y}_{\mathcal{N}}.\boldsymbol{Y}_{\mathcal{N}}^T \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\, K_{\mathcal{N}}^{-1}\, \mathbf{k}_{\mathcal{N}}^*.$$

Further,

$$\begin{aligned}\mathbb{E}\left[\mathcal{Y}_{n,i}\mathcal{Y}_{n,j} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] &= f(\mathcal{X}_{n,i})f(\mathcal{X}_{n,j}) + \mathbb{E}\left[2f(\mathcal{X}_{n,i})\Xi_{n,j} + \Xi_{n,i}\Xi_{n,j} \mid \mathcal{X}_{n,j}\right] \\ &= f(\mathcal{X}_{n,i})f(\mathcal{X}_{n,j}) + \left[\Sigma_{\boldsymbol{\xi}}\right]_{ij},\end{aligned}$$

where in the last equality we have used the noise model assumptions (mean-zero and independent given the nearest-neighbours). Thus,

$$\mathbb{E}\left[(\tilde{\mu}_{GPnn})^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \left(\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\right)^2 + \Gamma^2\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\, \Sigma_{\boldsymbol{\xi}}\, K_{\mathcal{N}}^{-1}\, \mathbf{k}_{\mathcal{N}}^*.$$

Putting together all these above formulae and using the definition of variance yields Equations (26).

In $NNGP$, we have $\mathbb{E}\left[\boldsymbol{Y}_{\mathcal{N}} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = T_{\mathcal{N}}\boldsymbol{b}$, thus

$$\mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \boldsymbol{t}_*^T.\hat{\boldsymbol{b}} + \Gamma\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} T_{\mathcal{N}} \left(\boldsymbol{b} - \hat{\boldsymbol{b}}\right).$$

Thus,

$$\tilde{\mu}_{NNGP} - \mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \left(\boldsymbol{w}\left(\boldsymbol{X}_{\mathcal{N}}\right) + \Xi_{\mathcal{N}}\right),$$

and the variance reads

$$\begin{aligned}&\mathrm{Var}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma^2\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\mathbb{E}\left[\left(\boldsymbol{w}\left(\boldsymbol{X}_{\mathcal{N}}\right) + \Xi_{\mathcal{N}}\right).\left(\boldsymbol{w}\left(\boldsymbol{X}_{\mathcal{N}}\right) + \Xi_{\mathcal{N}}\right)^T \mid \mathcal{X}_*, \boldsymbol{X}_n\right] \times \\ &\times K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* = \Gamma^2\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\mathbb{E}\left[\boldsymbol{w}\left(\boldsymbol{X}_{\mathcal{N}}\right).\boldsymbol{w}\left(\boldsymbol{X}_{\mathcal{N}}\right)^T + \Xi_{\mathcal{N}}.\Xi_{\mathcal{N}}^T \mid \mathcal{X}_*, \boldsymbol{X}_n\right] K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* \\ &= \Gamma^2\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\left(\tilde{K}_{\mathcal{N}} + \Sigma_\xi\right) K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^*,\end{aligned}$$

where we have used the fact that the sample path $w(\cdot)$ and the noise variables are independent. This proves Equations (27). The remaining components of the bias-variance decomposition for $NNGP$ are completely analogous, so we skip them. ■

In order to establish the desired $n \to \infty$ limits we will use the following result concerning the shrinking of nearest neighbour sets.

Lemma OA.C.2 *(Györfi et al., 2002, Lemma 6.1) Let $\boldsymbol{X}_n = (\mathcal{X}_i)_{i=1}^n$ be a sampling sequence of i.i.d. points from the distribution $P_{\mathcal{X}}$ and let $\boldsymbol{x}_* \in \mathrm{Supp}(P_{\mathcal{X}})$. Assume that the nearest neighbours are chosen according to the Euclidean distance. Allow the number of nearest neighbours m to change with n so that $\lim_{n\to\infty} m_n/n = 0$ (in particular, m can be fixed). Define d_m as the distance of the m-th nearest neighbour of $\boldsymbol{x}_*$ in $\boldsymbol{X}_n$ i.e.,*

$$d_m(\boldsymbol{x}_*, \boldsymbol{X}_n) := \|\mathcal{X}_{n,m}(\boldsymbol{x}_*) - \boldsymbol{x}_*\|_2 ,$$

where $\|\cdot\|_2$ deontes the Euclidean distance in $\mathbb{R}^d$. Then, $d_m(\boldsymbol{x}_*, X_n) \xrightarrow{n\to\infty} 0$ with probability one.

Remark OA.C.3 *Using a fully analogous technique to the one used in the proof of Lemma 6.1 in (Györfi et al., 2002), one can replace the Euclidean metric with the kernel-induced (pseudo)metric ρ_c, i.e. choose the nearest-neighbours according to the (pseudo)metric ρ_c. In result, we obtain that for every $\boldsymbol{x}_* \in \mathrm{Supp}_{\rho_c}(P_{\mathcal{X}})$ we have*

$$\epsilon_m(\boldsymbol{x}_*, \boldsymbol{X}_n) := \rho_c\left(\mathcal{X}_{n,m}(\boldsymbol{x}_*), \boldsymbol{x}_*\right) \xrightarrow{n\to\infty} 0$$

with probability one.

Corollary OA.C.4 *By Remark OA.C.3 and the triangle inequality*

$$\rho_c(\boldsymbol{x}_{n,i}, \boldsymbol{x}_{n,j}) \le \rho_c(\boldsymbol{x}_{n,i}, \boldsymbol{x}_*) + \rho_c(\boldsymbol{x}_{n,j}, \boldsymbol{x}_*) \le 2\rho_c(\boldsymbol{x}_{n,m}, \boldsymbol{x}_*)$$

we also have $\rho_c(\mathcal{X}_{n,i}, \mathcal{X}_{n,j}) \xrightarrow{n\to\infty} 0$ with probability one for all $1 \le i, j \le m$. Consequently, the kernel elements defined in Equation (3) have the following limits with probability one

$$[K_{\mathcal{N}}]_{ij} \xrightarrow{n\to\infty} \hat{\sigma}_f^2 + \hat{\sigma}_\xi^2\,\delta_{ij}, \quad [\boldsymbol{k}_{\mathcal{N}}^*]_j \xrightarrow{n\to\infty} \hat{\sigma}_f^2, \quad 1 \le i, j \le m.$$

Lemma OA.C.5 (Gram matrix limits.) *Under the assumptions (AC.1-3) with $m \in \mathbb{N}_{>0}$ fixed and $\boldsymbol{x}_* \in \mathrm{Supp}_{\rho_c}(P_{\mathcal{X}})$ fixed the following limits hold with probability one.*

$$K_{\mathcal{N}} \xrightarrow{n\to\infty} K_{\mathcal{N}}^\infty := \hat{\sigma}_\xi^2\,\mathbb{1} + \hat{\sigma}_f^2\,\mathbf{1}.\mathbf{1}^T, \quad \boldsymbol{k}_{\mathcal{N}}^* \xrightarrow{n\to\infty} \hat{\sigma}_f^2\,\mathbf{1}, \tag{28}$$

$$K_{\mathcal{N}}^{-1} \xrightarrow{n\to\infty} (K_{\mathcal{N}}^\infty)^{-1} = \frac{1}{\hat{\sigma}_\xi^2}\left(\mathbb{1} - \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\,\mathbf{1}.\mathbf{1}^T\right), \tag{29}$$

$$f\left(\boldsymbol{X}_{\mathcal{N}}(\boldsymbol{x}_*)\right) \xrightarrow{n\to\infty} f(\boldsymbol{x}_*)\,\mathbf{1}, \tag{30}$$

where $\mathbb{1}$ is the $m \times m$ identity matrix and $\mathbf{1}$ is the column vector of ones.

Proof The limits (28) follow straightforwardly form Corollary OA.C.4. The fact that $K_{\mathcal{N}}^{-1} \xrightarrow{n\to\infty} (K_{\mathcal{N}}^\infty)^{-1}$ follows from the continuity of matrix inverse. The RHS of Equation (29) is calculated using the Sherman–Morrison formula (Sherman and Morrison, 1950). The limit (30) follows directly from the assumption (AC.3) stating that f is almost continuous. ■

In the next Lemma we show that the estimators defined in Equations (5) and (6) are asymptotically unbiased (given the test point).

Lemma OA.C.6 (Pointwise limits of bias and variance.) *Under the assumptions (AC.1-4) with $m \in \mathbb{N}_{>0}$ fixed and test point $\boldsymbol{x}_* \in \mathrm{Supp}_{\rho_c}(P_{\mathcal{X}})$ fixed the following limit for the predictive noise variance defined in* (4) *holds with probability one.*

$$\sigma_{\mathcal{N}}^{*\,2} \xrightarrow{n\to\infty} \hat{\sigma}_\xi^2 \left(1 + \frac{1}{m\Gamma}\right), \quad \Gamma := \frac{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}{m\hat{\sigma}_f^2}. \tag{31}$$

We also have that the following limits for the estimators $\tilde{\mu}_{GPnn}$ and $\tilde{\mu}_{NNGP}$ defined in (5) *and* (6) *hold with probability one for their respective response models.*

$$\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} f(\boldsymbol{x}_*), \quad \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m}, \tag{32}$$

$$\begin{aligned} &\mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} \boldsymbol{t}^T(\boldsymbol{x}_*).\boldsymbol{b}, \\ &\mathrm{Var}\left[\tilde{\mu}_{NNGP} - \boldsymbol{t}^T(\boldsymbol{x}_*).\boldsymbol{b} - w(\boldsymbol{x}_*) \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m}. \end{aligned} \tag{33}$$

What is more, if $m \equiv m_n$ grows with n such that $\lim_{n\to\infty} m_n/n = 0$, we have that with probability one

$$\sigma_{\mathcal{N}}^{*\,2} \xrightarrow{n\to\infty} \hat{\sigma}_\xi^2$$

and

$$\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} f(\boldsymbol{x}_*), \quad \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} 0, \tag{34}$$

$$\begin{aligned} &\mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} \boldsymbol{t}^T(\boldsymbol{x}_*).\boldsymbol{b}, \\ &\mathrm{Var}\left[\tilde{\mu}_{NNGP} - \boldsymbol{t}^T(\boldsymbol{x}_*).\boldsymbol{b} - w(\boldsymbol{x}_*) \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \xrightarrow{n\to\infty} 0. \end{aligned} \tag{35}$$

Proof Let us start with *GPnn*. By Lemma OA.C.1, we have

$$\begin{aligned} \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] &= \Gamma\, \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}}), \\ \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] &= \Gamma^2\, \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} \Sigma_{\boldsymbol{\xi}} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^{*}. \end{aligned}$$

When m is fixed, we insert the limits from Lemma OA.C.5 and use the continuity of f from (AC.3) to get

$$\Gamma\, \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}}) \to \Gamma\, f(\boldsymbol{x}_*)\, \hat{\sigma}_f^2\, \mathbf{1}^T \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} = f(\boldsymbol{x}_*) = \mathbb{E}\left[\mathcal{Y} \mid \mathcal{X} = \boldsymbol{x}_*\right],$$

which shows that the *GPnn* bias term tends to zero. Because $\sigma_\xi^2(\boldsymbol{x})$ is assumed to be an almost continuous function we have that $\Sigma_{\boldsymbol{\xi}} \xrightarrow{n\to\infty} \sigma_\xi^2(\boldsymbol{x}_*)\mathbb{1}$ with probability one. Inserting the limits from Lemma OA.C.5 to the variance expression yields after some algebra

$$\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \to \frac{\sigma_\xi(\boldsymbol{x}_*)^2}{m}. \tag{36}$$

This proves Equations (32). The limit for $\sigma_{\mathcal{N}}^{*\,2}$ is derived in a fully analogous way.

When $m \equiv m_n$ grows with n, we decompose $\mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} = \hat{\sigma}_f^2 \mathbf{1}^T \left(K_{\mathcal{N}}^{\infty}\right)^{-1} + \Delta^T$ and $f(X_{\mathcal{N}}) = f(\boldsymbol{x}_*)\,\mathbf{1} + \delta$ to get

$$\begin{aligned} \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}}) &= \left(\hat{\sigma}_f^2 \mathbf{1}^T \left(K_{\mathcal{N}}^{\infty}\right)^{-1} + \Delta^T\right).\left(f(\boldsymbol{x}_*)\,\mathbf{1} + \delta\right) \\ &= f(\boldsymbol{x}_*)\, \hat{\sigma}_f^2\, \mathbf{1}^T \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} + \hat{\sigma}_f^2 \mathbf{1}^T \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \delta + f(\boldsymbol{x}_*)\, \Delta^T.\mathbf{1} + \Delta^T.\delta \end{aligned}$$

Using the formula from Equation (29) we get that the first term has the limit

$$\Gamma_n\, f(\boldsymbol{x}_*)\, \hat{\sigma}_f^2\, \mathbf{1}^T\, \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} = f(\boldsymbol{x}_*),$$

where Γ_n now depends on n (but tends to one, so the Γ-correction is inconsequential in the limit). It remains to show that the last three terms tend to zero as $n \to \infty$. First,

$$\begin{aligned}\Gamma_n \left|\hat{\sigma}_f^2 \mathbf{1}^T\, \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \delta\right| &\leq \Gamma_n \hat{\sigma}_f^2 \left\|\mathbf{1}^T\, \left(K_{\mathcal{N}}^{\infty}\right)^{-1}\right\|_1 \|\delta\|_1 = \frac{1}{m}\|\delta\|_1 \\ &\leq \max_{1\leq i\leq m} \left|f\left(\mathcal{X}_{n,i}(\boldsymbol{x}_*)\right) - f\left(\boldsymbol{x}_*\right)\right| \xrightarrow{n\to\infty} 0,\end{aligned}$$

where we have used the formula from Equation (29) and in the last step we have used Remark OA.C.3 together with the continuity of f from (AC.3). Next,

$$\Gamma_n \left|\Delta^T.\mathbf{1}\right| \leq m \left\|\Delta^T\right\|_1 \leq \frac{\epsilon_m + \epsilon_E}{1-\epsilon_E} \xrightarrow{n\to\infty} 0,$$

where we have used: i) Equation (19) from Lemma OA.B.3 to bound $\left\|\Delta^T\right\|_1$, ii) Remark OA.C.3 to get $\epsilon_m \xrightarrow{n\to\infty} 0$, iii) Corollary OA.C.4 to get $\epsilon_E \xrightarrow{n\to\infty} 0$. Similarly, we get

$$\left|\Delta^T.\delta\right| \leq \left\|\Delta^T\right\|_1 \|\delta\|_1 \leq \frac{1}{\Gamma_n}\frac{\epsilon_m + \epsilon_E}{1-\epsilon_E} \max_{1\leq i\leq m} \left|f\left(\boldsymbol{x}_{n,i}(\boldsymbol{x}_*)\right) - f\left(\boldsymbol{x}_*\right)\right| \xrightarrow{n\to\infty} 0.$$

This proves the first part of (34). To prove the variance-part of (34), we decompose

$$K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^* = \hat{\sigma}_f^2 \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} + \Delta = \frac{1}{m\Gamma_n}\mathbf{1} + \Delta, \quad \Sigma_{\boldsymbol{\xi}} = \sigma_\xi^2(\boldsymbol{x}_*)\mathbb{1} + \delta\Sigma_{\boldsymbol{\xi}}$$

to get

$$\begin{aligned}\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] =& \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m_n} + 2\Gamma_n \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m_n}\mathbf{1}^T.\Delta + \Gamma_n^2 \sigma_\xi^2(\boldsymbol{x}_*)\Delta^T.\Delta + \frac{1}{m_n^2}\mathbf{1}^T.\delta\Sigma_{\boldsymbol{\xi}}\mathbf{1} \\ &+ \frac{2\Gamma_n}{m_n}\mathbf{1}^T.\delta\Sigma_{\boldsymbol{\xi}}\Delta + \Gamma_n^2 \Delta^T.\delta\Sigma_{\boldsymbol{\xi}}\Delta\end{aligned}$$

Next, we show that all of the above terms tend to zero with probability one as $n \to \infty$. Since m_n grows with n, the first term vanishes as $n \to \infty$. Next,

$$\begin{aligned}\left|\frac{\Gamma_n}{m_n}\mathbf{1}^T.\Delta\right| &\leq \frac{\Gamma_n}{m_n}\|\mathbf{1}^T\|_1 \|\Delta\|_1 \leq \frac{1}{m_n}\frac{\epsilon_m + \epsilon_E}{1-\epsilon_E} \xrightarrow{n\to\infty} 0, \\ \Gamma_n^2 \left|\Delta^T.\Delta\right| &\leq \Gamma_n^2 \|\Delta^T\|_1 \|\Delta\|_1 \leq \frac{1}{m_n}\frac{\epsilon_m + \epsilon_E}{1-\epsilon_E} \xrightarrow{n\to\infty} 0,\end{aligned}$$

where we have used Equations (18) and (19) from Lemma OA.B.3 to bound $\|\Delta\|_1$ and $\left\|\Delta^T\right\|_1$ and Remark OA.C.3 with Corollary OA.C.4 to get $\epsilon_m \xrightarrow{n\to\infty} 0$ and $\epsilon_E \xrightarrow{n\to\infty} 0$. The remaining terms tend to zero using the same arguments together with the assumption that $\sigma_\xi^2(\boldsymbol{x})$ is an almost continuous function which implies that

$$\|\delta\Sigma_{\boldsymbol{\xi}}\|_1 = \max_{1\leq i\leq m} \left|\sigma_\xi^2(\boldsymbol{x}_*) - \sigma_\xi^2(\boldsymbol{x}_{n,i})\right| \xrightarrow{n\to\infty} 0$$

with probability one. This proves the second part of (34) for $GPnn$.

Let us next move to proving the bias and variance limits for $NNGP$. By Lemma OA.C.1, we have

$$\mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] = \boldsymbol{t}_*^T.(\boldsymbol{b} - \hat{\boldsymbol{b}}) - \Gamma\, k_{\mathcal{N}}^{*T}\, K_{\mathcal{N}}^{-1} T_{\mathcal{N}}(\boldsymbol{b} - \hat{\boldsymbol{b}}),$$

$$\begin{aligned}\mathrm{Var}\left[\tilde{\mu}_{NNGP} - \boldsymbol{t}^T(\boldsymbol{x}_*).\boldsymbol{b} - w(\boldsymbol{x}_*) \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] &= \sigma_w^2 + \Gamma^2\, k_{\mathcal{N}}^{*T}\, K_{\mathcal{N}}^{-1}\left(\tilde{K}_{\mathcal{N}} + \Sigma_\xi\right) K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* \\ &\quad - 2\Gamma\, k_{\mathcal{N}}^{*T}\, K_{\mathcal{N}}^{-1} \tilde{k}_{\mathcal{N}}^*.\end{aligned}$$

Decompose the variance into the noise-part and and random field (RF)-part

$$\mathrm{Var}_{noise} := \Gamma^2\, k_{\mathcal{N}}^{*T}\, K_{\mathcal{N}}^{-1} \Sigma_\xi K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^*, \quad \mathrm{Var}_{RF} := \sigma_w^2 + \Gamma^2\, k_{\mathcal{N}}^{*T}\, K_{\mathcal{N}}^{-1} \tilde{K}_{\mathcal{N}} K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* - 2\Gamma\, k_{\mathcal{N}}^{*T}\, K_{\mathcal{N}}^{-1} \tilde{k}_{\mathcal{N}}^*.$$

We recognise that $\mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right]$ and Var_{noise} are effectively bias and variance of $GPnn$ regression with the effective regression function $f_{eff}(\mathcal{X}) := \boldsymbol{t}(\mathcal{X})^T.(\boldsymbol{b} - \hat{\boldsymbol{b}})$. Thus, we can directly apply the $GPnn$-results (32) and (34) to show that $\mathbb{E}\left[\tilde{\mu}_{NNGP} \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n\right] \to 0$ both when m is fixed and when m grows with n. Similarly, we get that $\mathrm{Var}_{noise} \to \sigma_\xi^2(\boldsymbol{x}_*)$ when m is fixed and $\mathrm{Var}_{noise} \to 0$ when m grows with n.

What remains to show is that $\mathrm{Var}_{RF} \to 0$ with probability one both when m is fixed and when m grows with n. This is straightforward to prove when m is fixed. Then, we can plug in the limits from Lemma OA.C.5 and use assumption (AC.5) together with Remark OA.C.3 and Corollary OA.C.4 to obtain that all the nearest-neighbours of $\boldsymbol{x}_*$ tend to $\boldsymbol{x}_*$ as $n \to \infty$ in both (pseudo)metrics $\rho_{\tilde{c}}$ and ρ_c with probability one. Consequently, the following limits hold with probability one.

$$K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* \xrightarrow{n\to\infty} \frac{1}{m\Gamma}\mathbf{1}, \quad \tilde{K}_{\mathcal{N}} \xrightarrow{n\to\infty} \sigma_w^2\, \mathbf{1}.\mathbf{1}^T, \quad \tilde{k}_{\mathcal{N}}^* \xrightarrow{n\to\infty} \sigma_w^2\, \mathbf{1}.$$

This yields

$$\mathrm{Var}_{RF} \xrightarrow{n\to\infty} \sigma_w^2 + \frac{\sigma_w^2}{m^2}\mathbf{1}^T.\left(\mathbf{1}.\mathbf{1}^T\right)\mathbf{1} - 2\frac{\sigma_w^2}{m}\mathbf{1}^T.\mathbf{1} = 0$$

with probability one.

When $m \equiv m_n$ grows with n, we decompose

$$\Gamma\, K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^* = \frac{1}{m_n}\mathbf{1} + \Delta, \quad \tilde{K}_{\mathcal{N}} = \sigma_w^2\left(\mathbf{1}.\mathbf{1}^T + \tilde{\Delta}\right), \quad \tilde{k}_{\mathcal{N}}^* = \sigma_w^2\left(\mathbf{1} + \tilde{\delta}\right).$$

This yields

$$\begin{aligned}\frac{\mathrm{Var}_{RF}}{\sigma_w^2} &= 1 + \left(\frac{1}{m_n}\mathbf{1}^T + \Delta^T\right).\left(\mathbf{1}.\mathbf{1}^T + \tilde{\Delta}\right)\left(\frac{1}{m_n}\mathbf{1} + \Delta\right) - 2\left(\frac{1}{m_n}\mathbf{1}^T + \Delta^T\right)\left(\mathbf{1} + \tilde{\delta}\right) \\ &= \frac{1}{m_n^2}\mathbf{1}^T.\tilde{\Delta}\mathbf{1} + \frac{1}{m_n}\mathbf{1}^T.\tilde{\Delta}\Delta + \Delta^T\mathbf{1}.\mathbf{1}^T\Delta + \frac{1}{m}\Delta^T.\tilde{\Delta}\mathbf{1} + \Delta^T.\tilde{\Delta}\Delta - 2\Delta^T.\tilde{\delta} - \frac{2}{m_n}\mathbf{1}^T.\tilde{\delta}.\end{aligned}$$

Thus,

$$\begin{aligned}\left|\frac{\mathrm{Var}_{RF}}{\sigma_w^2}\right| \leq &\frac{1}{m_n}\|\tilde{\Delta}\|_1 + \frac{1}{m_n}\|\tilde{\Delta}\|_1\|\Delta\|_1 + m_n\|\Delta^T\|_1\|\Delta\|_1 + \|\Delta^T\|_1\|\tilde{\Delta}\|_1 \\ &+ \|\Delta^T\|_1\|\tilde{\Delta}\|_1\|\Delta\|_1 + 2\|\Delta^T\|_1\|\tilde{\delta}\|_1 + \frac{2}{m_n}\|\tilde{\delta}\|_1.\end{aligned}$$

Next, define

$$\tilde{\epsilon}_E := \frac{1}{m_n} \max_{1\le j\le m_n} \sum_{i=1}^{m_n} \rho_{\tilde{c}}^2 \left(\boldsymbol{x}_{n,i}(\boldsymbol{x}_*), \boldsymbol{x}_{n,j}(\boldsymbol{x}_*)\right), \quad \tilde{\epsilon}_m := \max_{1\le i\le m_n} \rho_{\tilde{c}}^2 \left(\boldsymbol{x}_*, \boldsymbol{x}_{n,i}(\boldsymbol{x}*)\right).$$

Note that

$$\frac{1}{m_n}\|\tilde{\Delta}\|_1 = \tilde{\epsilon}_E \le 4\tilde{\epsilon}_m \xrightarrow{n\to\infty} 0, \quad \frac{1}{m_n}\|\tilde{\delta}\|_1 \le \tilde{\epsilon}_m \xrightarrow{n\to\infty} 0$$

with probability one, where we have used assumption (AC.5) together with Remark OA.C.3 and Corollary OA.C.4. By Equations (18) and (19) from Lemma OA.B.3 we have

$$\|\Delta\|_1 \le \frac{\epsilon_E + \epsilon_m}{1-\epsilon_E}, \quad \|\Delta^T\|_1 \le \frac{1}{m_n}\frac{\epsilon_E + \epsilon_m}{1-\epsilon_E}.$$

By Remark OA.C.3 and Corollary OA.C.4, we have $\epsilon_m \xrightarrow{n\to\infty} 0$ and $\epsilon_E \xrightarrow{n\to\infty} 0$. Plugging this into the bound for Var_{RF}, we get

$$\left|\frac{\mathrm{Var}_{RF}}{\sigma_w^2}\right| \le (2\tilde{\epsilon}_m + \tilde{\epsilon}_E)\frac{\epsilon_m + 1}{1-\epsilon_E} + \left(\frac{\epsilon_E + \epsilon_m}{1-\epsilon_E}\right)^2 (1+\tilde{\epsilon}_E) \xrightarrow{n\to\infty} 0. \tag{37}$$

■

We are now ready to prove Theorem 7 which we repeat below for reader's convenience.

Theorem 7 (Universal Point-Wise Consistency). *Assume (AC.1-5). If the number of nearest neighbours m is fixed, the following limits hold for GPnn and NNGP with probability one (with respect to $\boldsymbol{X}_n \sim P_{\mathcal{X}}^n$) and for any test point $\boldsymbol{x}_* \in \mathrm{Supp}_{\rho_c}(P_{\mathcal{X}})$ (see Definition 6).*

$$MSE(\boldsymbol{x}_*, \boldsymbol{X}_n) \xrightarrow{n\to\infty} \sigma_\xi^2(\boldsymbol{x}_*)\left(1 + \frac{1}{m}\right), \tag{38}$$

$$CAL(\boldsymbol{x}_*, \boldsymbol{X}_n) \xrightarrow{n\to\infty} \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{\hat{\sigma}_\xi^2}\left(1 + \mathcal{O}\left(m^{-2}\right)\right), \tag{39}$$

$$NLL(\boldsymbol{x}_*, \boldsymbol{X}_n) \xrightarrow{n\to\infty} \frac{1}{2}\left(\log\left(2\pi\,\hat{\sigma}_\xi^2\right) + \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{\hat{\sigma}_\xi^2} + \frac{1}{m}\right) + \mathcal{O}\left(m^{-2}\right). \tag{40}$$

What is more, if m grows with n so that $\lim_{n\to\infty} m_n/n = 0$, the following limits hold with probability one and for any text point $\boldsymbol{x}_* \in \mathrm{Supp}_{\rho_c}(P_{\mathcal{X}})$.

$$MSE(\boldsymbol{x}_*, \boldsymbol{X}_n) \xrightarrow{n\to\infty} \sigma_\xi^2(\boldsymbol{x}_*), \quad CAL(\boldsymbol{x}_*, \boldsymbol{X}_n) \xrightarrow{n\to\infty} \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{\hat{\sigma}_\xi^2} \tag{41}$$

$$NLL(\boldsymbol{x}_*, \boldsymbol{X}_n) \xrightarrow{n\to\infty} \frac{1}{2}\left(\log\left(2\pi\,\hat{\sigma}_\xi^2\right) + \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{\hat{\sigma}_\xi^2}\right). \tag{42}$$

Proof The limit (38) follows from plugging the limits (32) and (33) into the bias-variance decomposition of the MSE from Lemma OA.C.1. The limit (39) follows from plugging the

MSE-limit (38) and the predictive variance limit (31) to the definition of the calibration coefficient (10) to obtain

$$CAL(\boldsymbol{x}_*, X_n) \xrightarrow{n\to\infty} \frac{\sigma_\xi(\boldsymbol{x}_*)^2}{\hat{\sigma}_\xi^2}\left(1+\frac{1}{m}\right)\left(1+\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2+m\hat{\sigma}_f^2}\right)^{-1}.$$

The final form of the limit (39) is obtained by expanding the above expression with respect to powers of $1/m$.

Finally, the limit (40) follows from plugging the CAL-limit (39) and the predictive variance limit (31) into the definition of NLL (11) and using the continuity of the logarithm. In the final step we have expanded the resulting expressions with respect to powers of $1/m$.

The limits (41) and (42) are obtained in a fully analogous way using (34) and (35) from Lemma OA.C.1. ■

Having established the pointwise limits of the performance measures, we are now ready to prove Theorem 9 which we repeat below.

Theorem 9 (Approximate Universal Consistency). *Let $\boldsymbol{X}_n$ be a sampling sequence of i.i.d. points from the distribution $P_\mathcal{X}$ and m be a fixed number of nearest-neughbours. Let $\mathcal{X}_* \sim P_\mathcal{X}$ be a test point. Apply the following assumptions:*

- *(AC.1-5),*
- *function f in the GPnn response model* (1) *satisfies $\|f(\cdot)\|_\infty < B_f < \infty$,*
- *functions t_i, $i = 1, \ldots, d_T$ in the NNGP response model* (2) *satisfy $\|t_i(\cdot)\|_\infty < B_T < \infty$,*
- *$\|\sigma_\xi^2(\cdot)\|_\infty < \infty$, where $\sigma_\xi^2(\boldsymbol{x}) := \mathbb{E}\left[\Xi^2 \mid \mathcal{X} = \boldsymbol{x}\right]$.*

Then we have the following limit for the risk for both GPnn and NNGP.

$$\mathbb{E}_{\mathcal{X}_*,\boldsymbol{X}_n}\left[MSE(\mathcal{X}_*, \boldsymbol{X}_n)\right] = \mathcal{R}_n + \mathbb{E}_{\mathcal{X}_*}\left[\sigma_\xi^2(\mathcal{X}_*)\right] \xrightarrow{n\to\infty} \mathbb{E}_{\mathcal{X}_*}\left[\sigma_\xi^2(\mathcal{X}_*)\right]\left(1+\frac{1}{m}\right), \tag{43}$$

where $\mathcal{R}_n$ is the risk defined in (6)*. Analogous limits hold for CAL and NLL, i.e.,*

$$\begin{aligned}
&\mathbb{E}_{\mathcal{X}_*,\boldsymbol{X}_n}\left[CAL(\mathcal{X}_*, \boldsymbol{X}_n)\right] \xrightarrow{n\to\infty} \frac{\mathbb{E}_{\mathcal{X}_*}\left[\sigma_\xi^2(\mathcal{X}_*)\right]}{\hat{\sigma}_\xi^2}\left(1+\mathcal{O}\left(m^{-2}\right)\right),\\
&\mathbb{E}_{\mathcal{X}_*,\boldsymbol{X}_n}\left[NLL(\mathcal{X}_*, \boldsymbol{X}_n)\right] \xrightarrow{n\to\infty} \frac{1}{2}\left(\log\left(2\pi\,\hat{\sigma}_\xi^2\right) + \frac{\mathbb{E}_{\mathcal{X}_*}\left[\sigma_\xi^2(\mathcal{X}_*)\right]}{\hat{\sigma}_\xi^2} + \frac{1}{m}\right) + \mathcal{O}\left(m^{-2}\right).
\end{aligned} \tag{44}$$

Proof We will prove the desired convergence in expectation using the dominated convergence theorem (DCT) (see Stein and Shakarchi, 2005, Theorem 1.13). From Theorem 7 we know that for both $GPnn$ and $NNGP$ the positive function

$$f_n(\boldsymbol{x}_*, X_\infty) := |MSE(\boldsymbol{x}_*, X_n) - MSE_\infty(\boldsymbol{x}_*)| \xrightarrow{n\to\infty} 0, \quad MSE_\infty(\boldsymbol{x}_*) := \sigma_\xi^2(\boldsymbol{x}_*)\left(1+\frac{1}{m}\right)$$

for almost every $(\boldsymbol{x}_*, X_\infty)$ with respect to the measure $P_{\mathcal{X}} \otimes P_{\mathcal{X}}^{\otimes\infty}$. In the above formula we treat X_n as the first n elements of the infinite sampling sequence X_∞. According to DCT, it suffices to find a measurable function $g(\boldsymbol{x}_*, X_\infty)$ such that

$$\mathbb{E}\left[g(\mathcal{X}_*, \boldsymbol{X}_\infty)\right] < \infty, \quad f_n(\boldsymbol{x}_*, X_\infty) \leq g(\boldsymbol{x}_*, X_\infty)$$

for every n and almost every $(\boldsymbol{x}_*, X_\infty)$ with respect to the measure $P_{\mathcal{X}} \otimes P_{\mathcal{X}}^{\otimes\infty}$.

Let us first prove the risk limit (43) for $GPnn$. Define

$$B_f := \|f\|_\infty < \infty, \quad B_\xi := \|\sigma_\xi^2(\boldsymbol{x})\|_\infty < \infty.$$

If m is held fixed, we will show that the function $f_n(\boldsymbol{x}_*, X_\infty)$ is upper-bounded by a constant independent of n. To see this, note first that

$$f_n(\boldsymbol{x}_*, X_\infty) \leq |MSE(\boldsymbol{x}_*, X_n)| + B_\xi\left(1 + \frac{1}{m}\right) \leq |MSE(\boldsymbol{x}_*, X_n)| + 2B_\xi.$$

The key observation is that $|MSE(\boldsymbol{x}_*, X_n)|$ is bounded. By the bias-variance decomposition of MSE from Lemma OA.C.1 we have

$$\begin{aligned} |MSE(\boldsymbol{x}_*, X_n)| \leq & \left(\frac{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}{m\hat{\sigma}_f^2}\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| + B_f\right)^2 \\ & + B_\xi\left(\frac{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}{m\hat{\sigma}_f^2}\right)^2 \left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \mathbf{k}_{\mathcal{N}}^*\right|. \end{aligned} \tag{45}$$

Since the spectrum of $K_{\mathcal{N}}$ is lower-bounded by $\hat{\sigma}_\xi^2$ (recall that $K_{\mathcal{N}}$ is the shifted Gram matrix), we have that every eigenvalue of $K_{\mathcal{N}}^{-2}$ satisfies

$$\lambda_i\left(K_{\mathcal{N}}^{-2}\right) = \frac{1}{\lambda_i\left(K_{\mathcal{N}}\right)^2} \leq \frac{1}{\hat{\sigma}_\xi^2}\frac{1}{\lambda_i\left(K_{\mathcal{N}}\right)} = \frac{1}{\hat{\sigma}_\xi^2}\lambda_i\left(K_{\mathcal{N}}^{-1}\right).$$

This means that the matrix $\frac{1}{\hat{\sigma}_\xi^2}K_{\mathcal{N}}^{-1} - K_{\mathcal{N}}^{-2}$ is positive semi-definite. Consequently,

$$\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \mathbf{k}_{\mathcal{N}}^* \leq \frac{1}{\hat{\sigma}_\xi^2}\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^* \leq \frac{1}{\hat{\sigma}_\xi^2}k(\boldsymbol{x}_*, \boldsymbol{x}_*) = \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}, \tag{46}$$

where in the second inequality we have used the fact that the GP predictive variance at $\boldsymbol{x}_*$ is non-negative, i.e. $k(\boldsymbol{x}_*, \boldsymbol{x}_*) - \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^* \geq 0$.

To bound $\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right|$, we use the submultiplicativity of the 2-norm as follows.

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| = \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2} K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2 \leq \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2.$$

Using the non-negativity of the GP predictive variance we get

$$\left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2^2 = \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^* \leq \hat{\sigma}_f^2.$$

By bounding the spectrum of $K_{\mathcal{N}}$ from below by $\hat{\sigma}_\xi^2$ and using the boundedness of f we get

$$\left\| K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}}) \right\|_2 \leq \left\| K_{\mathcal{N}}^{-1/2} \right\|_2 \|f(X_{\mathcal{N}})\|_2 \leq \sqrt{m}\, \frac{B_f}{\hat{\sigma}_\xi}.$$

Plugging all of the above results into (45) we obtain the final upper bound

$$f_n(\boldsymbol{x}_*, X) \leq B_f^2 \left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f \hat{\sigma}_\xi} \sqrt{m} + 1 \right)^2 + B_\xi \left(2 + \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2} \left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f^2} \right)^2 \right). \tag{47}$$

which is the last ingredient of DCT.

Let next prove the risk limit (43) for $NNGP$. From Lemma OA.C.1 we have that

$$MSE_{NNGP}(\boldsymbol{x}_*, X_n) = MSE_{GPnn}(\boldsymbol{x}_*, X_n) + \mathrm{Var}_{RF}(\boldsymbol{x}_*, X_n),$$

where

$$\begin{aligned} MSE_{GPnn} &= \sigma_\xi^2(\boldsymbol{x}_*) + \Gamma^2\, {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \Sigma_{\boldsymbol{\xi}} K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* + \left(\boldsymbol{t}_*^T.(\boldsymbol{b} - \hat{\boldsymbol{b}}) - \Gamma\, {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} T_{\mathcal{N}} (\boldsymbol{b} - \hat{\boldsymbol{b}}) \right)^2, \\ \mathrm{Var}_{RF} &= \sigma_w^2 + \Gamma^2\, {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \tilde{K}_{\mathcal{N}} K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* - 2\Gamma\, {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \tilde{k}_{\mathcal{N}}^*. \end{aligned}$$

Using the earlier upper bound for $GPnn$ MSE (47), we immediately get an upper bound for the MSE_{GPnn}-component of MSE_{NNGP}. To see this, define $B_T := \max_{1 \leq i \leq d_T} \|t_i\|_\infty < \infty$. By Cauchy-Schwarz we have

$$\left| \boldsymbol{t}(\boldsymbol{x})^T.(\boldsymbol{b} - \hat{\boldsymbol{b}}) \right| \leq \|\boldsymbol{t}(\boldsymbol{x})\|_2\, \|\boldsymbol{b} - \hat{\boldsymbol{b}}\|_2 \leq B_T\, \|\boldsymbol{b} - \hat{\boldsymbol{b}}\|_2.$$

Then, replacing B_f by $B_T\, \|\boldsymbol{b} - \hat{\boldsymbol{b}}\|_2$ in the inequality (47) we get

$$MSE_{GPnn} \leq B_T^2 \|\boldsymbol{b} - \hat{\boldsymbol{b}}\|_2^2 \left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f \hat{\sigma}_\xi} \sqrt{m} + 1 \right)^2 + B_\xi \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2} \left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f^2} \right)^2. \tag{48}$$

It remains to bound MSE_{RF}.

$$\mathrm{Var}_{RF} \leq \sigma_w^2 + \Gamma^2 \left| {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \tilde{K}_{\mathcal{N}} K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* \right| + 2\Gamma \left| {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \tilde{k}_{\mathcal{N}}^* \right|.$$

Further by the submultiplicativity of the 2-norm,

$$\left| {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \tilde{k}_{\mathcal{N}}^* \right| \leq \left\| {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1/2} \right\|_2 \left\| K_{\mathcal{N}}^{-1/2} \right\|_2 \left\| \tilde{k}_{\mathcal{N}}^* \right\|_2 \leq \sigma_w^2 \sqrt{m}\, \frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}.$$

Similarly, we get

$$\begin{aligned} \left| {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1} \tilde{K}_{\mathcal{N}} K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* \right| \leq & \left\| {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1/2} \right\|_2 \left\| K_{\mathcal{N}}^{-1/2} \right\|_2 \left\| \tilde{K}_{\mathcal{N}} \right\|_2 \left\| K_{\mathcal{N}}^{-1/2} \right\|_2 \left\| K_{\mathcal{N}}^{-1/2} k_{\mathcal{N}}^* \right\|_2 \\ \leq & \sigma_w^2\, m\, \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}, \end{aligned}$$

where we have used the standard inequality $\left\|\tilde{K}_{\mathcal{N}}\right\|_2 \leq \sigma_w^2\, m$ which stems from the fact that the entries of $\tilde{K}_{\mathcal{N}}$ are bounded from above by σ_w^2. Summing up, we have

$$\mathrm{Var}_{RF} \leq \sigma_w^2 \left(1 + 2\Gamma\sqrt{m}\,\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi} + \Gamma^2 m\,\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\right) = \sigma_w^2 \left(1 + \sqrt{m}\,\Gamma\,\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}\right)^2. \tag{49}$$

This ends the derivation of the upper bound for MSE_{NNGP} which is independent of n and thus by DCT proves the risk limit (43) for $NNGP$.

The limits (44) are proved in a fully analogous way by using the definitions of CAL (10) and NLL (11) and additionally utilising standard inequality

$$\hat{\sigma}_\xi^2 \leq {\sigma_{\mathcal{N}}^*}^2 \leq \hat{\sigma}_\xi^2 + \hat{\sigma}_f^2.$$

■

OA.C.1 Universal Consistency

We start by establishing some preliminary ingredients. The m_n-nearest-neighbour (m_n-NN) regression function estimate is defined as follows (using notation from Section 3).

$$\mu_{NN}(\boldsymbol{x}_*) = \frac{1}{m_n}\sum_{j=1}^{m_n} y_{n,j},$$

where $y_{n,j}$ is the observed response at the j-th nearest neighbour $\boldsymbol{x}_{n,j}(\boldsymbol{x}_*)$.

Theorem OA.C.7 (Györfi et al. (2002, Theorem 6.1)) *Let the number of nearest - neighbours m_n grow with n such that $m_n/n \xrightarrow{n\to\infty} 0$. Then the m_n-NN regression function estimate is universally consistent for all distributions of $(\mathcal{X}, \mathcal{Y})$ where nearest-neighbour ties occur with probability zero and $\mathbb{E}\left[\mathcal{Y}^2\right] < \infty$.*

Lemma OA.C.8 *Let $\boldsymbol{X}_n$ be a random sampling sequence of n i.i.d. points from the distribution $P_{\mathcal{X}}$ and assume nearest neighbours are selected according to their Euclidean distance from the test point. Let $\mathcal{X}_* \sim P_{\mathcal{X}}$ be a test point. Fix $\nu > 0$ and assume that there exists $\beta > \frac{2\nu d_{\mathcal{X}}}{d_{\mathcal{X}} - 2\nu}$ for which $\mathbb{E}\left[\|\mathcal{X}\|_2^\beta\right] < \infty$ under the probability distribution $P_{\mathcal{X}}$ on $\mathbb{R}^{d_{\mathcal{X}}}$ with $d_{\mathcal{X}} > 2\nu$. Then, for any $0 < R \leq 1$ the following inequality holds*

$$\sqrt{P\left[\min\left\{d_m(\mathcal{X}_*, \boldsymbol{X}_n), 1\right\} \geq R\right]} \leq \frac{1}{R^\nu}\sqrt{c}\, 2^{\frac{\nu}{d_{\mathcal{X}}} + \frac{1}{2}} \left(\frac{m}{n}\right)^{\nu/d_{\mathcal{X}}},$$

where $d_m(\boldsymbol{x}_*, X_n)$ is the distance between $\boldsymbol{x}_*$ and it's m-th nearest neighbour in X_n and the positive constant $c < \infty$ depends on $d_{\mathcal{X}}, \nu, \beta$ and $\mathbb{E}\left[\|\mathcal{X}\|_2^\beta\right]$.

Proof Apply Markov's inequality which states that for any non-negative random variable U and $\lambda > 0$ we have

$$P\left[U \geq \lambda\right] \leq \frac{\mathbb{E}\left[U\right]}{\lambda}.$$

Take $U \equiv \min\{d_m^{2\nu}, 1\}$ and $\lambda = R^{2\nu}$. This yields

$$P\left[\min\{d_m, 1\} \geq R\right] = P\left[U \geq R^{2\nu}\right] \leq \frac{\mathbb{E}\left[U\right]}{R^{2\nu}}.$$

Next, we use Lemma OA.D.2 applied to $\min\left\{d_m^{2\nu}, 1\right\}$ to upper bound $\mathbb{E}\left[U\right]$ as follows.

$$\mathbb{E}\left[U\right] \leq c\, 2^{\frac{2\nu}{d_{\mathcal{X}}}+1} \left(\frac{m}{n}\right)^{2\nu/d_{\mathcal{X}}}.$$

Taking the square root of both sides, we prove the lemma. ■

Theorem 11 (Universal Consistency). *Let $\boldsymbol{X}_n$ be a random sampling sequence of i.i.d. points from the distribution $P_{\mathcal{X}}$ and let $\mathcal{X}_* \sim P_{\mathcal{X}}$ be a test point. Let the number of nearest - neighbours m_n grow as $m_n \propto n^{\gamma}$ with $0 < \gamma < 1/3$. Apply the following assumptions:*

- *there exists $\beta > \frac{2\gamma d_{\mathcal{X}}}{1-3\gamma}$ for which $\mathbb{E}\left[\|\mathcal{X}\|_2^{\beta}\right] < \infty$ under the probability distribution $P_{\mathcal{X}}$ on $\mathbb{R}^{d_{\mathcal{X}}}$.*
- *(AC.1-5) and (AR.1-2),*
- *function f in the GPnn response model (1) satisfies $\|f(\cdot)\|_\infty \leq B_f < \infty$ for some $B_f > 0$,*
- *functions t_i, $i = 1, \ldots, d_T$ in the NNGP response model (2) satisfy $\|t_i(\cdot)\|_\infty < B_T < \infty$ for some $B_T > 0$,*
- *$\|\sigma_\xi^2(\cdot)\|_\infty < \infty$, where $\sigma_\xi^2(\boldsymbol{x}) := \mathbb{E}\left[\Xi^2 \mid \mathcal{X} = \boldsymbol{x}\right]$.*

Then we have the following limit for the risk of GPnn and NNGP.

$$\mathbb{E}_{\mathcal{X}_*, \boldsymbol{X}_n}\left[MSE(\mathcal{X}_*, \boldsymbol{X}_n)\right] \xrightarrow{n\to\infty} \mathbb{E}_{\mathcal{X}_*}\left[\sigma_\xi^2(\mathcal{X}_*)\right]. \tag{50}$$

Proof (for *GPnn*) At a test point $\boldsymbol{x}_*$, we write the *GPnn* predictor as a weighted sum of the nearest-neighbour responses

$$\tilde{\mu}_{GPnn}(\boldsymbol{x}_*) = \sum_{j=1}^{m} w_{n,j}(\boldsymbol{x}_*, X_n)\, y_{n,j} = \boldsymbol{w}_n^T.\boldsymbol{y}_{n,m}, \quad \boldsymbol{w}_n(\boldsymbol{x}_*, X_n) = \Gamma\, K_{\mathcal{N}}^{-1}\, \boldsymbol{k}_{\mathcal{N}}^*.$$

Define the difference between the *GPnn* and m-NN estimators

$$D_n(\boldsymbol{x}_*) := \tilde{\mu}_{GPnn}(\boldsymbol{x}_*) - \mu_{NN}(\boldsymbol{x}_*) = \boldsymbol{a}_n^T.\boldsymbol{y}_{n,m}, \quad \boldsymbol{a}_n = \boldsymbol{w}_n - \frac{1}{m}\mathbf{1}.$$

Then, the *GPnn* squared error at test point $\boldsymbol{x}_*$ can be written as

$$\begin{aligned}(f(\boldsymbol{x}_*) - \tilde{\mu}_{GPnn}(\boldsymbol{x}_*))^2 =& (f(\boldsymbol{x}_*) - \mu_{NN}(\boldsymbol{x}_*) - D_n(\boldsymbol{x}_*))^2 \\ \leq& 2\,(f(\boldsymbol{x}_*) - \mu_{NN}(\boldsymbol{x}_*))^2 + 2\, D_n(\boldsymbol{x}_*)^2,\end{aligned}$$

where in the last line we have use the fact that for any real a, b we have $(a-b)^2 \leq 2a^2+2b^2$. Take expectations (w.r.t. all the training and test data) of both sides to get

$$\mathcal{R}_n^{(GPnn)} = \mathbb{E}\left[(f(\boldsymbol{x}_*) - \tilde{\mu}_{GPnn}(\boldsymbol{x}_*))^2\right] \leq 2\mathbb{E}\left[(f(\boldsymbol{x}_*) - \mu_{NN}(\boldsymbol{x}_*))^2\right] + 2\mathbb{E}\left[D_n(\boldsymbol{x}_*)^2\right]$$

By Theorem OA.C.7, we have that the m-NN risk tends to zero as the training data size grows

$$\mathcal{R}_n^{(NN)} = \mathbb{E}\left[(f(\boldsymbol{x}_*) - \mu_{NN}(\boldsymbol{x}_*))^2\right] \xrightarrow{n\to\infty} 0.$$

The remaining part of this proof is devoted to showing that

$$\mathbb{E}\left[D_n(\boldsymbol{x}_*)^2\right] \xrightarrow{n\to\infty} 0, \tag{51}$$

which is the last ingredient needed to prove the universal consistency. To this end, we first use the Cauchy-Schwarz inequality $\left(\boldsymbol{u}^T.\boldsymbol{v}\right)^2 \leq \|\boldsymbol{u}\|_2^2 \|\boldsymbol{v}\|_2^2$ with

$$u_j = \sqrt{|a_{n,j}|}\,\mathrm{sign}(a_{n,j}), \quad v_j = y_{n,j}\sqrt{|a_{n,j}|},$$

to get

$$D_n^2 = \left(\sum_{j=1}^m a_{n,j}\, y_{n,j}\right)^2 \leq \left(\sum_{j=1}^m |a_{n,j}|\right)\left(\sum_{j=1}^m |a_{n,j}|\, y_{n,j}^2\right) = \|\boldsymbol{a}_n\|_1 \left(\sum_{j=1}^m |a_{n,j}|\, y_{n,j}^2\right).$$

Thus, the conditional expectation is bounded as

$$\mathbb{E}\left[D_n^2 \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n = X_n\right] \leq \|\boldsymbol{a}_n\|_1 \left(\sum_{j=1}^m |a_{n,j}|\, \mathbb{E}\left[\mathcal{Y}_{n,j}^2 \mid \mathcal{X}_{n,j} = \boldsymbol{x}_{n,j}(\boldsymbol{x}_*)\right]\right). \tag{52}$$

By the assumption of the boundedness of $f(\cdot)$ and the boundedness of the noise variance

$$\mathbb{E}\left[\mathcal{Y}^2 \mid \mathcal{X} = \boldsymbol{x}\right] \leq 2f(\boldsymbol{x})^2 + 2\mathbb{E}\left[\Xi^2 \mid \mathcal{X} = \boldsymbol{x}\right] \leq C_{Y,2} < \infty$$

for some constant $C_{Y,2} > 0$. To derive this, we have used the inequality $(a+b)^2 \leq 2a^2 + 2b^2$ again. Moreover, by the inequality (18) from Lemma OA.B.3 and the identity $\hat{\sigma}_f^2 \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} = \frac{1}{m\Gamma}\mathbf{1}$, we get that for each $\boldsymbol{x}_*, X_n$

$$\|\boldsymbol{a}_n(\boldsymbol{x}_*, X_n)\|_1 \leq \frac{\epsilon_m(\boldsymbol{x}_*, X_n) + \epsilon_E(\boldsymbol{x}_*, X_n)}{1 - \epsilon_E(\boldsymbol{x}_*, X_n)}.$$

The functions ϵ_m and ϵ_E are defined in Lemma OA.B.3. Plugging this into (52) yields

$$\mathbb{E}\left[D_n^2 \mid \mathcal{X}_* = \boldsymbol{x}_*, \boldsymbol{X}_n = X_n\right] \leq C_{Y,2}\, \|\boldsymbol{a}_n\|_1^2 \leq C_{Y,2} \left(\frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E}\right)^2. \tag{53}$$

To proceed, we need to take the expectation over the training and test data $\boldsymbol{X}_n, \mathcal{X}_*$. This requires handling the possible blowup of the above upper bound when ϵ_E approaches 1. To this end, we define the good event in the space of training and test data

$$G_n := \left\{\min\left\{d_m(\mathcal{X}_*, \boldsymbol{X}_n), 1\right\} < \min\left\{R, 1\right\}\right\}, \quad R = \left(\frac{1}{8\max\{L_c, 1\}}\right)^{1/2p},$$

where L_c is defined in (AR.2). Then, we use the tower property of the expectations and split the expectation of D_n^2 as

$$\mathbb{E}\left[D_n^2\right] = \mathbb{E}\left[\mathbb{E}\left[D_n^2\,\chi_{G_n} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\right] + \mathbb{E}\left[\mathbb{E}\left[D_n^2\,\chi_{G_n^c} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\right],$$

where $\chi_{\mathcal{S}}$ is the indicator function of the set $\mathcal{S}$ and $\mathcal{S}^c$ is the complement of $\mathcal{S}$. By (AR.2) we have that

$$\epsilon_m \leq \min\left\{L_c\, d_m^{2p}, 1\right\} \leq \max\{L_c, 1\}\min\{d_m^{2p}, 1\},$$

where in the second inequality we have used the fact that for any $a, b \geq 0$ we have $\min\{ab, 1\} \leq \max\{a, 1\}\min\{b, 1\}$. Thus,

$$\text{if} \quad \min\{d_m(\boldsymbol{x}_*, X_n), 1\} < \min\{R, 1\}, \quad \text{then} \quad \epsilon_m(\boldsymbol{x}_*, X_n) < 1/8.$$

Furthermore, by Lemma OA.B.4 we get that $\epsilon_E \leq 4\epsilon_m < 1/2$, thus $1/(1-\epsilon_E) < 2$ and we get from (53) that

$$\mathbb{E}\left[D_n^2\,\chi_{G_n}\right] < 100\, C_{Y,2}\,\mathbb{E}\left[\epsilon_m^2\,\chi_{G_n}\right] \xrightarrow{n\to\infty} 0.$$

This follows from the dominated convergence theorem since Remark OA.C.3 implies that

$$\epsilon_m^2\,\chi_{G_n} \xrightarrow{n\to\infty} 0 \quad a.s.$$

and ϵ_m is upper-bounded by 1.

Finally, we need to show that

$$\mathbb{E}\left[D_n^2\,\chi_{G_n^c}\right] \xrightarrow{n\to\infty} 0.$$

By Cauchy-Schwarz inequality we have

$$\begin{aligned}\mathbb{E}\left[\mathbb{E}\left[D_n^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\,\chi_{G_n^c}\right] \leq &\sqrt{\mathbb{E}\left[\left(\mathbb{E}\left[D_n^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\right)^2\right]}\times \\ &\times\sqrt{P\left[\min\left\{d_m(\mathcal{X}_*, \boldsymbol{X}_n), 1\right\} \geq \min\{R, 1\}\right]}\end{aligned} \tag{54}$$

Next, we find a suitable upper bound on $\|\boldsymbol{a}_n\|_1$ as follows

$$\begin{aligned}\|\boldsymbol{a}_n\|_1 \leq \|\boldsymbol{w}_n\|_1 + \frac{1}{m}\|\mathbf{1}\|_1 &= \Gamma\left\|K_{\mathcal{N}}^{-1}\,\boldsymbol{k}_{\mathcal{N}}^*\right\|_1 + 1 \leq \sqrt{m}\Gamma\left\|K_{\mathcal{N}}^{-1}\,\boldsymbol{k}_{\mathcal{N}}^*\right\|_2 + 1 \\ &\leq \sqrt{m}\Gamma\left\|K_{\mathcal{N}}^{-1/2}\right\|_2\left\|K_{\mathcal{N}}^{-1/2}\,\boldsymbol{k}_{\mathcal{N}}^*\right\|_2 + 1 \leq \sqrt{m}\Gamma\frac{\hat{\sigma}_2}{\hat{\sigma}_\xi} + 1 \leq \sqrt{m}\left(\Gamma\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi} + 1\right),\end{aligned}$$

where in the last line we have used the facts that i) the minimum eigenvalue of $K_{\mathcal{N}}$ is bounded from below by $\hat{\sigma}_\xi^2$ which implies $\left\|K_{\mathcal{N}}^{-1/2}\right\|_2 \leq 1/\hat{\sigma}_\xi$ and ii) $\left\|K_{\mathcal{N}}^{-1/2}\,\boldsymbol{k}_{\mathcal{N}}^*\right\|_2 \leq \hat{\sigma}_f$ which follows from the non-negativity of the GP predictive covariance

$$\hat{\sigma}_f^2 \geq {\boldsymbol{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-1}\,\boldsymbol{k}_{\mathcal{N}}^* = \left\|K_{\mathcal{N}}^{-1/2}\,\boldsymbol{k}_{\mathcal{N}}^*\right\|_2^2.$$

Plugging this into (53), we get

$$\mathbb{E}\left[D_n^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] \leq C_{Y,2}\left(\Gamma\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi} + 1\right)^2 m.$$

Next, we use the above inequality in combination with Lemma OA.C.8 for $\nu > d_{\mathcal{X}}\gamma/(1-\gamma)$ which by (54) yields

$$\mathbb{E}\left[D_n^2\,\chi_{G_n^c}\right] \leq C_{Y,2}\left(\Gamma\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}+1\right)^2 \frac{1}{\min\{R^\nu,1\}}\sqrt{c}\,2^{\frac{\nu}{d_{\mathcal{X}}}+\frac{3}{2}}\,m\,\left(\frac{m}{n}\right)^{\nu/d_{\mathcal{X}}}.$$

The condition $\nu > d_{\mathcal{X}}\gamma/(1-\gamma)$ ensures that by taking $m = An^\gamma$ for some $A > 0$, the above bound implies that

$$\mathbb{E}\left[D_n^2\,\chi_{G_n^c}\right] \leq C_{Y,2}\left(\Gamma\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}+1\right)^2 \frac{1}{\min\{R^\nu,1\}}\sqrt{c}\,2^{\frac{\nu}{d_{\mathcal{X}}}+\frac{3}{2}}\,\tilde{A}\,n^{\gamma-\frac{\nu}{d_{\mathcal{X}}}(1-\gamma)} \xrightarrow{n\to\infty} 0,$$

where $\tilde{A} > 0$ depends on $A, \gamma, d_{\mathcal{X}}, \nu$. The above upper bound tends to zero as $n \to \infty$, because $\gamma - \frac{\nu}{d_{\mathcal{X}}}(1-\gamma) < 0$ for our choice of ν. Note that the condition $d_{\mathcal{X}} > 2\nu$ of Lemma OA.C.8 is then guaranteed for any $d_{\mathcal{X}}$ by the fact that $\nu > d_{\mathcal{X}}\gamma/(1-\gamma)$ and $0 < \gamma < 1/3$ (this can be straightforwardly verified by substitution). Finally, note that when $\nu > d_{\mathcal{X}}\gamma/(1-\gamma)$, then we have

$$\frac{2\nu d_{\mathcal{X}}}{d_{\mathcal{X}}-2\nu} > \frac{2d_{\mathcal{X}}^2\frac{\gamma}{1-\gamma}}{d_{\mathcal{X}}-2d_{\mathcal{X}}\frac{\gamma}{1-\gamma}} = \frac{2\gamma d_{\mathcal{X}}}{1-3\gamma}.$$

Thus, for any $\beta > \frac{2\gamma d_{\mathcal{X}}}{1-3\gamma}$, we can find $\nu > d_{\mathcal{X}}\gamma/(1-\gamma)$ which additionally satisfies $\frac{2\nu d_{\mathcal{X}}}{d_{\mathcal{X}}-2\nu} < \beta$, thereby satisfying the moment-condition $\beta > \frac{2\nu d_{\mathcal{X}}}{d_{\mathcal{X}}-2\nu}$ of Lemma OA.C.8. This finishes the proof. ■

Proof (for $NNGP$) The goal is to prove that

$$\mathbb{E}\left[\left(\mu_{NNGP}\left(\mathcal{X}_*\right)-g\left(\mathcal{X}_*\right)\right)^2\right] \xrightarrow{n\to\infty} 0, \quad g\left(\mathcal{X}_*\right) = \boldsymbol{t}\left(\mathcal{X}_*\right)^T.\boldsymbol{b} + w\left(\mathcal{X}_*\right),$$

where g is the noise-free part of the $NNGP$ response from Equation (2) and the expectation is over the noise and the random GP sample paths w. Then,

$$\begin{aligned}\left(\mu_{NNGP}\left(\mathcal{X}_*\right)-g\left(\mathcal{X}_*\right)\right)^2 &= \left(\mu_{NNGP}\left(\mathcal{X}_*\right)-\mu_{NN}\left(\mathcal{X}_*\right)+\mu_{NN}\left(\mathcal{X}_*\right)-g\left(\mathcal{X}_*\right)\right)^2\\ &\leq 2\left(\mu_{NN}\left(\mathcal{X}_*\right)-g\left(\mathcal{X}_*\right)\right)^2 + 2\left(\mu_{NNGP}\left(\mathcal{X}_*\right)-\mu_{NN}\left(\mathcal{X}_*\right)\right)^2.\end{aligned}$$

Firstly, using m-NN universal consistency we show that

$$\mathbb{E}\left[\left(\mu_{NN}\left(\mathcal{X}_*\right)-g\left(\mathcal{X}_*\right)\right)^2\right] \xrightarrow{n\to\infty} 0.$$

To this end, we decompose $g\left(\mathcal{X}_*\right) - \mu_{NN}\left(\mathcal{X}_*\right) = A_n + B_n$, where

$$A_n = \boldsymbol{t}\left(\mathcal{X}_*\right)^T.\boldsymbol{b} - \frac{1}{m}\sum_{j=1}^m\left(\boldsymbol{t}\left(\mathcal{X}_{n,j}\right)^T.\boldsymbol{b}+\Xi_{n,j}\right), \quad B_n = w\left(\mathcal{X}_*\right) - \frac{1}{m}\sum_{j=1}^m w\left(\mathcal{X}_{n,j}\right).$$

Then,

$$\mathbb{E}\left[\left(\mu_{NN}\left(\mathcal{X}_*\right)-g\left(\mathcal{X}_*\right)\right)^2\right] \leq 2\mathbb{E}\left[A_n^2\right] + 2\mathbb{E}\left[B_n^2\right].$$

The term $\mathbb{E}\left[A_n^2\right] \xrightarrow{n\to\infty} 0$ due to the universal consistency of m-NN applied to the regression function $f(\boldsymbol{x}) = \boldsymbol{t}(\boldsymbol{x})^T.\boldsymbol{b}$.

To see that the term $\mathbb{E}\left[B_n^2\right] \xrightarrow{n\to\infty} 0$, note first that $(\sum_j z_j/m)^2 \leq (\sum_j z_j^2)/m$ with $z_j = w(\mathcal{X}_*) - w(\mathcal{X}_{n,j})$ implies the upper bound

$$B_n^2 \leq \frac{1}{m}\sum_{j=1}^{m}\left(w(\mathcal{X}_*) - w(\mathcal{X}_{n,j})\right)^2.$$

Thus,

$$\mathbb{E}\left[B_n^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] \leq \frac{2\sigma_w^2}{m}\sum_{j=1}^{m}\left(1 - \tilde{c}(\mathcal{X}_*, \mathcal{X}_{n,j})\right),$$

where we have used the fact that

$$\begin{aligned}\mathbb{E}\left[\left(w(\mathcal{X}_*) - w(\mathcal{X}_{n,j})\right)^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] =& \sigma_w^2\left(\tilde{c}(\mathcal{X}_*, \mathcal{X}_*) + \tilde{c}(\mathcal{X}_{n,j}, \mathcal{X}_{n,j}) - 2\tilde{c}(\mathcal{X}_*, \mathcal{X}_{n,j})\right)\\ =& 2\sigma_w^2\left(1 - \tilde{c}(\mathcal{X}_*, \mathcal{X}_{n,j})\right).\end{aligned}$$

Note that $1-\tilde{c}(\mathcal{X}_*, \mathcal{X}_{n,j}) = \rho_{\tilde{c}}^2(\mathcal{X}_*, \mathcal{X}_{n,j})$. From Remark OA.C.3 we know that $\rho_c^2(\mathcal{X}_*, \mathcal{X}_{n,j}) \to 0$ with probability one. By assumption (AC.5) this implies that $\rho_{\tilde{c}}^2(\mathcal{X}_*, \mathcal{X}_{n,j}) \to 0$ with probability one and thus $\mathbb{E}\left[B_n^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] \to 0$ with probability one. Since $\mathbb{E}\left[B_n^2 \mid \mathcal{X}_*, \boldsymbol{X}_n\right] \leq 2\sigma_w^2$, dominated convergence theorem implies that $\mathbb{E}\left[B_n^2\right] \to 0$.

To complete the proof it remains to show that

$$\mathbb{E}\left[\left(\mu_{NNGP}(\mathcal{X}_*) - \mu_{NN}(\mathcal{X}_*)\right)^2\right] \xrightarrow{n\to\infty} 0.$$

To this end, we rewrite the $NNGP$-estimator (5) as

$$\mu_{NNGP}(\mathcal{X}_*) = \boldsymbol{w}_n^T.\boldsymbol{y}_{\mathcal{N}} + \left(\boldsymbol{t}(\mathcal{X}_*)^T - \boldsymbol{w}_n^T T_{\mathcal{N}}\right)\hat{\boldsymbol{b}}, \quad \boldsymbol{w}_n = \Gamma\, K_{\mathcal{N}}^{-1}\boldsymbol{k}_{\mathcal{N}}^*, \quad \boldsymbol{a}_n = \boldsymbol{w}_n - \frac{1}{m}\mathbf{1}.$$

Hence

$$\mu_{NNGP}(\mathcal{X}_*) - \mu_{NN}(\mathcal{X}_*) = D_n + \tilde{B}_n, \quad D_n := \boldsymbol{a}_n^T.\boldsymbol{y}_{\mathcal{N}}, \quad \tilde{B}_n := \left(\boldsymbol{t}(\mathcal{X}_*)^T - \boldsymbol{w}_n^T T_{\mathcal{N}}\right)\hat{\boldsymbol{b}}.$$

Next, we use the upper bound

$$\mathbb{E}\left[\left(\mu_{NNGP}(\mathcal{X}_*) - \mu_{NN}(\mathcal{X}_*)\right)^2\right] \leq 2\mathbb{E}\left[D_n^2\right] + 2\mathbb{E}\left[\tilde{B}_n^2\right]$$

We next show that $\mathbb{E}\left[D_n^2\right] \to 0$ using the methods established in the $GPnn$-part of the proof and that $\mathbb{E}\left[\tilde{B}_n^2\right] \to 0$ using continuity of $\boldsymbol{t}$ and the fact that $\mathbb{E}\left[\|\boldsymbol{a}_n\|_1^2\right] \to 0$.

By the assumption of the boundedness of $\boldsymbol{t}$ and the boundedness of the noise variance we have that the conditional second moment of the $NNGP$ responses is bounded, i.e.,

$$\begin{aligned}\mathbb{E}\left[\mathcal{Y}^2 \mid \mathcal{X} = \boldsymbol{x}\right] = \mathbb{E}\left[(\boldsymbol{t}(\boldsymbol{x})^T.\boldsymbol{b} + w(\mathcal{X}) + \Xi)^2 \mid \mathcal{X} = \boldsymbol{x}\right] &\leq 3\|\boldsymbol{t}(\boldsymbol{x})\|_2^2\,\|\boldsymbol{b}\|_2^2 + 3\sigma_\xi^2(\boldsymbol{x}) + 3\sigma_w^2\\ &\leq \tilde{C}_{Y,2} < \infty\end{aligned}$$

for some positive constant $\tilde{C}_{Y,2}$. In the above inequality we have used the fact that $(a+b+c)^2 \leq 3a^2+3b^2+3c^2$. As explained in the $GPnn$-part of the proof, the boundedness of $\mathbb{E}\left[\mathcal{Y}^2 \mid \mathcal{X}=\boldsymbol{x}\right]$ implies that

$$\mathbb{E}\left[D_n^2 \mid \mathcal{X}_*=\boldsymbol{x}_*, \boldsymbol{X}_n=X_n\right] \leq \tilde{C}_{Y,2}\left\|\boldsymbol{a}_n\right\|_1^2 \leq \tilde{C}_{Y,2}\left(\frac{\epsilon_m+\epsilon_E}{1-\epsilon_E}\right)^2.$$

Using the method for handling the possible blowup of the above upper bound via the good-bad event split and bounding $\|\boldsymbol{a}_n\|_1$ described in the $GPnn$-part of the proof, we get that $\mathbb{E}\left[D_n^2\right] \to 0$.

The last part of the proof is to show that $\mathbb{E}\left[\tilde{B}_n^2\right] \to 0$. To this end, we use the submultiplicativity of the 2-nom to get

$$\tilde{B}_n^2=\left\|\hat{\boldsymbol{b}}^T.\left(\boldsymbol{t}\left(\mathcal{X}_*\right)-T_{\mathcal{N}}^T \boldsymbol{w}_n\right)\right\|_2^2 \leq\left\|\hat{\boldsymbol{b}}^T\right\|_2^2\left\|\boldsymbol{t}\left(\mathcal{X}_*\right)-T_{\mathcal{N}}^T \boldsymbol{w}_n\right\|_2^2$$

Next, we split

$$\boldsymbol{t}\left(\mathcal{X}_*\right)-T_{\mathcal{N}}^T \boldsymbol{w}_n=\boldsymbol{t}\left(\mathcal{X}_*\right)-\frac{1}{m} T_{\mathcal{N}}^T \mathbf{1}+\frac{1}{m} T_{\mathcal{N}}^T \mathbf{1}-T_{\mathcal{N}}^T \boldsymbol{w}_n.$$

Then,

$$\left\|\boldsymbol{t}\left(\mathcal{X}_*\right)-T_{\mathcal{N}}^T \boldsymbol{w}_n\right\|_2^2 \leq 2\left\|\boldsymbol{t}\left(\mathcal{X}_*\right)-\frac{1}{m} T_{\mathcal{N}}^T \mathbf{1}\right\|_2^2+2\left\|T_{\mathcal{N}}^T\left(\frac{1}{m} \mathbf{1}-\boldsymbol{w}_n\right)\right\|_2^2.$$

Note that

$$\boldsymbol{t}\left(\mathcal{X}_*\right)-\frac{1}{m} T_{\mathcal{N}}^T \mathbf{1}=\boldsymbol{t}\left(\mathcal{X}_*\right)-\frac{1}{m} \sum_{j=1}^m \boldsymbol{t}\left(\mathcal{X}_{n,j}\right),$$

thus we can use the universal consistency of m-NN applied to each function $t_i(\boldsymbol{x})$, $i=1,\ldots,d_T$ to conclude that

$$\mathbb{E}\left[\left\|\boldsymbol{t}\left(\mathcal{X}_*\right)-\frac{1}{m} T_{\mathcal{N}}^T \mathbf{1}\right\|_2^2\right] \xrightarrow{n \to \infty} 0.$$

Finally,

$$\begin{aligned}\left\|T_{\mathcal{N}}^T\left(\frac{1}{m} \mathbf{1}-\boldsymbol{w}_n\right)\right\|_2 &=\left\|\sum_{j=1}^m \boldsymbol{t}\left(\mathcal{X}_{n,j}\right)\left(\frac{1}{m}-w_{n,j}\right)\right\|_2 \leq \sum_{j=1}^m\left\|\boldsymbol{t}\left(\mathcal{X}_{n,j}\right)\left(\frac{1}{m}-w_{n,j}\right)\right\|_2 \\ &\leq \sum_{j=1}^m\left\|\boldsymbol{t}\left(\mathcal{X}_{n,j}\right)\right\|_2\left|\left(\frac{1}{m}-w_{n,j}\right)\right| \leq d_T B_T\|\boldsymbol{a}_n\|_1,\end{aligned}$$

where in the last inequality we have used the boundedness of $t_i(\boldsymbol{x})$. Thus,

$$\mathbb{E}\left[\left\|T_{\mathcal{N}}^T\left(\frac{1}{m} \mathbf{1}-\boldsymbol{w}_n\right)\right\|_2^2\right] \leq d_T^2 B_T^2 \mathbb{E}\left[\|\boldsymbol{a}_n\|_1^2\right] \xrightarrow{n \to \infty} 0,$$

where we have used the fact that $\mathbb{E}\left[\|\boldsymbol{a}_n\|_1^2\right] \to 0$ which has been explained in the $GPnn$-part of the proof. In summary, this shows that $\mathbb{E}\left[\tilde{B}_n^2\right] \to 0$ and finishes the entire proof. ■

Appendix OA.D. Convergence Rates of $\mathbb{E}[d_m]$, $\mathbb{E}[\epsilon_m]$

One of the key ingredients in the derivation of the convergence rates of the MSE and its derivatives is the knowledge of the convergence rate of the expectation of the functions $d_m(\mathcal{X}_*, \boldsymbol{X}_n)$ and $\epsilon_m(\mathcal{X}_*, \boldsymbol{X}_n)$ as $n \to \infty$. We derive these rates in this section relying following lemma.

Lemma OA.D.1 *(Kohler et al., 2006, Lemma 1) Assume (AC.1), and that the nearest neighbours are chosen according to the Euclidean metric. Let $\boldsymbol{X}_n$ be training data sampled i.i.d. from $P_{\mathcal{X}}$ and let $\mathcal{X}_* \sim P_{\mathcal{X}}$. Let $r > 0$ and assume that $d > 2r$ and that there exists $\beta > 2r\frac{d_{\mathcal{X}}}{d_{\mathcal{X}}-2r}$ such that $\mathbb{E}\left[\|\boldsymbol{x}\|_2^{\beta}\right] < \infty$. Define*

$$d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n) := \min_{\mathcal{X}\in\boldsymbol{X}_n} \|\mathcal{X} - \mathcal{X}_*\|_2.$$

Then,

$$\mathbb{E}\left[\min\left\{d_{\min}^{2r}, 1\right\}\right] \leq c\, n^{-2r/d_{\mathcal{X}}},$$

where the constant $c > 0$ depends on $d_{\mathcal{X}}, r, \beta$ and $\mathbb{E}\left[\|\mathcal{X}\|_2^{\beta}\right]$.

Lemma OA.D.2 *Under the same assumptions as in Lemma OA.D.1 define $\mathcal{X}_{n,j}(\mathcal{X}_*, \boldsymbol{X}_n)$ as the j-th nearest neighbour of $\mathcal{X}_*$ in the sample $\boldsymbol{X}_n$ (assuming that ties occur with probability zero). Let*

$$d_j(\mathcal{X}_*, \boldsymbol{X}_n) := \|\mathcal{X}_{n,j}(\mathcal{X}_*, \boldsymbol{X}_n) - \mathcal{X}_*\|_2, \quad \langle d\rangle_{m,r}(\mathcal{X}_*, \boldsymbol{X}_n) := \frac{1}{m}\sum_{j=1}^{m} \min\left\{d_j(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}, 1\right\}.$$

Then, we have the following bounds

$$\mathbb{E}\left[\langle d\rangle_{m,r}\right] \leq c\left(\frac{m}{n}\right)^{2r/d_{\mathcal{X}}}, \quad 1 < m \leq n, \tag{55}$$

$$\mathbb{E}\left[\min\left\{d_m^{2r}, 1\right\}\right] \leq 2^{\frac{2r}{d_{\mathcal{X}}}+1} c\left(\frac{m}{n}\right)^{2r/d_{\mathcal{X}}}, \quad 1 < m \leq n/2. \tag{56}$$

Proof First prove the bound for $\mathbb{E}\left[\langle d\rangle_{m,r}\right]$ applying a technique from (Györfi et al., 2002, Proof of Theorem 6.2). Namely, we randomly split the training set $\boldsymbol{X}_n$ into $m+1$ disjoint subsets, such that the first m subsets have $\lfloor \frac{n}{m} \rfloor$ elements. Denote by $\tilde{\mathcal{X}}_j$ the nearest neighbour to $\mathcal{X}_*$ in the j-th subset, $j = 1, \ldots, m$. Clearly,

$$\|\mathcal{X}_{n,j} - \mathcal{X}_*\|_2^{2r} \leq \|\tilde{\mathcal{X}}_j - \mathcal{X}_*\|_2^{2r}, \quad j = 1, \ldots, m.$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\langle d\rangle_{m,r}\right] &= \frac{1}{m}\mathbb{E}\left[\sum_{j=1}^{m} \min\left\{\|\mathcal{X}_{n,j} - \mathcal{X}_*\|_2^{2r}, 1\right\}\right] \leq \frac{1}{m}\mathbb{E}\left[\sum_{j=1}^{m} \min\left\{\left\|\tilde{\mathcal{X}}_j - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] \\
&= \frac{1}{m}\sum_{j=1}^{m} \mathbb{E}\left[\min\left\{\left\|\tilde{\mathcal{X}}_j - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] = \mathbb{E}\left[\min\left\{\left\|\tilde{\mathcal{X}}_1 - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] \\
&= \mathbb{E}\left[\min\left\{\left\|\mathcal{X}_{\lfloor \frac{n}{m} \rfloor, 1} - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] \leq c\left(\frac{n}{m}\right)^{-2r/d_{\mathcal{X}}},
\end{aligned}$$

where in the last line we have applied Lemma OA.D.1. Finally, we proceed to prove (56). We have

$$\begin{aligned}\min\left\{d_m^{2r},1\right\}+\langle d\rangle_{m,r} &= \frac{1}{m}\left(m\min\left\{d_m^{2r},1\right\}+\sum_{i=1}^{m}\min\left\{d_i^{2r},1\right\}\right)\\ &\leq \frac{1}{m}\left(\sum_{i=1}^{m}\min\left\{d_{i+m}^{2r},1\right\}+\sum_{i=1}^{m}\min\left\{d_i^{2r},1\right\}\right)=2\langle d\rangle_{2m,r}.\end{aligned}$$

Using the above inequality in combination with (55), we get (56) as follows.

$$\mathbb{E}\left[\min\left\{d_m^{2r},1\right\}\right]\leq\mathbb{E}\left[\min\left\{d_m^{2r},1\right\}\right]+\mathbb{E}\left[\langle d\rangle_{m,r}\right]\leq 2\mathbb{E}\left[\langle d\rangle_{2m,r}\right]\leq 2^{\frac{2r}{d_{\mathcal{X}}}+1}\,c\left(\frac{m}{n}\right)^{2r/d_{\mathcal{X}}}.$$

■

Lemma OA.D.3 *Let $\boldsymbol{X}_n$ be training data sampled i.i.d. from $P_{\mathcal{X}}$ and let $\mathcal{X}_*\sim P_{\mathcal{X}}$. Under the assumptions (AC.2), (AR.1) and (AR.2) define $\mathcal{X}_{n,j}(\mathcal{X}_*,\boldsymbol{X}_n)$ as the j-th nearest neighbour of $\mathcal{X}_*$ in the sample $\boldsymbol{X}_n$ (assuming that ties occur with probability zero). Define the following distances in terms of the kernel-induced metric ρ_c.*

$$\epsilon_{\min}\left(\mathcal{X}_*,\boldsymbol{X}_n\right):=\min_{\mathcal{X}\in\boldsymbol{X}_n}\rho_c^2\left(\mathcal{X},\mathcal{X}_*\right),\quad \epsilon_j\left(\mathcal{X}_*,\boldsymbol{X}_n\right):=\rho_c^2\left(\mathcal{X}_{n,j},\mathcal{X}_*\right),$$

$$\langle\epsilon\rangle_m\left(\mathcal{X}_*,\boldsymbol{X}_n\right):=\frac{1}{m}\sum_{j=1}^{m}\rho_c^2\left(\mathcal{X}_{n,j},\mathcal{X}_*\right).$$

Assume that $d>2p$ (with p defined (AR.2)) in and that there exists $\beta>2p\frac{d_{\mathcal{X}}}{d_{\mathcal{X}}-2p}$ such that $\mathbb{E}\left[\|\mathcal{X}\|_2^{\beta}\right]<\infty$. Then, we have the following bounds

$$\mathbb{E}\left[\epsilon_{\min}\right]\leq C\,n^{-2p/d_{\mathcal{X}}},\tag{57}$$

$$\mathbb{E}\left[\langle\epsilon\rangle_m\right]\leq C\left(\frac{m}{n}\right)^{2p/d_{\mathcal{X}}},\quad 1<m\leq n,\tag{58}$$

$$\mathbb{E}\left[\epsilon_m\right]\leq 2^{\frac{2p}{d_{\mathcal{X}}}+1}\,C\left(\frac{m}{n}\right)^{2p/d_{\mathcal{X}}},\quad 1<m\leq n/2,\tag{59}$$

where

$$C=\max\left\{\frac{L_c}{\hat{\ell}^{2p}},1\right\}c$$

with c defined in Lemma OA.D.1 and L_c defined in (AR.2).

Proof By the assumption (AR.1) the ordering of the nearest neighbour set under the Euclidean metric is the same as the ordering of the nearest neighbour set under the kernel-induced metric. Since $\epsilon_{\min}=1-c(\mathcal{X}_{n,1}/\hat{\ell},\mathcal{X}_*/\hat{\ell})$, by (AR.2) we have that

$$\epsilon_{\min}\leq\min\left\{\frac{L_c}{\hat{\ell}^{2p}}\|\mathcal{X}_{n,1}-\mathcal{X}_*\|_2^{2p},1\right\}\leq\max\left\{\frac{L_c}{\hat{\ell}^{2p}},1\right\}\min\left\{\|\mathcal{X}_{n,1}-\mathcal{X}_*\|_2^{2p},1\right\},$$

where in the second inequality we have used the fact that for any $a, b \geq 0$ we have $\min\{ab, 1\} \leq \max\{a, 1\} \min\{b, 1\}$. Thus, by Lemma OA.D.1 we get that

$$\mathbb{E}\left[\epsilon_{\min}\right] \leq \max\left\{\frac{L_c}{\hat{\ell}^{2p}}, 1\right\} \mathbb{E}\left[\min\left\{\|\mathcal{X}_{n,1} - \mathcal{X}_*\|_2^{2p}, 1\right\}\right] \leq C\, n^{-2p/d_{\mathcal{X}}},$$

where

$$C = \max\left\{\frac{L_c}{\hat{\ell}^{2p}}, 1\right\} c$$

with c defined in Lemma OA.D.1.

Next, we prove the bound for $\mathbb{E}\left[\langle\epsilon\rangle_m\right]$ applying a technique from (Györfi et al., 2002, Proof of Theorem 6.2). Namely, we randomly split the training set X_n into $m+1$ disjoint subsets so that the first m subsets contain $\lfloor \frac{n}{m} \rfloor$ elements. Denote by $\tilde{\mathcal{X}}_j$ the nearest neighbour to $\mathcal{X}_*$ in the j-th subset. Then,

$$\begin{aligned}\mathbb{E}\left[\langle\epsilon\rangle_m\right] &= \frac{1}{m}\mathbb{E}\left[\sum_{i=1}^{m} \rho_c^2\left(\mathcal{X}_{n,i}, \mathcal{X}_*\right)\right] \leq \frac{1}{m}\mathbb{E}\left[\sum_{j=1}^{m} \rho_c^2\left(\tilde{\mathcal{X}}_j, \mathcal{X}_*\right)\right] = \frac{1}{m}\sum_{j=1}^{m}\mathbb{E}\left[\rho_c^2\left(\tilde{\mathcal{X}}_j, \mathcal{X}_*\right)\right] \\ &= \mathbb{E}\left[\rho_c^2\left(\tilde{\mathcal{X}}_1, \mathcal{X}_*\right)\right] = \mathbb{E}\left[\rho_c^2\left(\mathcal{X}_{1,\lfloor\frac{n}{m}\rfloor}, \mathcal{X}_*\right)\right] \leq C\left(\frac{n}{m}\right)^{-2p/d}.\end{aligned}$$

Finally, we prove (59). We have

$$\epsilon_m + \langle\epsilon\rangle_m = \frac{1}{m}\left(m\epsilon_m + \sum_{i=1}^{m}\epsilon_i\right) \leq \frac{1}{m}\left(\sum_{i=1}^{m}\epsilon_{i+m} + \sum_{i=1}^{m}\epsilon_i\right) = 2\langle\epsilon\rangle_{2m}.$$

Thus,

$$\mathbb{E}\left[\epsilon_m\right] \leq \mathbb{E}\left[\epsilon_m\right] + \mathbb{E}\left[\langle\epsilon\rangle_m\right] \leq 2\mathbb{E}\left[\langle\epsilon\rangle_{2m}\right] \leq 2C\left(\frac{n}{2m}\right)^{-2p/d_{\mathcal{X}}}.$$

■

Next, we state some auxiliary results needed for establishing the asymptotic convergence rate in Proposition 14.

Lemma OA.D.4 (Asymptotics of $d_{\min}$ - compact case.) *Let $\boldsymbol{X}_n = (\mathcal{X}_1, \ldots, \mathcal{X}_n)$ i.i.d. from $P_{\mathcal{X}}$ on $\mathbb{R}^d$, and let $\mathcal{X}_* \sim P_{\mathcal{X}}$ be independent of $\boldsymbol{X}_n$. Assume that $P_{\mathcal{X}}$ has a density q supported on a compact convex set $C \subset \mathbb{R}^{d_{\mathcal{X}}}$, where q is smooth and strictly positive on C. Then for every $r > 0$,*

$$n^{2r/d}\,\mathbb{E}\big[\min\left\{d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}, 1\right\}\big] \xrightarrow{n\to\infty} V_{d_{\mathcal{X}}}^{-2r/d_{\mathcal{X}}}\Gamma\left(1 + \frac{2r}{d_{\mathcal{X}}}\right)\int_C q(x)^{\,1-2r/d_{\mathcal{X}}}\,dx,$$

where $V_{d_{\mathcal{X}}}$ denotes the volume of the Euclidean unit ball in $\mathbb{R}^{d_{\mathcal{X}}}$.

Proof We begin with the compact-support asymptotic of Evans and Jones (2002) for nearest-neighbour moments. In the notation of the present paper, it states that if $\boldsymbol{U}_K := (\mathcal{U}_1, \ldots, \mathcal{U}_K)$ are i.i.d. with density q as above and

$$\delta_{\min}(\mathcal{U}_i, \boldsymbol{U}_K) := \min_{j\in\{1,\ldots,K\}\setminus\{i\}} \|\mathcal{U}_i - \mathcal{U}_j\|_2,$$

then, for every fixed $s > 0$,

$$K^{s/d_{\mathcal{X}}}\,\mathbb{E}[\delta_{\min}(\mathcal{U}_i, \boldsymbol{U}_K)^s] \xrightarrow{K\to\infty} V_{d_{\mathcal{X}}}^{-s/d_{\mathcal{X}}}\Gamma\left(1+\frac{s}{d_{\mathcal{X}}}\right)\int_C q(x)^{\,1-s/d_{\mathcal{X}}}\,dx.$$

Applying this with $s = 2r$ and $K = n+1$ yields

$$(n+1)^{2r/d_{\mathcal{X}}}\,\mathbb{E}\big[\delta_{\min}(\mathcal{U}_i, \boldsymbol{U}_{n+1})^{2r}\big] \xrightarrow{n\to\infty} V_{d_{\mathcal{X}}}^{-2r/d}\Gamma\left(1+\frac{2r}{d}\right)\int_C q(x)^{\,1-2r/d}\,dx.$$

We now translate this result to the independent-$\mathcal{X}_*$ setup. Let $\mathcal{U}_0 \equiv \mathcal{X}_*$ and $\mathcal{U}_i := \mathcal{X}_i$ for $i = 1, \ldots, n$. Then $\mathcal{U}_0, \ldots, \mathcal{U}_n$ are i.i.d. from $P_{\mathcal{X}}$. Moreover,

$$\delta_{\min}(\mathcal{U}_0, (\mathcal{U}_0, \ldots, \mathcal{U}_n)) = d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n).$$

By variable exchange symmetry, the random variables $\delta_{\min}(\mathcal{U}_i, (\mathcal{U}_0, \ldots, \mathcal{U}_n))$, $i = 0, \ldots, n$, are identically distributed. Hence

$$\mathbb{E}\big[d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}\big] = \mathbb{E}\big[\delta_{\min}(Z_0, (Z_0, \ldots, Z_n))^{2r}\big] = \mathbb{E}\big[\delta_{\min}(Z_i, (Z_0, \ldots, Z_n))^{2r}\big]$$

for any $i = 1, \ldots, n$. Therefore,

$$(n+1)^{2r/d}\,\mathbb{E}\big[d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}\big] \xrightarrow{n\to\infty} V_{d_{\mathcal{X}}}^{-2r/d}\Gamma\left(1+\frac{2r}{d}\right)\int_C q(x)^{\,1-2r/d}\,dx.$$

Since $(n+1)^{2r/d} \sim n^{2r/d}$, this is equivalent to

$$n^{2r/d_{\mathcal{X}}}\,\mathbb{E}\big[d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}\big] \xrightarrow{n\to\infty} V_{d_{\mathcal{X}}}^{-2r/d_{\mathcal{X}}}\Gamma\left(1+\frac{2r}{d_{\mathcal{X}}}\right)\int_C q(x)^{\,1-2r/d_{\mathcal{X}}}\,dx.$$

It remains to show that replacing $d_{\min}^{2r}$ with $\min\{d_{\min}^{2r}, 1\}$ does not affect the asymptotics. Since C is compact, $D := \sup_{x,y\in C}\|x-y\|_2 < \infty$, and thus $d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n) \le D$ almost surely. Furthermore, because q is continuous and strictly positive on the compact set C, and C is compact and convex, the function

$$x \mapsto P_{\mathcal{X}}[B(x,1)]$$

is continuous and strictly positive on C. Hence

$$\eta := \inf_{x\in C} P_{\mathcal{X}}[B(x,1)] > 0.$$

Conditioning on $\mathcal{X}_*$ therefore gives

$$P\left[d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n) > 1 \;\middle|\; \mathcal{X}_* = x\right] = [1 - P_{\mathcal{X}}[B(x,1)]]^n \le (1-\eta)^n,$$

and so

$$P\left[d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n) > 1\right] \le (1-\eta)^n.$$

Consequently,

$$0 \le \mathbb{E}\big[d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}\big] - \mathbb{E}\big[\min\big\{d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}, 1\big\}\big] \le D^{2r}(1-\eta)^n.$$

The right-hand side decays exponentially, hence is negligible compared with $n^{-2r/d}$. Therefore $d_{\min}^{2r}$ and $\min\{d_{\min}^{2r}, 1\}$ have the same asymptotic behaviour, and

$$n^{2r/d_{\mathcal{X}}}\,\mathbb{E}\big[\min\big\{d_{\min}(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}, 1\big\}\big] \longrightarrow V_{d_{\mathcal{X}}}^{-2r/d_{\mathcal{X}}}\Gamma\Big(1+\frac{2r}{d_{\mathcal{X}}}\Big)\int_C q(x)^{\,1-2r/d_{\mathcal{X}}}\,dx.$$

This proves the Lemma. ■

Lemma OA.D.5 (Asymptotics of d_m – compact case.) *Let $\boldsymbol{X}_n = (\mathcal{X}_1, \ldots, \mathcal{X}_n)$ i.i.d. from $P_{\mathcal{X}}$ on $\mathbb{R}^d$, and let $\mathcal{X}_* \sim P_{\mathcal{X}}$ be independent of $\boldsymbol{X}_n$. Assume that $P_{\mathcal{X}}$ has a density q supported on a compact convex set $C \subset \mathbb{R}^d$, where q is smooth and strictly positive on C. Then for every $r > 0$, $1 < m \le n/2$, and n large enough we have*

$$\mathbb{E}\big[\min\big\{d_m(\mathcal{X}_*, \boldsymbol{X}_n)^{2r}, 1\big\}\big] \le c_1\,\Big(\frac{m}{n}\Big)^{-2r/d}$$

where $0 < c_1 < \infty$ depends on r, $d_{\mathcal{X}}$ and $P_{\mathcal{X}}$.

Proof We randomly split the training set $\boldsymbol{X}_n$ into $m+1$ disjoint subsets, such that the first m subsets have $\lfloor \frac{n}{m} \rfloor$ elements. Denote by $\tilde{\mathcal{X}}_j$ the nearest neighbour to $\mathcal{X}_*$ in the j-th subset, $j = 1, \ldots, m$. Clearly,

$$\|\mathcal{X}_{n,j} - \mathcal{X}_*\|_2^{2r} \le \|\tilde{\mathcal{X}}_j - \mathcal{X}_*\|_2^{2r}, \quad j = 1, \ldots, m.$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\langle d\rangle_{m,r}\right] &= \frac{1}{m}\mathbb{E}\left[\sum_{j=1}^{m}\min\left\{\|\mathcal{X}_{n,j} - \mathcal{X}_*\|_2^{2r}, 1\right\}\right] \le \frac{1}{m}\mathbb{E}\left[\sum_{j=1}^{m}\min\left\{\left\|\tilde{\mathcal{X}}_j - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] \\
&= \frac{1}{m}\sum_{j=1}^{m}\mathbb{E}\left[\min\left\{\left\|\tilde{\mathcal{X}}_j - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] = \mathbb{E}\left[\min\left\{\left\|\tilde{\mathcal{X}}_1 - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] \\
&= \mathbb{E}\left[\min\left\{\left\|\mathcal{X}_{\lfloor \frac{n}{m}\rfloor,1} - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right].
\end{aligned}$$

To prove the compact-$P_{\mathcal{X}}$ case we directly apply Lemma OA.D.4 with n replaced by n/m which implies that for n/m large enough

$$\mathbb{E}\left[\langle d\rangle_{m,r}\right] \le \mathbb{E}\left[\min\left\{\left\|\mathcal{X}_{\lfloor \frac{n}{m}\rfloor,1} - \mathcal{X}_*\right\|_2^{2r}, 1\right\}\right] \le c_0\,\Big(\frac{n}{m}\Big)^{-2r/d},$$

where $c_0 = V_{d_{\mathcal{X}}}^{-2r/d_{\mathcal{X}}}\Gamma\Big(1+\frac{2r}{d_{\mathcal{X}}}\Big)\int_C q(x)^{\,1-2r/d_{\mathcal{X}}}\,dx$. Finally, we have

$$\begin{aligned}
\min\left\{d_m^{2r}, 1\right\} + \langle d\rangle_{m,r} &= \frac{1}{m}\left(m\min\left\{d_m^{2r}, 1\right\} + \sum_{i=1}^{m}\min\left\{d_i^{2r}, 1\right\}\right) \\
&\le \frac{1}{m}\left(\sum_{i=1}^{m}\min\left\{d_{i+m}^{2r}, 1\right\} + \sum_{i=1}^{m}\min\left\{d_i^{2r}, 1\right\}\right) = 2\langle d\rangle_{2m,r}.
\end{aligned}$$

Using the above inequality in combination with the previous bound for $\mathbb{E}\left[\langle d\rangle_{m,r}\right]$, we get the result of the Lemma as follows.

$$\mathbb{E}\left[\min\left\{d_m^{2r},1\right\}\right] \leq \mathbb{E}\left[\min\left\{d_m^{2r},1\right\}\right] + \mathbb{E}\left[\langle d\rangle_{m,r}\right] \leq 2\mathbb{E}\left[\langle d\rangle_{2m,r}\right] \leq 2^{\frac{2r}{d_{\mathcal{X}}}+1} c_0 \left(\frac{m}{n}\right)^{2r/d_{\mathcal{X}}}.$$

■

Appendix OA.E. Convergence Rates Proof

Lemma OA.E.1 *$\Omega \subset \mathbb{R}^D$ be a probability space with probability measure P and let $S \subsetneq \Omega$ satisfy $0 < \mathcal{P}[S] < 1$. Let $g : \mathbb{R}^D \to \mathbb{R}_{\geq 0}$ be a measurable function such that $c_{g,2} := \mathbb{E}\left[g(\mathcal{X})^2\right] < \infty$. Define the conditional expectation*

$$\mathbb{E}\left[g(\mathcal{X}) \mid \mathcal{X} \in S\right] := \frac{1}{\mathcal{P}[S]} \int_S g\, dP.$$

Let $S^c := \Omega - S$. Then, we have

$$\mathbb{E}\left[g(\mathcal{X})\right] \leq \mathbb{E}\left[g(\mathcal{X}) \mid \mathcal{X} \in S^c\right] + \sqrt{c_{g,2}}\sqrt{\mathcal{P}[S]},$$

Proof Decompose the expectation as follows

$$\mathbb{E}\left[g(\mathcal{X})\right] = \int_\Omega g\, dP = \int_{\Omega - S} g\, dP + \int_S g\, dP \leq \mathbb{E}\left[g(\mathcal{X}) \mid \mathcal{X} \in S^c\right] + \int_S g\, dP,$$

where we have used the fact that $\mathcal{P}(S^c) \leq 1$ and the definition of the conditional expectation. What is more, by the Cauchy-Schwarz inequality we have

$$\int_S g\, dP = \mathbb{E}\left[g(\mathcal{X})\mathcal{I}_S(\mathcal{X})\right] \leq \sqrt{\mathbb{E}\left[g(\mathcal{X})^2\right]}\sqrt{\mathbb{E}\left[\mathcal{I}_S(\mathcal{X})\right]} = \sqrt{c_{g,2}}\sqrt{\mathcal{P}[S]},$$

where $\mathcal{I}_S$ is the indicator function of S. Combining the above two inequalities we get the desired result. ■

Lemma OA.E.2 *Let $\epsilon_m(\mathcal{X}_*, \boldsymbol{X}_n)$ and $d_m(\mathcal{X}_*, \boldsymbol{X}_n)$ be as defined in Lemma OA.D.3 and Lemma OA.D.2 and let $\mathcal{X}_* \sim P_{\mathcal{X}}$ and $\boldsymbol{X}_n \sim P_{\mathcal{X}}^n$. For any $s > 0$, $0 < R \leq 1$ we have*

$$\mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m < R\right] \leq \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*\right], \tag{60}$$

$$\mathbb{E}\left[d_m^s \mid \mathcal{X}_*, d_m < R\right] \leq \mathbb{E}\left[\min\left\{d_m^s, 1\right\} \mid \mathcal{X}_*\right]. \tag{61}$$

Proof Using the definition of conditional expectation we have

$$\begin{aligned}
\mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*\right] &= \int \epsilon_m(X_n, \mathcal{X}_*) dP_{\mathcal{X}}^n(X_n) = \int_{\epsilon_m < R} \epsilon_m(X_n, \mathcal{X}_*) dP_{\mathcal{X}}^n(X_n) \\
&+ \int_{\epsilon_m \geq R} \epsilon_m(X_n, \mathcal{X}_*) dP_{\mathcal{X}}^n(X_n) = \mathcal{P}[\epsilon_m < R \mid \mathcal{X}_*]\, \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m < R\right] \\
&+ \mathcal{P}[\epsilon_m \geq R \mid \mathcal{X}_*]\, \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m \geq R\right] \\
&= \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m < R\right] \\
&+ \mathcal{P}[\epsilon_m \geq R \mid \mathcal{X}_*]\, \left(\mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m \geq R\right] - \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m < R\right]\right),
\end{aligned}$$

where in the last line we substituted $\mathcal{P}[\epsilon_m < R \mid \mathcal{X}_*] = 1 - \mathcal{P}[\epsilon_m \geq R \mid \mathcal{X}_*]$. Clearly,

$$\mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m < R\right] \leq R \leq \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m \geq R\right],$$

which implies that

$$\mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*\right] \geq \mathbb{E}\left[\epsilon_m \mid \mathcal{X}_*, \epsilon_m < R\right].$$

The inequality (61) can be derived in a fully analogous way. ■

Theorem 13 (Convergence Rates). *Let n be the size of the GPnn/NNGP training set which is i.i.d. sampled from the distribution $P_{\mathcal{X}}$ and let the test point be also sampled from $P_{\mathcal{X}}$. Let m be the (fixed) number of nearest-neighbours used in GPnn/NNGP. Assume (AC.5) and (AR.1-6). Define $\alpha := \min\{p, q\}$ for GPnn and $\alpha := \min\{p, q_0, q_1, \ldots, q_{d_T}\}$ for NNGP. Then, if $d_{\mathcal{X}} > 4(\alpha + p)$, we have*

$$\mathcal{R}_n \leq \frac{\sigma_\xi^2}{m} + A_1 \left(\frac{m}{n}\right)^{2\alpha/d_{\mathcal{X}}} + A_2\, m \left(\frac{m}{n}\right)^{2(\alpha+p)/d_{\mathcal{X}}}, \tag{62}$$

where $\mathcal{R}_n$ is the GPnn/NNGP risk defined in (6) and $A_1, A_2 > 0$ depend on p, q, $d_{\mathcal{X}}$, B_f, B_T, L_f, L_c, σ_ξ and the GPnn/NNGP hyper-paramaters. Taking $m_n = n^{\frac{2p}{2p+d_{\mathcal{X}}}}$ we obtain the following optimal minimax convergence rate.

$$\mathcal{R}_n \leq \left(\sigma_\xi^2 + A_1 + A_2\right) n^{-\frac{2\alpha}{2p+d_{\mathcal{X}}}}. \tag{63}$$

Proof Let us start with proving the *GPnn* part of the theorem. The *NNGP* case is addressed at the end. Recall from (8) that for fixed $(\boldsymbol{x}_*, X_n)$,

$$\begin{aligned} MSE(\mathcal{X}_*, \boldsymbol{X}_n) &= \mathbb{E}\left[(\mathcal{Y}_* - \tilde{\mu}_{GPnn}(\boldsymbol{x}_*))^2 \mid \mathcal{X}_*,\, \boldsymbol{X}_n\right] \\ &= \sigma_\xi^2 + \mathbb{E}\left[(f(\mathcal{X}_*) - \tilde{\mu}_{GPnn}(\mathcal{X}_*))^2 \mid \mathcal{X}_*,\, \boldsymbol{X}_n\right]. \end{aligned}$$

Averaging over $\mathcal{X}_* \sim P_{\mathcal{X}}$ and $\boldsymbol{X}_n \sim P_{\mathcal{X}}^n$ yields

$$\mathcal{R}_n = \mathbb{E}[MSE(\mathcal{X}_*, \boldsymbol{X}_n)] - \sigma_\xi^2,$$

where $\mathcal{R}_n$ is the risk (6). Define

$$f_{MSE}(X_n, \boldsymbol{x}_*) := |MSE(\boldsymbol{x}_*, X_n) - MSE_\infty|, \quad MSE_\infty := \sigma_\xi^2 \left(1 + \frac{1}{m}\right).$$

Note that

$$\begin{aligned} MSE(\boldsymbol{x}_*, X_n) &= |MSE(\boldsymbol{x}_*, X_n) - MSE_\infty + MSE_\infty| \\ &\leq |MSE(\boldsymbol{x}_*, X_n) - MSE_\infty| + MSE_\infty = f_{MSE}(\boldsymbol{x}_*, X_n) + \sigma_\xi^2 \left(1 + \frac{1}{m}\right). \end{aligned}$$

Taking expectations and subtracting σ_ξ^2 gives

$$\mathcal{R}_n \leq \frac{\sigma_\xi^2}{m} + \mathbb{E}[f_{MSE}(\mathcal{X}_*, \boldsymbol{X}_n)]. \tag{64}$$

Next, in order to upper bound $\mathbb{E}\left[f_{MSE}\right]$ we use Lemma OA.E.1 for $g \equiv f_{MSE}$ and $S \equiv \Omega_{m,n}(R)$, where

$$\Omega_{m,n}(R) := \left\{(\boldsymbol{x}_*, X_n) \mid d_m(\boldsymbol{x}_*, X_n) \geq R\right\}, \quad 0 < R \leq 1.$$

By Lemma OA.E.1,

$$\mathbb{E}\left[f_{MSE}\right] \leq \mathbb{E}\left[f_{MSE} \mid d_m < R\right] + \sqrt{c_{m,n}^{(2)}}\sqrt{P\left[\Omega_{m,n}(R)\right]}, \tag{65}$$

where $P\left[\Omega_{m,n}(R)\right]$ is the probability of the event $\Omega_{m,n}(R)$ under the probability measure $P_{\mathcal{X}} \otimes P_{\mathcal{X}}^n$ and

$$c_{m,n}^{(2)} = \mathbb{E}\left[f_{MSE}(\mathcal{X}_*, \boldsymbol{X}_n)^2\right].$$

Our goal is to show that the terms in inequality (65) have the following upper bounds when $d > 4(\alpha + p)$ with $\alpha = \min\{p, q\}$.

$$\begin{aligned} &\sqrt{c_{m,n}^{(2)}}\sqrt{P\left[\Omega_{m,n}(R)\right]} \leq A_2\, m \left(\frac{m}{n}\right)^{2(\alpha+p)/d_{\mathcal{X}}}, \quad A_2 > 0, \\ &\mathbb{E}_{X_n, \boldsymbol{x}_*}\left[f_{MSE}(X_n, \boldsymbol{x}_*) | d_m < R\right] \leq A_1 \left(\frac{m}{n}\right)^{2\alpha/d_{\mathcal{X}}}, \quad A_1 > 0. \end{aligned} \tag{66}$$

Let us start with proving the first statement of (66). To this end, we first apply Lemma OA.C.8 with $\nu = 2(\alpha + p)$ which gives

$$\sqrt{P\left[\Omega_{m,n}(R)\right]} = \sqrt{P\left[\min\left\{d_m(\mathcal{X}_*, \boldsymbol{X}_n), 1\right\} \geq R\right]} \leq \frac{1}{R^{2(\alpha+p)}}\sqrt{c}\, 2^{\frac{2(\alpha+p)}{d_{\mathcal{X}}} + \frac{1}{2}} \left(\frac{m}{n}\right)^{2(\alpha+p)/d_{\mathcal{X}}}.$$

What is more, $\sqrt{c_{m,n}^{(2)}}$ is bounded, since using the results from the proof of Theorem 9 we have that $f_{MSE}(X_n, \boldsymbol{x}_*)^2$ is bounded. In particular, Equation (47) states that

$$f_{MSE}(X_n, \boldsymbol{x}_*)^2 \leq \left(B_f^2 \left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f \hat{\sigma}_\xi}\sqrt{m} + 1\right)^2 + B_\xi \left(2 + \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f^2}\right)^2\right)\right)^2.$$

Thus, the product $\sqrt{c_{m,n}^{(2)}}\sqrt{\Pi_{m,n}(R)}$ is upper bounded as

$$\sqrt{c_{m,n}^{(2)}}\sqrt{P\left[\Omega_{m,n}(R)\right]} \leq A_2\, m \left(\frac{m}{n}\right)^{2(\alpha+p)/d} \tag{67}$$

with

$$A_2 = \frac{1}{R^{2(\alpha+p)}}\sqrt{c}\, 2^{\frac{2(\alpha+p)}{d_{\mathcal{X}}} + \frac{1}{2}} \left(B_f^2 \left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f \hat{\sigma}_\xi} + 1\right)^2 + B_\xi \left(2 + \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\left(\frac{\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_f^2}\right)^2\right)\right)$$

whenever $d > 4(\alpha + p)$. This proves the first statement of (66).

Let us next move to the proof of the second statement of (66). We use the fact that $f_{MSE}(X_n, \boldsymbol{x}_*)$ has an upper bound given by Theorem OA.F.2 which we repeat below for reader's convenience (we put $L_\xi = 0$ since σ_ξ^2 is constant).

$$f_{MSE}(X_n, \boldsymbol{x}_*) \leq \Big(\left(|f(\boldsymbol{x}_*)| + 2B_f L_f \min\{d_m^q, 1\}\right) \frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} + 2B_f L_f \min\{d_m^q, 1\} \Big)^2 + \sigma_\xi^2 \frac{3}{m} \frac{\epsilon_m + \epsilon_E}{(1 - \epsilon_E)^2}.$$

Next, we assume that $d_m < R$ with

$$R = \min\left\{ \hat{\ell} \left(8L_c\right)^{-\frac{1}{2p}}, 1 \right\}.$$

By (AR.2) this implies that $\epsilon_m < 1/8$ which combined with the upper bounds $\epsilon_E \leq 4\epsilon_m$ (see Lemma OA.B.4) and $\epsilon_m, \epsilon_E \leq 1$ gives

$$\frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} \leq 10\epsilon_m, \quad \frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} \leq \frac{2}{1 - 4\epsilon_m} \leq 4, \quad \frac{1}{1 - \epsilon_E} \leq \frac{1}{1 - 4\epsilon_m} \leq 2.$$

This in turn allows us to further upper bound $f_{MSE}(X_n, \boldsymbol{x}_*)$ as follows.

$$f_{MSE}(X_n, \boldsymbol{x}_*) \leq \left(10 \left(|f(\boldsymbol{x}_*)| + 2B_f L_f R^q\right) \epsilon_m + 2B_f L_f \min\{d_m^q, 1\}\right)^2 + 60 \frac{\sigma_\xi^2}{m} \epsilon_m.$$

Next, we expand the squared term and apply the bounds $d_m^q \leq R^q$, $\epsilon_m \leq 1$, $R \leq 1$ and $|f(\boldsymbol{x}_*)| \leq B_f$ in suitable places. This yields

$$f_{MSE}(X_n, \boldsymbol{x}_*) \leq 20 \left(B_f \left(1 + 2L_f\right)\left(5 + 12L_f\right) + 3 \frac{\sigma_\xi^2}{m} \right) \epsilon_m + (2B_f L_f)^2 \min\left\{d_m^{2q}, 1\right\}. \quad (68)$$

Next, we evaluate the conditional expectation $\mathbb{E}\left[* \mid \mathcal{X}_*, d_m < R\right]$ of the both sides of the inequality (68). Assumption (AR.1) implies that ϵ_m is the squared kernel-metric distance from $\mathcal{X}_*$ to it's m-th nearest neighbour in $\boldsymbol{X}_n$ and d_m is is the Euclidean distance of *the same* m-th nearest neighbour from $\mathcal{X}_*$. Furthermore, by assumption (AR.2) we have

$$\epsilon_m \leq \max\left\{ \frac{L_c}{\hat{\ell}^{2p}} d_m^{2p}, 1 \right\} \leq \max\left\{ \frac{L_c}{\hat{\ell}^{2p}}, 1 \right\} \min\left\{d_m^{2p}, 1\right\},$$

where we have used the fact that for any $a, b \geq 0$ we have $\min\{ab, 1\} \leq \max\{a, 1\} \min\{b, 1\}$.

By Lemma OA.E.2 we have the inequality

$$\mathbb{E}\left[\min\{d_m^{2q}, 1\} \mid \mathcal{X}_*, d_m < R\right] \leq \mathbb{E}\left[\min\{d_m^{2q}, 1\} \mid \mathcal{X}_*\right].$$

After plugging the above bounds into inequality (68) and applying Lemma OA.D.2 and Lemma OA.D.3 (after taking the conditional expectation $\mathbb{E}\left[* \mid d_m < R\right]$ of both sides), we get

$$\mathbb{E}\left[f_{MSE}(X_n, \boldsymbol{x}_*) \mid d_m < R\right] \leq C_p \left(\frac{m}{n}\right)^{2p/d} + C_q \left(\frac{m}{n}\right)^{2q/d}, \quad (69)$$

where

$$C_p = 2^{\frac{2p}{d}+3}\, 5\left(B_f\,(1+2L_f)\,(5+12L_f) + 3\,\sigma_\xi^2\right)\,\max\left\{\frac{L_c}{\hat{\ell}^{2p}}, 1\right\}\,c,$$

$$C_q = 2^{\frac{2q}{d}+3}\,(B_f L_f)^2\,c$$

with c defined in Lemma OA.D.1. Combining (64) with (65), (67) and (69) gives

$$\mathcal{R}_n \le \frac{\sigma_\xi^2}{m} + A_1\Big(\frac{m}{n}\Big)^{\frac{2\alpha}{d_{\mathcal{X}}}} + A_2\, m\Big(\frac{m}{n}\Big)^{\frac{2(\alpha+p)}{d_{\mathcal{X}}}},$$

which is (62).

$NNGP$ **case.** For $NNGP$, use the bias–variance decomposition (see Lemma OA.C.1) which splits the $NNGP$ MSE into a $GPnn$-type term plus a random-field term. The $GPnn$-type term is controlled exactly as above (with the relevant Hölder exponent), while the random-field term

$$\mathrm{Var}_{RF} = \sigma_w^2 + \Gamma^2\, {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1}\tilde{K}_{\mathcal{N}} K_{\mathcal{N}}^{-1} k_{\mathcal{N}}^* - 2\Gamma\, {k_{\mathcal{N}}^*}^T\, K_{\mathcal{N}}^{-1}\tilde{k}_{\mathcal{N}}^*$$

is bounded on the good region $\{d_m < R\}$ by using the inequality (37)

$$\left|\frac{\mathrm{Var}_{RF}}{\sigma_w^2}\right| \le (2\tilde{\epsilon}_m + \tilde{\epsilon}_E)\,\frac{\epsilon_m + 1}{1-\epsilon_E} + \left(\frac{\epsilon_E + \epsilon_m}{1-\epsilon_E}\right)^2 (1+\tilde{\epsilon}_E)$$

derived in the proof of Lemma OA.C.6 together with (AR.3). This produces contributions of order $(m/n)^{2q_0/d}$ and $(m/n)^{2q_i/d}$ and hence the overall good-region rate $(m/n)^{2\alpha/d}$ with $\alpha = \min\{p, q_0, q_1, \ldots, q_{d_T}\}$. The bad-region term is treated identically using Lemma OA.C.8 with $\nu = 2(\alpha + p)$, making use of the upper bound (48) for the random-field term

$$\mathrm{Var}_{RF} \le \sigma_w^2\left(1 + \sqrt{m}\,\Gamma\,\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}\right)^2 \le m\,\sigma_w^2\left(1 + \Gamma\,\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}\right)^2.$$

derived in the proof of Theorem 9. This completes the proof. ■

Proposition 14 (Asymptotic Convergence Rates). *Let n be the size of the training set which is i.i.d. sampled from the distribution $P_{\mathcal{X}}$ and let the test point be also sampled from $P_{\mathcal{X}}$. Define α for $GPnn/NNGP$ as in Theorem 13. Assume (AC.5), (AR.1-6) and*

- *$P_{\mathcal{X}}$ is supported on a compact convex set and has density which is smooth and strictly positive.*

Then taking $m_n = n^{\frac{2p}{2p+d_{\mathcal{X}}}}$ we have for sufficiently large n

$$\mathcal{R}_n \le A\, n^{-\frac{2\alpha}{2p+d_{\mathcal{X}}}}$$

where $0 < A < \infty$ depends on $P_{\mathcal{X}}$, p, q, $d_{\mathcal{X}}$, B_f, B_T, L_f, L_c, σ_ξ and the $GPnn$ or $NNGP$ hyper-paramaters.

Proof From the proof of Theorem 13 and the proof of Lemma OA.C.8 with $\nu = 2(\alpha + p)$ we have that

$$\mathcal{R}_n \leq \frac{\sigma_\xi^2}{m} + \tilde{A}_1 \mathbb{E}\left[\min\left\{d_m^{2\alpha}, 1\right\}\right] + \tilde{A}_2\, m \sqrt{\mathbb{E}\left[\min\left\{d_m^{4(\alpha+p)}, 1\right\}\right]}.$$

Applying Lemma OA.D.5 twice for $r = \alpha$ and $r = 4(\alpha + p)$ separately we get that for n large enough and $1 < m < n/2$

$$\mathcal{R}_n \leq \frac{\sigma_\xi^2}{m} + \tilde{A}_1 c_{1,1} \left(\frac{m}{n}\right)^{-2\alpha/d_{\mathcal{X}}} + \tilde{A}_2 \sqrt{c_{1,2}}\, m \left(\frac{m}{n}\right)^{-2(\alpha+p)/d_{\mathcal{X}}},$$

where $c_{1,1}$ depends on $d_{\mathcal{X}}$, $P_{\mathcal{X}}$ and α and $c_{1,2}$ depends on $d_{\mathcal{X}}$, $P_{\mathcal{X}}$, α and p. Taking $m = n^{\frac{2p}{2p+d_{\mathcal{X}}}}$ proves the Proposition. ■

Appendix OA.F. A key bound for MSE

Lemma OA.F.1 *Assume (AC.1-2), (AR.4) and (AC.4*).*

(AC.4*) *The noise variance $\sigma_\xi^2(\boldsymbol{x})$ is bounded by some constant $B_\xi \geq 1$ and is s-Hölder-continuous, i.e., there exist constants $L_\xi \geq 1$ and $0 < s \leq 1$ such that for every $\boldsymbol{x}, \boldsymbol{x}'$*

$$\left|\sigma_\xi^2(\boldsymbol{x}) - \sigma_\xi^2(\boldsymbol{x}')\right| \leq L_\xi \left\|\boldsymbol{x} - \boldsymbol{x}'\right\|_2^s.$$

Let ϵ_m and d_m be defined as in Remark OA.C.3 and Lemma OA.C.2 respectively and define

$$\epsilon_E := \frac{1}{m} \|E\|_1$$

with E is the matrix of pairwise distances between the nearest neighbours, i.e., $E_{i,j} = \epsilon_{i,j} := \rho_c^2\left(\boldsymbol{x}_{n,i}(\boldsymbol{x}*), \boldsymbol{x}_{n,j}(\boldsymbol{x}*)\right)$, $1 \leq i, j \leq m$. The following inequalities hold.

$$\begin{aligned} \left|\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - f(\boldsymbol{x}_*)\right| \leq \left(|f(\boldsymbol{x}_*)| + 2B_f L_f \min\{d_m^q, 1\}\right) \frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} \\ + 2B_f L_f \min\{d_m^q, 1\}, \end{aligned} \tag{70}$$

$$\begin{aligned} \left|\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m}\right| \leq \frac{2L_\xi B_\xi}{m} \min\{d_m^s, 1\} + \frac{3}{m} \frac{\epsilon_m + \epsilon_E}{(1 - \epsilon_E)^2} \times \\ \times \left(\sigma_\xi^2(\boldsymbol{x}_*) + 2B_\xi L_\xi \min\{d_m^s, 1\}\right). \end{aligned} \tag{71}$$

Proof Let us start with the proof of (70). Denote

$$\boldsymbol{\Delta} := K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^* - \hat{\sigma}_f^2 \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1}$$

and $\Delta f(X) := f(X) - f(\boldsymbol{x}_*)\mathbf{1}$. Then,

$$\begin{aligned} \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] &= \Gamma f(X)^T K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^* = \Gamma \left(f(\boldsymbol{x}_*)\mathbf{1} + \Delta f(X)\right)^T \left(\hat{\sigma}_f^2 \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} + \boldsymbol{\Delta}\right) \\ &= f(\boldsymbol{x}_*) + \Gamma f(\boldsymbol{x}_*)\, \mathbf{1}^T \boldsymbol{\Delta} + \Gamma \hat{\sigma}_f^2\, \Delta f(X)^T \left(K_{\mathcal{N}}^{\infty}\right)^{-1} \mathbf{1} + \Gamma \Delta f(X)^T \boldsymbol{\Delta}, \end{aligned}$$

Using the boundedness and the Hölder property of f (assumption AR.4), we get

$$||\Delta f(X)^T||_1 \leq \min\{L_f d_m^q, 2B_f\} \leq 2B_f L_f \min\{d_m^q, 1\}.$$

Furthermore, taking the 1-norms of the both sides and using the triangle inequality together with the fact that the matrix 1-norm is submultiplicative, we obtain

$$\begin{aligned}|\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - f(\boldsymbol{x}_*)| \leq &\left(|f(\boldsymbol{x}_*)| + 2B_f L_f \min\{d_m^q, 1\}\right) \Gamma \|\boldsymbol{\Delta}\|_1 \\ &+ \hat{\sigma}_f^2\, 2B_f L_f\, \Gamma \left\|(K_{\mathcal{N}}^{\infty})^{-1}\, \mathbf{1}\right\|_1 \min\{d_m^q, 1\}.\end{aligned}$$

The final result follows from the application of Equation (18) from Lemma OA.B.3 in Appendix OA.B and by noting that

$$\left\|\hat{\sigma}_f^2\, (K_{\mathcal{N}}^{\infty})^{-1}\, \mathbf{1}\right\|_1 = \frac{m\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2} = \frac{1}{\Gamma}. \tag{72}$$

Next, we proceed to the proof of (71). As shown in the proof of Lemma OA.C.1 we have

$$\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] = \Gamma^2 \mathbf{k}_{\mathcal{N}}^{*\,T}\, K_{\mathcal{N}}^{-1} \Sigma_{\boldsymbol{\xi}} K_{\mathcal{N}}^{-1}\, \mathbf{k}_{\mathcal{N}}^*,$$

where $\Sigma_{\boldsymbol{\xi}} := \mathrm{diag}\{\sigma_\xi^2(\boldsymbol{x}_{n,1}), \ldots, \sigma_\xi^2(\boldsymbol{x}_{n,m})\}$. Define $\delta\Sigma_\xi = \Sigma_{\boldsymbol{\xi}} - \sigma_\xi(\boldsymbol{x}_*)^2 \mathbb{1}$. Then, we have

$$\begin{aligned}\frac{1}{\Gamma^2}\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] &= \left(\boldsymbol{\Delta} + \hat{\sigma}_f^2\, (K_{\mathcal{N}}^{\infty})^{-1}\, \mathbf{1}\right)^T \left(\delta\Sigma_\xi + \sigma_\xi(\boldsymbol{x}_*)^2 \mathbb{1}\right) \left(\boldsymbol{\Delta} + \hat{\sigma}_f^2\, (K_{\mathcal{N}}^{\infty})^{-1}\, \mathbf{1}\right) \\ &= \frac{1}{\Gamma^2}\mathrm{Var}_{\mathcal{N},\infty} + 2\sigma_\xi^2(\boldsymbol{x}_*)\, \hat{\sigma}_f^2\, \mathbf{1}^T\, (K_{\mathcal{N}}^{\infty})^{-1}\, \boldsymbol{\Delta} + \sigma_\xi^2(\boldsymbol{x}_*)\, \boldsymbol{\Delta}^T\, \boldsymbol{\Delta} \\ &+ \hat{\sigma}_f^4\, \mathbf{1}^T\, (K_{\mathcal{N}}^{\infty})^{-1}\, \delta\Sigma_\xi\, (K_{\mathcal{N}}^{\infty})^{-1}\, \mathbf{1} + 2\hat{\sigma}_f^2\, \mathbf{1}^T\, (K_{\mathcal{N}}^{\infty})^{-1}\, \delta\Sigma_\xi \boldsymbol{\Delta} + \boldsymbol{\Delta}^T \delta\Sigma_\xi \boldsymbol{\Delta},\end{aligned}$$

where $\mathrm{Var}_{\mathcal{N},\infty} = \lim_{n\to\infty} \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]$ (see Equation (36) in the proof of Lemma OA.C.6). Taking the one-norm of the both sides we obtain

$$\begin{aligned}\frac{1}{\Gamma^2}\left|\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - \mathrm{Var}_{\mathcal{N},\infty}\right| \leq &\, 2\sigma_\xi^2(\boldsymbol{x}_*)\, \hat{\sigma}_f^2 \left\|\mathbf{1}^T\, (K_{\mathcal{N}}^{\infty})^{-1}\right\|_1 \|\boldsymbol{\Delta}\|_1 \\ &+ \sigma_\xi^2(\boldsymbol{x}_*)\, \left\|\boldsymbol{\Delta}^T\right\|_1 \|\boldsymbol{\Delta}\|_1 \\ &+ \|\delta\Sigma_\xi\|_1 \left(\frac{1}{m}\left(\frac{m\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\right)^2 + 2\hat{\sigma}_f^2 \left\|\mathbf{1}^T\, (K_{\mathcal{N}}^{\infty})^{-1}\right\|_1 \|\boldsymbol{\Delta}\|_1 + \left\|\boldsymbol{\Delta}^T\right\|_1 \|\boldsymbol{\Delta}\|_1\right).\end{aligned}$$

Next, we plug in the inequalities from Equations (18) and (19) from Lemma OA.B.3 in Appendix OA.B. We also use Equations (72) and the fact that

$$\left\|\hat{\sigma}_f^2 \mathbf{1}^T\, (K_{\mathcal{N}}^{\infty})^{-1}\right\|_1 = \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}, \quad \|\delta\Sigma_\xi\|_1 \leq 2B_\xi L_\xi \min\{d_m^s, 1\}.$$

Then, after some algebra we get the following inequality for the variance of the estimator $\tilde{\mu}_{\mathcal{N}}^*$.

$$\begin{aligned}\left|\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m}\right| \leq &\, \frac{2B_\xi L_\xi}{m} \min\{d_m^s, 1\} + \frac{1}{m}\frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} \times \\ &\times \left(2 + \frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E}\right) \left(\sigma_\xi^2(\boldsymbol{x}_*) + 2B_\xi L_\xi \min\{d_m^s, 1\}\right).\end{aligned}$$

The final result is obtained by applying the inequality $\epsilon_E, \epsilon_m \leq 1$ to get

$$2 + \frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} \leq 3 \frac{1}{1 - \epsilon_E}.$$

■

Theorem OA.F.2 (A key bound for MSE.) *Assume (AC.1-2), (AR.4), (AR.6) and (AC.4*). Let d_m, ϵ_m and ϵ_E be as defined in Lemma OA.C.2, Remark OA.C.3 and Lemma OA.F.1 respectively. Denote*

$$MSE_\infty(\boldsymbol{x}_*) := \sigma_\xi^2(\boldsymbol{x}_*) \left(1 + \frac{1}{m}\right).$$

The following bound holds for GPnn.

$$\begin{aligned} |MSE(\boldsymbol{x}_*, X) - MSE_\infty(\boldsymbol{x}_*)| \leq \Big(&(|f(\boldsymbol{x}_*)| + 2B_f L_f \min\{d_m^q, 1\}) \frac{\epsilon_m + \epsilon_E}{1 - \epsilon_E} \\ &+ 2B_f L_f \min\{d_m^q, 1\}\Big)^2 \\ + \frac{2B_\xi L_\xi}{m} \min\{d_m^s, 1\} &+ \frac{3}{m} \frac{\epsilon_m + \epsilon_E}{(1 - \epsilon_E)^2} \left(\sigma_\xi^2(\boldsymbol{x}_*) + 2B_\xi L_\xi \min\{d_m^s, 1\}\right). \end{aligned}$$

Proof Using the bias-variance decomposition of MSE and the triangle inequality for the absolute value we get

$$\begin{aligned} |MSE(\boldsymbol{x}_*, X) - MSE_\infty(\boldsymbol{x}_*)| \leq &\left(\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - f(\boldsymbol{x}_*)\right)^2 \\ &+ \left| \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right] - \frac{\sigma_\xi^2(\boldsymbol{x}_*)}{m} \right|. \end{aligned}$$

The final form of the inequality follows from combining the results of Lemma OA.F.1. ■

Appendix OA.G. Uniform flatness of risk landscape and asymptotic vanishing of MSE derivatives

OA.G.1 The Uniform Convergence of MSE in the Hyper-parameter Space

Theorem 15 (Uniform convergence of MSE in the hyper-parameter space). *Let $X = (\boldsymbol{x}_1, \boldsymbol{x}_2, \dots)$ be an infinite sequence of i.i.d. points sampled from $P_\mathcal{X}$ and denote by X_n its truncation to the first n points. Assume (AC.1-3), (AC.5), (AR.6) and (AR.1), (AR.4). Then, for almost every sampling sequence X and test point $\boldsymbol{x}_* \in \mathrm{Supp}(P_\mathcal{X})$ and any compact subset S of the hyper-parameters $\Theta = \left(\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\ell}\right) \in S \subset \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0} \times \mathbb{R}_{>0}$ we have that*

$$MSE(\boldsymbol{x}_*, X_n; \Theta) \xrightarrow{n \to \infty} MSE_\infty(\boldsymbol{x}_*; \Theta) := \sigma_\xi^2(\boldsymbol{x}_*) \left(1 + \frac{1}{m}\right)$$

and this convergence is uniform as a function of $\Theta \in S$.

Proof (For $GPnn$ – the $NNGP$ counterpart follows straightforwardly using the same techniques.) Fix $\boldsymbol{x}_* \in \text{Supp}(P_{\mathcal{X}})$ and an (infinite) sampling sequence X such that X is dense $\text{Supp}(P_{\mathcal{X}})$. Fix a compact subset of the hyper-parameter space S. Using Theorem OA.F.2, we will construct a non-negative function $h_\Theta(\boldsymbol{x}_*, X_n)$ that is continuous as a function of Θ and that bounds the MSE as follows

$$f_{MSE}(X_n, \boldsymbol{x}_*; \Theta) := |MSE(\boldsymbol{x}_*, X_n; \Theta) - MSE_\infty(\boldsymbol{x}_*; \Theta)| \leq h_\Theta(\boldsymbol{x}_*, X_n) \quad \text{for all} \quad \Theta \in S,$$

and that forms a monotonically decreasing sequence of functions of Θ with respect to n, i.e.,

$$h_\Theta(\boldsymbol{x}_*, X_{n+1}) \leq h_\Theta(\boldsymbol{x}_*, X_n), \quad h_\Theta(\boldsymbol{x}_*, X_n) \xrightarrow{n\to\infty} 0 \quad \text{for all} \quad \Theta \in S. \tag{73}$$

By Dini's theorem (Rudin, 1976) we have that $h_\Theta(\boldsymbol{x}_*, X_n) \xrightarrow{n\to\infty} 0$ uniformly on S. Since $f_{MSE}(X_n, \boldsymbol{x}_*; \Theta)$ is sandwiched between $h_\Theta(\boldsymbol{x}_*, X_n)$ and the constant zero function it follows that $f_{MSE}(X_n, \boldsymbol{x}_*; \Theta) \xrightarrow{n\to\infty} 0$ uniformly on S.

It remains to construct $h_\Theta(\boldsymbol{x}_*, X_n)$. To this end, define

$$\begin{aligned} h^{(0)}_\Theta(\boldsymbol{x}_*, X_n) := &\Big((|f(\boldsymbol{x}_*)| + 2B_f L_f \min\{d_m^q, 1\}) \frac{\epsilon_m + \epsilon_E}{1-\epsilon_E} + 2B_f L_f \min\{d_m^q, 1\} \Big)^2 \\ &+ \frac{2B_\xi L_\xi}{m} \min\{d_m^s, 1\} + \frac{3}{m} \frac{\epsilon_m + \epsilon_E}{(1-\epsilon_E)^2} \left(\sigma_\xi^2(\boldsymbol{x}_*) + 2B_\xi L_\xi \min\{d_m^s, 1\} \right). \end{aligned}$$

By Theorem OA.F.2 we have that $f_{MSE}(X_n, \boldsymbol{x}_*; \Theta) \leq h^{(0)}_\Theta(\boldsymbol{x}_*, X_n)$ for all $\Theta \in S$. The function $\epsilon_m(\boldsymbol{x}_*, X_n)$ decreases monotonically with n since the nearest neighbours are chosen with respect to the kernel-induced metric and $\epsilon_m(\boldsymbol{x}_*, X_n)$ is just the distance between $\boldsymbol{x}_*$ and its m-th nearest neighbour in X_n. However, $\epsilon_E(\boldsymbol{x}_*, X_n)$ may not decrease with n, so $h^{(0)}_\Theta(\boldsymbol{x}_*, X_n)$ may not monotonically decrease with n as well. To fix this, we will find an upper bound for $\epsilon_E(\boldsymbol{x}_*, X_n)$ which does decrease monotonically with n.

By (AR.1) the nearest neighbours can be equivalently chosen according to the Euclidean metric, thus the nearest-neighbour set is independent of the length scale choice $\hat{\ell}$ which enters the kernel-induced metric. Thus, we have

$$\epsilon_m(\boldsymbol{x}_*, X_n) = \rho_c^2 \left(\frac{d_m(\boldsymbol{x}_*, X_n)}{\hat{\ell}} \right), \quad d_m(\boldsymbol{x}_*, X_n) = \|\boldsymbol{x}_* - \boldsymbol{x}_{n,m}(\boldsymbol{x}_*)\|_2.$$

What is more, by (AR.1) the kernel metric is a strictly increasing function of the Euclidean distance. Thus, $\epsilon_E(\boldsymbol{x}_*, X_n)$ is upper bounded by the ϵ_E calculated for the nearest neighbour configuration where the nearest neighbours are grouped on the antipodal points of the Euclidean ball of the radius $d_m(\boldsymbol{x}_*, X_n)$, i.e.

$$\boldsymbol{x}_{n,1} = \cdots = \boldsymbol{x}_{n,m-1} = 2\boldsymbol{x}_* - \boldsymbol{x}_{n,m}.$$

In other words,

$$\epsilon_E(\boldsymbol{x}_*, X_n) \leq \tilde{\epsilon}_E(\boldsymbol{x}_*, X_n) := (m-1)\, \rho_c^2 \left(\frac{2d_m(\boldsymbol{x}_*, X_n)}{\hat{\ell}} \right).$$

The so-defined function $\tilde{\epsilon}_E(\boldsymbol{x}_*, X_n)$ is now monotonically decreasing with n. Hence, we put

$$\begin{aligned} h_\Theta(\boldsymbol{x}_*, X_n) := \Big((|f(\boldsymbol{x}_*)| + 2B_f L_f \min\{d_m^q, 1\}) \frac{\epsilon_m + \tilde{\epsilon}_E}{1 - \tilde{\epsilon}_E} + 2B_f L_f \min\{d_m^q, 1\} \Big)^2 \\ + \frac{2B_\xi L_\xi}{m} \min\{d_m^s, 1\} + \frac{3}{m} \frac{\epsilon_m + \tilde{\epsilon}_E}{(1 - \tilde{\epsilon}_E)^2} \left(\sigma_\xi^2(\boldsymbol{x}_*) + 2B_\xi L_\xi \min\{d_m^s, 1\} \right). \end{aligned} \tag{74}$$

Clearly, we have

$$f_{MSE}(X_n, \boldsymbol{x}_*; \Theta) \leq h_\Theta^{(0)}(\boldsymbol{x}_*, X_n) \leq h_\Theta(\boldsymbol{x}_*, X_n).$$

The upper bound $h_\Theta(\boldsymbol{x}_*, X_n)$ satisfies all the conditions of Dini's theorem: it is monotonically decreasing with n and is also a continuous function of Θ. This is because only the length scale $\hat{\ell}$ enters the formula (74) explicitly through ϵ_m and $\tilde{\epsilon}_E$ which are computed via the kernel metric and the kernel function is assumed to be continuous with respect to its arguments. ■

OA.G.2 The limits and the convergence rates of MSE derivatives

Using the MSE bias-variance expansion formula, we get that for any of the hyperparamteters $\phi \in \{\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\ell}\}$

$$\begin{aligned} \frac{\partial MSE_{GPnn}(\boldsymbol{x}_*, X)}{\partial \phi} = & 2\,\mathrm{Bias}_{GPnn}(\mathcal{X}_*, \boldsymbol{X}_n) \frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \phi} \\ & + \frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \phi}, \end{aligned} \tag{75}$$

where

$$\mathrm{Bias}_{GPnn}(\mathcal{X}_*, \boldsymbol{X}_n) = \Gamma\, \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*).$$

For simplicity, throughout this Section we adapt the homoscedastic noise model from (AR.6), however some of the results can be extended to encompass heteroscedastic noise. Using the well-known formulas for matrix derivatives, we further obtain

$$\frac{1}{\Gamma} \frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \phi} = \left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \phi} \right)^T K_{\mathcal{N}}^{-1} f(X) - \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} \frac{\partial K_{\mathcal{N}}}{\partial \phi} K_{\mathcal{N}}^{-1} f(X), \tag{76}$$

$$\frac{1}{\Gamma^2} \frac{1}{2\sigma_\xi^2} \frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \phi} = \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-2} \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \phi} - \mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-2} \frac{\partial K_{\mathcal{N}}}{\partial \phi} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^*. \tag{77}$$

Lemma OA.G.1 *The derivatives of the expected value and the variance of the estimator $\tilde{\mu}_{GPnn}$ with respect to the noise variance and the kernel scale read*

$$\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial\left(\hat{\sigma}_\xi^2\right)} = \frac{1}{m\hat{\sigma}_f^2}\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\left(f(X_{\mathcal{N}}) - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right), \tag{78}$$

$$\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial\left(\hat{\sigma}_\xi^2\right)} = \frac{2\sigma_\xi^2}{m}\frac{\Gamma}{\hat{\sigma}_f^2}\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2}\left(\mathbf{k}_{\mathcal{N}}^* - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\right), \tag{79}$$

$$\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial\left(\hat{\sigma}_f^2\right)} = -\frac{\hat{\sigma}_\xi^2}{m\hat{\sigma}_f^4}\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\left(f(X_{\mathcal{N}}) - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right), \tag{80}$$

$$\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial\left(\hat{\sigma}_f^2\right)} = -\frac{2\sigma_\xi^2}{m}\frac{\hat{\sigma}_\xi^2\Gamma}{\hat{\sigma}_f^4}\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2}\left(\mathbf{k}_{\mathcal{N}}^* - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\right). \tag{81}$$

Consequently, under the assumptions (AC.1-3), (AC.5) and (AR.6) we have the following limits holding for every test point $\boldsymbol{x}_* \in \mathrm{Supp}_{\rho_c}(P_{\mathcal{X}})$ and almost every sampling sequence $\boldsymbol{X}_n \sim P_{\mathcal{X}}^n$.

$$\frac{\partial MSE(\boldsymbol{x}_*, \boldsymbol{X}_n)}{\partial\left(\hat{\sigma}_\xi^2\right)} \xrightarrow{n\to\infty} 0, \quad \frac{\partial MSE(\boldsymbol{x}_*, \boldsymbol{X}_n)}{\partial\left(\hat{\sigma}_f^2\right)} \xrightarrow{n\to\infty} 0, \tag{82}$$

$$\left\|\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP}(\boldsymbol{x}_*, \boldsymbol{X}_n)\right\|_2 \xrightarrow{n\to\infty} 0. \tag{83}$$

Moreover, under (AC.1-3), (AC.5) and (AR.6) and the assumptions that

- *the function f in the GPnn response model (1) satisfies $\|f(\cdot)\|_\infty < B_f < \infty$,*
- *the functions t_i, $i = 1, \ldots, d_T$ in the NNGP response model (2) satisfy $\|t_i(\cdot)\|_\infty < B_T < \infty$,*

and fixed number of nearest neighbours m we have

$$\mathbb{E}\left[\frac{\partial MSE(\mathcal{X}_*, \boldsymbol{X}_n)}{\partial\left(\hat{\sigma}_\xi^2\right)}\right] \xrightarrow{n\to\infty} 0, \quad \mathbb{E}\left[\frac{\partial MSE(\mathcal{X}_*, \boldsymbol{X}_n)}{\partial\left(\hat{\sigma}_f^2\right)}\right] \xrightarrow{n\to\infty} 0, \tag{84}$$

$$\mathbb{E}\left[\left\|\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP}(\mathcal{X}_*, \boldsymbol{X}_n)\right\|_2\right] \xrightarrow{n\to\infty} 0. \tag{85}$$

Proof (For $GPnn$ – the $NNGP$ counterpart follows straightforwardly using the same techniques.) First, note that the derivatives of the kernel elements with respect to the kernel scale parameter read

$$\frac{\partial k(\boldsymbol{x}_1, \boldsymbol{x}_2)}{\partial(\hat{\sigma}_f^2)} = \frac{\partial(\hat{\sigma}_f^2\, c(\boldsymbol{x}_1/\hat{\ell}, \boldsymbol{x}_2/\hat{\ell}))}{\partial\hat{\sigma}_f^2} = c(\boldsymbol{x}_1/\hat{\ell}, \boldsymbol{x}_2/\hat{\ell}) = \frac{1}{\hat{\sigma}_f^2} k(\boldsymbol{x}_1, \boldsymbol{x}_2).$$

Thus, we have the following matrix and vector derivatives

$$\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial(\hat{\sigma}_\xi^2)} = 0, \quad \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial(\hat{\sigma}_f^2)} = \frac{1}{\hat{\sigma}_f^2}\mathbf{k}_{\mathcal{N}}^*, \quad \frac{\partial K_{\mathcal{N}}}{\partial(\hat{\sigma}_\xi^2)} = \mathbb{1}, \quad \frac{\partial K_{\mathcal{N}}}{\partial(\hat{\sigma}_f^2)} = \frac{1}{\hat{\sigma}_f^2}\left(K_{\mathcal{N}} - \hat{\sigma}_\xi^2\mathbb{1}\right) \tag{86}$$

The derivation of the formulas (78) - (81) boils down to applying the chain rule and using the generic derivative formulas (76) and (77) together with derivatives (86). The resulting calculation is a straightforward but tedious task, thus we skip its details.

Finally, in order to prove the limits (82), note that the expressions in the brackets in the equations (78) - (81) vanish as $n \to \infty$ because

$$\mathbf{1} - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)\left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1} = 0$$

due to Lemma OA.C.5. The proof of the limits (84) stating that the MSE derivatives tend to zero in expectation follows the same lines as the proof of Theorem 11. Namely, using the same methods one can show that the expressions from Equations (78)-(81) are bounded from above for any n by the respective constants independent of n and that depend only on the hyper-parameters and m. In particular, we have that

$$\begin{aligned}
&\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\left(f(X_{\mathcal{N}}) - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right)\right| \leq \left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| + \\
&+ \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} f(X_{\mathcal{N}})\right| \leq \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2 \\
&+ \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)\left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \left\|K_{\mathcal{N}}^{-1}\right\|_2 \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2 \\
&\leq \hat{\sigma}_f\sqrt{m}\frac{B_f}{\hat{\sigma}_\xi} + \hat{\sigma}_f\frac{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\sqrt{m}\frac{B_f}{\hat{\sigma}_\xi} \leq m\sqrt{m}\,\frac{B_f\hat{\sigma}_f\left(2\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2\right)}{\hat{\sigma}_\xi^3},
\end{aligned}$$

where we have used the facts that

$$\left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \leq \hat{\sigma}_f, \quad \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2 \leq \sqrt{m}\frac{B_f}{\hat{\sigma}_\xi}, \quad \left\|K_{\mathcal{N}}^{-1}\right\|_2 \leq \frac{1}{\hat{\sigma}_\xi^2}.$$

Similarly, we get

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2}\left(\mathbf{k}_{\mathcal{N}}^{*} - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^{*}\right)\right| \leq m\,\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\frac{2\hat{\sigma}_\xi^2 + \hat{\sigma}_f^2}{\hat{\sigma}_\xi^3}.$$

In particular, this yields uniform bounds in $\boldsymbol{x}_*, \boldsymbol{X}_n$ for the MSE-derivatives with the leading m-dependence of $\mathcal{O}(m)$. This allows us to use the dominated convergence theorem to obtain (84).

To prove (83) and (85), we use the bias-variance decomposition of MSE in $NNGP$ from Lemma OA.C.1 and find that the MSE depends only quadratically on $(\hat{\boldsymbol{b}} - \boldsymbol{b})$. Consequently, we have

$$\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP} = -2\left(\boldsymbol{v}^T.(\boldsymbol{b} - \hat{\boldsymbol{b}})\right)\boldsymbol{v}, \quad \boldsymbol{v}(\mathcal{X}_*, \boldsymbol{X}_n) := \boldsymbol{t}_*^T - \Gamma\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} T_{\mathcal{N}}.$$

and thus $\left\|\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP}\right\|_2 \leq 2\|\boldsymbol{v}\|_2^2\,\|\boldsymbol{b} - \hat{\boldsymbol{b}}\|_2$. Note that $\|\boldsymbol{v}\|_2^2$ can be bounded using the same techniques as the one used in previous proofs to bound the $GPnn$ bias term for the regression function $f \equiv t_i$, since

$$\|\boldsymbol{v}\|_2^2 = \sum_{i=1}^{d_T}\left(t_i\left(\mathcal{X}_*\right) - \Gamma\, k_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} t_i\left(X_{\mathcal{N}}\right)\right)^2 \leq d_T B_T^2\left(1 + \frac{\Gamma\hat{\sigma}_f}{\hat{\sigma}_\xi}\sqrt{m}\right)^2.$$

Thus, $\|\boldsymbol{v}\|_2^2 \to 0$ with probability one as $n \to \infty$ and $\mathbb{E}\left[\|\boldsymbol{v}\|_2^2\right] \to 0$ which proves (83) and (85). ■

Let us next move to proving convergence results for the $\hat{\ell}$-derivative.

Lemma OA.G.2 *Let $k(\boldsymbol{x}_1, \boldsymbol{x}_2) = \sigma_f^2\, c\left(\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2/\hat{\ell}\right)$ be an isotropic kernel function such that $c(u)$ is differentiable for $u > 0$, the limit $\lim_{u\to 0^+} c'(u)$ exists (but may not be finite), and $0 \le c(u) \le 1$ for all $u \ge 0$, and $c(0) = 1$. Assume that the corresponding normalised kernel metric $\rho_c(u) = \sqrt{1 - c(u)}$ satisfies the condition*

$$\rho_c(u) \le L\, u^p, \quad L > 0, \quad 0 < p \le 1. \tag{87}$$

Then,

$$\lim_{u\to 0^+} u\, c'(u) = 0. \tag{88}$$

Proof First, note that $c'(u) = -(\rho^2(u))'$, thus it suffices to show that $\lim_{u\to 0^+} u\,(\rho^2(u))' = 0$. What is more,

$$\lim_{u\to 0^+} (u\,\rho^2(u))' = \lim_{u\to 0^+} u\,(\rho^2(u))' + \lim_{u\to 0^+} \rho^2(u) = \lim_{u\to 0^+} u\,(\rho^2(u))',$$

since $\rho(0) = 0$. Let $f(u) = u\,\rho^2(u)$ and $g(u) = L^2\, u^{2p+1}$, thus $f(0) = g(0) = 0$. Because $\rho(u)$ satisfies (87), we also have $0 \le f(u) \le g(u)$. This implies the following bound for the right-sided derivative of $f(u)$

$$f'(0^+) = \lim_{h\to 0^+} \frac{f(h)}{h} \le \lim_{h\to 0^+} \frac{g(h)}{h} = g'(0^+).$$

Therefore,

$$\lim_{u\to 0^+} (u\,\rho^2(u))' \le L^2 \lim_{u\to 0^+} (u^{2p+1})' = L^2(2p+1) \lim_{u\to 0^+} u^{2p} = 0.$$

We also have that $\lim_{u\to 0^+} (u\,\rho^2(u))' \ge 0$, since $\rho'(0) \ge 0$ as $\rho(u)$ achieves its global minimum at $u = 0$. This proves (88). ■

Lemma OA.G.3 *Under the assumptions of Lemma OA.G.2 and (AD.1) and (AR.2) we have*

$$\frac{\partial MSE(\boldsymbol{x}_*, X_n)}{\partial \hat{\ell}} \xrightarrow{n\to\infty} 0 \tag{89}$$

with probability one. Moreover, under (AC.1-3), (AC.5), (AR.6), (AD.1-2) and (AR.2) and the assumptions that

- *the function f in the GPnn response model (1) satisfies $\|f(\cdot)\|_\infty < B_f < \infty$,*
- *the functions t_i, $i = 1, \ldots, d_T$ in the NNGP response model (2) satisfy $\|t_i(\cdot)\|_\infty < B_T < \infty$,*

we have that for $(\mathcal{X}_*, \boldsymbol{X}_n) \sim P_{\mathcal{X}} \otimes P_{\mathcal{X}}^n$

$$\mathbb{E}\left[\left|\frac{\partial MSE(\boldsymbol{x}_*, X_n)}{\partial \hat{\ell}}\right|\right] \xrightarrow{n\to\infty} 0. \tag{90}$$

Proof (For $GPnn$ – the $NNGP$ counterpart follows straightforwardly using the same techniques.) Fully analogous to the proof of Lemma OA.G.1. The pointwise limit is shown using Equations (75)-(77) together with Lemma OA.G.2. Note that the derivative of an isotropic kernel $k(r)$ with respect to the lengthscale reads

$$\frac{\partial k(r/\hat{\ell})}{\partial \hat{\ell}} = -\frac{1}{\hat{\ell}^2} r\, k'(r/\hat{\ell}).$$

Thus, by Lemma OA.G.2 we have $\frac{\partial k(r/\hat{\ell})}{\partial \hat{\ell}} \xrightarrow{n\to\infty} 0$. The limit in expectation (90) is shown using the dominated convergence theorem. To this end, we derive an upper bound on the MSE that is independent of $\boldsymbol{x}_*, X_n$ using the Equations (75)-(77) and assumption (AD.2) together with the bounds used in the proof of Theorem 11.

Starting from the bias-variance expansion (75) with $\phi = \hat{\ell}$ and taking absolute values, we have

$$\begin{aligned}\left|\frac{\partial MSE_{GPnn}(\mathcal{X}_*, \boldsymbol{X}_n)}{\partial \hat{\ell}}\right| &= 2\left|\Gamma\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right| \left|\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \\ &\quad + \left|\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right|,\end{aligned} \tag{91}$$

We now bound each factor on the right-hand side.

First, we bound $\left|\Gamma\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right|$ using $|f(\boldsymbol{x}_*)| \leq B_f$ and the Cauchy-Schwarz bound as follows

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}})\right| = \left|\left(\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right)\left(K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right)\right| \leq \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2.$$

By the non-negativity of the GP predictive variance (as used in the derivation leading to (47)) we have

$$\left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \leq \hat{\sigma}_f, \qquad \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2 \leq \sqrt{m}\, \frac{B_f}{\hat{\sigma}_\xi}.$$

Therefore,

$$\left|\Gamma\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right| \leq B_f + \Gamma\, \hat{\sigma}_f \sqrt{m}\, \frac{B_f}{\hat{\sigma}_\xi} = B_f\left(1 + \frac{\Gamma\, \hat{\sigma}_f}{\hat{\sigma}_\xi}\sqrt{m}\right). \tag{92}$$

Next, we bound $\left|\partial_{\hat{\ell}} \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]\right|$ using using (76) as follows.

$$\frac{1}{\Gamma}\left|\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \leq \left|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| + \left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right|.$$

We bound the two RHS-terms separately. Using Cauchy-Schwarz we get

$$\left|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| = \left|\left(K_{\mathcal{N}}^{-1/2} \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T \left(K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right)\right| \leq \left\|K_{\mathcal{N}}^{-1/2} \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_2 \left\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\right\|_2.$$

By (AD.2) and the chain rule, for any pair of inputs,

$$\left|\frac{\partial k(\boldsymbol{x}_1,\boldsymbol{x}_2)}{\partial \hat{\ell}}\right| = \left|\frac{\hat{\sigma}_f^2}{\hat{\ell}}\, u\, c'(u)\right| \le \frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}, \qquad u = \frac{\|\boldsymbol{x}_1-\boldsymbol{x}_2\|_2}{\hat{\ell}}.$$

Hence every entry of $\partial_{\hat{\ell}}\mathbf{k}_{\mathcal{N}}^*$ has magnitude at most $\hat{\sigma}_f^2 B_c/\hat{\ell}$, so

$$\left\|\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_2 \le \sqrt{m}\,\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}.$$

Together with $\|K_{\mathcal{N}}^{-1/2}\|_2 \le 1/\hat{\sigma}_\xi$ and $\|K_{\mathcal{N}}^{-1/2} f(X_{\mathcal{N}})\|_2 \le \sqrt{m}\, B_f/\hat{\sigma}_\xi$, we obtain

$$\left|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| \le \frac{1}{\hat{\sigma}_\xi}\left(\sqrt{m}\,\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}\right)\left(\sqrt{m}\,\frac{B_f}{\hat{\sigma}_\xi}\right) = \frac{B_f \hat{\sigma}_f^2 B_c}{\hat{\ell}\,\hat{\sigma}_\xi^2}\, m. \tag{93}$$

Next, we apply Cauchy–Schwarz again as follows.

$$\begin{aligned}\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| &= \left|\left(\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right)\left(K_{\mathcal{N}}^{-1/2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right)\right| \\ &\le \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\right\|_2 \left\|K_{\mathcal{N}}^{-1/2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right\|_2.\end{aligned}$$

Using $\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1/2}\|_2 \le \hat{\sigma}_f$ and

$$\left\|K_{\mathcal{N}}^{-1/2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f\right\|_2 \le \|K_{\mathcal{N}}^{-1/2}\|_2 \left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_2 \|K_{\mathcal{N}}^{-1} f\|_2, \quad \|K_{\mathcal{N}}^{-1} f\|_2 \le \|K_{\mathcal{N}}^{-1/2}\|_2 \|K_{\mathcal{N}}^{-1/2} f\|_2,$$

we get

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| \le \hat{\sigma}_f \cdot \frac{1}{\hat{\sigma}_\xi}\left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_2 \left(\frac{1}{\hat{\sigma}_\xi}\cdot\sqrt{m}\,\frac{B_f}{\hat{\sigma}_\xi}\right) = \hat{\sigma}_f \left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_2 \frac{B_f\sqrt{m}}{\hat{\sigma}_\xi^3}.$$

Finally, by (AD.2) we have the uniform entrywise bound $\left|\partial_{\hat{\ell}} k(\boldsymbol{x}_i,\boldsymbol{x}_j)\right| \le \hat{\sigma}_f^2 B_c/\hat{\ell}$ for all i,j, so the maximal row sum satisfies

$$\left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_2 \le \left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_\infty \le m\,\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}.$$

Hence,

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right| \le \hat{\sigma}_f\left(m\,\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}\right)\frac{B_f\sqrt{m}}{\hat{\sigma}_\xi^3} = \frac{B_f\hat{\sigma}_f^3 B_c}{\hat{\ell}\,\hat{\sigma}_\xi^3}\, m^{3/2}. \tag{94}$$

Combining (93) and (94) with (109) yields

$$\left|\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \le \Gamma\,\frac{B_f B_c}{\hat{\ell}}\left(\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\, m + \frac{\hat{\sigma}_f^3}{\hat{\sigma}_\xi^3}\, m^{3/2}\right). \tag{95}$$

As the final step, we bound $|\partial_{\hat{\ell}}\mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]|$ using (77) as follows

$$\frac{1}{\Gamma^2}\frac{1}{2\sigma_\xi^2}\left|\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \leq \left|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-2}\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right| + \left|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1}\,\mathbf{k}_{\mathcal{N}}^*\right|.$$

For the first term,

$$\left|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-2}\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right| = \left|(K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*)^T\left(K_{\mathcal{N}}^{-1}\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)\right| \leq \|K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\|_2\left\|K_{\mathcal{N}}^{-1}\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_2.$$

Using

$$\|K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\|_2 \leq \|K_{\mathcal{N}}^{-1/2}\|_2\,\|K_{\mathcal{N}}^{-1/2}\mathbf{k}_{\mathcal{N}}^*\|_2 \leq \frac{1}{\hat{\sigma}_\xi}\,\hat{\sigma}_f,$$

$$\left\|K_{\mathcal{N}}^{-1}\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_2 \leq \|K_{\mathcal{N}}^{-1}\|_2\left\|\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_2 \leq \frac{1}{\hat{\sigma}_\xi^2}\left(\sqrt{m}\,\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}\right),$$

we obtain

$$\left|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-2}\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right| \leq \frac{\hat{\sigma}_f}{\hat{\sigma}_\xi}\cdot\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}\,\hat{\sigma}_\xi^2}\sqrt{m} = \frac{\hat{\sigma}_f^3 B_c}{\hat{\ell}\,\hat{\sigma}_\xi^3}\sqrt{m}. \tag{96}$$

For the second term, insert $K_{\mathcal{N}}^{-1/2}$ and use Cauchy–Schwarz:

$$\begin{aligned}\left|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1}\,\mathbf{k}_{\mathcal{N}}^*\right| &= \left|\left({\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-1/2}\right)\left(K_{\mathcal{N}}^{-3/2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1}\,\mathbf{k}_{\mathcal{N}}^*\right)\right| \\ &\leq \left\|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-1/2}\right\|_2\left\|K_{\mathcal{N}}^{-3/2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1}\,\mathbf{k}_{\mathcal{N}}^*\right\|_2 \\ &\leq \hat{\sigma}_f\cdot\|K_{\mathcal{N}}^{-3/2}\|_2\left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_2\|K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\|_2.\end{aligned}$$

Using $\|K_{\mathcal{N}}^{-3/2}\|_2 \leq 1/\hat{\sigma}_\xi^3$, $\|K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\|_2 \leq \|K_{\mathcal{N}}^{-1/2}\|_2\|K_{\mathcal{N}}^{-1/2}\mathbf{k}_{\mathcal{N}}^*\|_2 \leq \hat{\sigma}_f/\hat{\sigma}_\xi$, and $\|\partial_{\hat{\ell}}K_{\mathcal{N}}\|_2 \leq m\hat{\sigma}_f^2 B_c/\hat{\ell}$, we get

$$\left|{\mathbf{k}_{\mathcal{N}}^*}^T K_{\mathcal{N}}^{-2}\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1}\,\mathbf{k}_{\mathcal{N}}^*\right| \leq \hat{\sigma}_f\cdot\frac{1}{\hat{\sigma}_\xi^3}\left(m\frac{\hat{\sigma}_f^2 B_c}{\hat{\ell}}\right)\frac{\hat{\sigma}_f}{\hat{\sigma}_\xi} = \frac{\hat{\sigma}_f^4 B_c}{\hat{\ell}\,\hat{\sigma}_\xi^4}\,m. \tag{97}$$

Combining (96) and (97) gives

$$\left|\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \leq 2\sigma_\xi^2\Gamma^2\frac{B_c}{\hat{\ell}}\left(\frac{\hat{\sigma}_f^3}{\hat{\sigma}_\xi^3}\sqrt{m} + \frac{\hat{\sigma}_f^4}{\hat{\sigma}_\xi^4}m\right). \tag{98}$$

Substituting (92), (95), and (98) into (91) yields the following explicit uper bound for $\left|\partial_{\hat{\ell}}MSE(\boldsymbol{x}_*, X_n)\right|$ that is uniform in $(\boldsymbol{x}_*, X_n)$ for fixed hyper-parameters.

$$\begin{aligned}\left|\frac{\partial MSE_{GPnn}(\mathcal{X}_*, \boldsymbol{X}_n)}{\partial \hat{\ell}}\right| \leq{}& 2\,\frac{B_f^2 B_c\Gamma}{\hat{\ell}}\left(1 + \frac{\Gamma\,\hat{\sigma}_f}{\hat{\sigma}_\xi}\sqrt{m}\right)\left(\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2}\,m + \frac{\hat{\sigma}_f^3}{\hat{\sigma}_\xi^3}\,m^{3/2}\right) \\ &+2\sigma_\xi^2\frac{B_c\Gamma}{\hat{\ell}}\left(\frac{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}{m\hat{\sigma}_f^2}\right)^2\left(\frac{\hat{\sigma}_f^3}{\hat{\sigma}_\xi^3}\sqrt{m} + \frac{\hat{\sigma}_f^4}{\hat{\sigma}_\xi^4}m\right).\end{aligned} \tag{99}$$

In particular, the leading m-dependence of the right-hand side is $\mathcal{O}(m^2)$. ■

Lemma OA.G.4 below proves MSE bounds for $GPnn$ that are a crucial component in proving Theorem 17 concerning the convergence rates of the derivatives. We skip the derivation of the corresponding bounds for $NNGP$, because they have the same general forms and follow directly from the derivations presented below.

Lemma OA.G.4 (Bounds for MSE derivatives.) *Assume (AC.1-2) and (AC.5) and (AR.1-6). Consider a sequence of sampling points $X_n = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n)$ and let $\boldsymbol{x}_* \in \text{Support}_{\rho_c}(P_{\mathcal{X}})$, $\boldsymbol{x}_* \notin X_n$ be a test point. Then,*

$$\begin{aligned}\left|\frac{\partial MSE_{GPnn}(\boldsymbol{x}_*, X_n)}{\partial(\hat{\sigma}_\xi^2)}\right| \leq & 48 B_f^2 L_f^2 \frac{1}{m\hat{\sigma}_f^2}\frac{1}{(1-\epsilon_E)^3} \min\left\{d_m^{2q}, 1\right\} \\ & + \frac{1}{m\hat{\sigma}_f^2}\left(24\sigma_\xi^2 \frac{1}{1-\epsilon_{E,2}}\frac{1}{1-\epsilon_E} + \frac{2B_f^2}{(1-\epsilon_E)^3}(78L_f+5)\right)\epsilon_m\end{aligned} \tag{100}$$

$$\begin{aligned}\left|\frac{\partial MSE_{GPnn}(\boldsymbol{x}_*, X_n)}{\partial(\hat{\sigma}_f^2)}\right| \leq & 48 B_f^2 L_f^2 \frac{\hat{\sigma}_\xi^2}{m\hat{\sigma}_f^2}\frac{1}{(1-\epsilon_E)^3} \min\left\{d_m^{2q}, 1\right\} \\ & + \frac{\hat{\sigma}_\xi^2}{m\hat{\sigma}_f^2}\left(24\sigma_\xi^2 \frac{1}{1-\epsilon_{E,2}}\frac{1}{1-\epsilon_E} + \frac{2B_f^2}{(1-\epsilon_E)^3}(78L_f+5)\right)\epsilon_m.\end{aligned} \tag{101}$$

$$\left\|\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP}\right\|_2 \leq 2\, d_T \|\boldsymbol{b} - \hat{\boldsymbol{b}}\|_2 \left(\frac{10B_T(1+2L_T)}{1-\epsilon_E}\epsilon_m + 4B_f L_f \min\{d_m^{\overline{q}}, 1\}\right)^2, \tag{102}$$

where $\overline{q} := \min_i\{q_i\}$ and $L_T := \max_i L_i$ with the relevant constants defined in (AR.4). Additionally, under (AR.2) and (AD.1-2) we have that the following upper bound holds for every $\boldsymbol{x}_*, X_n$.

$$\begin{aligned}\left|\frac{\partial MSE_{GPnn}(\boldsymbol{x}_*, X)}{\partial\hat{\ell}}\right| \leq \frac{1}{\hat{\ell}}\Bigg(& \frac{6B_f^2 B_c L_c'(1+2L_f)}{(1-\epsilon_E)^2}\left(\frac{5(1+2L_f)}{1-\epsilon_E}\epsilon_m + 2L_f \min\{d_m^q, 1\}\right) \\ & + \frac{8\sigma_\xi^2\,\hat{\sigma}_f^2 B_c L_c'}{(1-\epsilon_E)(1-\epsilon_{E,2})}\Bigg)\max\left\{\hat{\ell}^{-2p'}, 1\right\}\min\left\{d_m^{2p'}, 1\right\}.\end{aligned} \tag{103}$$

Proof The proof repeats the techniques that have been used in the previous parts of this paper.

We first prove the bound (100). The proof of the bound (101) is almost identical. According to the Equation (75),

$$\begin{aligned}\left|\frac{\partial MSE_{GPnn}(\mathcal{X}_*, \boldsymbol{X}_n)}{\partial\hat{\sigma}_\xi^2}\right| = & 2\left|\Gamma\,\mathbf{k}_{\mathcal{N}}^{*\,T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right|\left|\frac{\partial\mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial\hat{\sigma}_\xi^2}\right| \\ & + \left|\frac{\partial\text{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial\hat{\sigma}_\xi^2}\right|,\end{aligned}$$

The expression $\left|\Gamma\, \mathbf{k}_{\mathcal{N}}^{*\,T}\, K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right|$ can be suitably upper bounded using the inequality (70) from Lemma OA.F.1.

Next, we use Equations (78) and (79) from Lemma OA.G.1 in order to find bounds on the relevant derivatives of the expectation and the variance of $\tilde{\mu}$. To this end, we use the following bounds.

$$\begin{aligned}
\left\|f(X_{\mathcal{N}}) - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1} f(X_{\mathcal{N}})\right\|_1 &= \Big\|f(X_{\mathcal{N}}) - f(\boldsymbol{x}_*)\mathbf{1} + f(\boldsymbol{x}_*)\mathbf{1} + \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) \times \\
&\times \left(K_{\mathcal{N}}^{-1}\, f\left(X_{n,m}\right) - f\left(\boldsymbol{x}_*\right)\left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1}\right) - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) f\left(\boldsymbol{x}_*\right)\left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1}\Big\|_1 \\
&= \left\|f(X_{\mathcal{N}}) - f(\boldsymbol{x}_*)\mathbf{1} + \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)\left(K_{\mathcal{N}}^{-1}\, f\left(X_{\mathcal{N}}\right) - f\left(\boldsymbol{x}_*\right)\left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1}\right)\right\|_1 \\
&\leq \left\|f(X_{\mathcal{N}}) - f(\boldsymbol{x}_*)\mathbf{1}\right\|_1 + \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right)\left\|K_{\mathcal{N}}^{-1}\, f\left(X_{\mathcal{N}}\right) - f\left(\boldsymbol{x}_*\right)\left(K_{\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1}\right\|_1 \\
&\leq 2m\, L_f B_f\, \min\{d_m^q, 1\} + mB_f \frac{2L_f \min\{d_m^q, 1\} + \epsilon_E}{1-\epsilon_E} \qquad (104)
\end{aligned}$$

In the last line we have used (AR.4) and the inequality (21) proved in the Appendix OA.B. In a similar way, we apply suitable triangle inequalities together with the inequality (18) proved in the Appendix OA.B to obtain

$$\left\|\mathbf{k}_{\mathcal{N}}^* - \left(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2\right) K_{\mathcal{N}}^{-1}\mathbf{k}_{\mathcal{N}}^*\right\|_1 \leq m\, \hat{\sigma}_f^2\, \epsilon_m \left(1 + \frac{5}{1-\epsilon_E}\right). \qquad (105)$$

We can also use the triangle inequality together with the inequality (19) to derive the following bound

$$\begin{aligned}
\left\|\mathbf{k}_{\mathcal{N}}^{*\,T}\, K_{\mathcal{N}}^{-1}\right\|_1 &\leq \left\|\mathbf{k}_{\mathcal{N}}^{*\,T}\, K_{\mathcal{N}}^{-1} - \hat{\sigma}_f^2\, \mathbf{1}^T\, \left(K_{\mathcal{N}}^{\infty}\right)^{-1}\right\|_1 + \hat{\sigma}_f^2 \left\|\mathbf{1}^T\, \left(K_{\mathcal{N}}^{\infty}\right)^{-1}\right\|_1 \\
&\leq \hat{\sigma}_f^2 \left\|\mathbf{1}^T\, \left(K_{\mathcal{N}}^{\infty}\right)^{-1}\right\|_1 \left(1 + \frac{\epsilon_m + \epsilon_E}{1-\epsilon_E}\right) = \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1+\epsilon_m}{1-\epsilon_E} \leq \frac{2}{m}\frac{1}{1-\epsilon_E} \qquad (106)
\end{aligned}$$

Similarly, we can use the triangle inequality together with the inequality (22) to derive the following bound

$$\left\|\mathbf{k}_{\mathcal{N}}^{*\,T}\, K_{\mathcal{N}}^{-2}\right\|_1 \leq \frac{\hat{\sigma}_f^2}{(\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2)^2}\frac{1+\epsilon_m}{1-\epsilon_{E,2}} \leq \frac{2}{m}\frac{1}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1}{1-\epsilon_{E,2}}. \qquad (107)$$

Combining the inequalities (104) and (106) yields the bounds on the derivatives of the expectation of $\tilde{\mu}$, while combining the inequalities (105) and (107) yields the bounds on the derivatives of the variance of $\tilde{\mu}$. The final form of the inequality (100) follows after some algebra and by applying the bounds $\epsilon_m \leq 1$ and $\min\left\{d_m^{2q}, 1\right\} \leq 1$ is suitable places.

Next, we move on to the proof of the inequality (103). Let us start with the expansion (75) which implies

$$\begin{aligned}
\left|\frac{\partial MSE_{GPnn}(\mathcal{X}_*, \boldsymbol{X}_n)}{\partial \hat{\ell}}\right| &= 2\left|\Gamma\, \mathbf{k}_{\mathcal{N}}^{*\,T}\, K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right| \left|\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \\
&+ \left|\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right|,
\end{aligned}$$

Using Lemma OA.F.1 and some suitable upper bounds on $|f(\boldsymbol{x}_*)| \leq B_f$, $\epsilon_m \leq 1$ we have

$$2\left|\Gamma\, \mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} f(\boldsymbol{X}_{\mathcal{N}}) - f(\mathcal{X}_*)\right| \leq \frac{10B_f(1+2L_f)}{1-\epsilon_E}\,\epsilon_m + 4B_f L_f \min\{d_m^q, 1\}. \tag{108}$$

Next, using (76) we have

$$\frac{m\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\left|\frac{\partial \mathbb{E}_{\boldsymbol{y}|X_n}\left[\tilde{\mu}_{\mathcal{N}}^*\right]}{\partial \hat{\ell}}\right| \leq \left|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T K_{\mathcal{N}}^{-1} f(X)\right| + \left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X)\right|. \tag{109}$$

The first term in the above sum can be bounded as

$$\left|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T K_{\mathcal{N}}^{-1} f(X)\right| \leq \left\|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T\right\|_1 \left\|K_{\mathcal{N}}^{-1} f(X)\right\|_1.$$

By assumption (AD.2) combined with the chain rule, we have

$$\left\|\left(\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right)^T\right\|_1 \leq \frac{\hat{\sigma}_f^2}{\hat{\ell}} \max_{1\leq i\leq m} \min\left\{\left|\frac{d_i}{\hat{\ell}} c'\left(\frac{d_i}{\hat{\ell}}\right)\right|, B_c\right\} \leq B_c\, L_c' \frac{\hat{\sigma}_f^2}{\hat{\ell}} \min\left\{\left(\frac{d_m}{\hat{\ell}}\right)^{2p'}, 1\right\}$$

Next, using Equation (21) from Lemma OA.B.3 in Appendix OA.B we get

$$\begin{aligned}\left\|K_{\mathcal{N}}^{-1} f(X)\right\|_1 &\leq \left\|K_{\mathcal{N}}^{-1} f(X) - f(\boldsymbol{x}_*)\left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1}\right\|_1 + \left\|f(\boldsymbol{x}_*)\left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-1}\mathbf{1}\right\|_1 \\ &\leq \frac{m\, B_f}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1+2L_f\min\{d_m^q,1\}}{1-\epsilon_E} \leq \frac{m\, B_f}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1+2L_f}{1-\epsilon_E}.\end{aligned} \tag{110}$$

Next, we move to the second term of the inequality (109).

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1} \frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} f(X)\right| \leq \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\right\|_1 \left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_1 \left\|K_{\mathcal{N}}^{-1} f(X)\right\|_1.$$

Using the triangle inequality and Equation (19) (Lemma OA.B.3, Appendix OA.B) we get

$$\begin{aligned}\left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-1}\right\|_1 &\leq \left\|(\mathbf{k}_{\mathcal{N}}^*)^T K_{\mathcal{N}}^{-1} - \hat{\sigma}_f^2\,\mathbf{1}^T\left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-1}\right\|_1 + \hat{\sigma}_f^2\left\|\mathbf{1}^T\left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-1}\right\|_1 \\ &\leq \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1+\epsilon_m}{1-\epsilon_E} \leq \frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1+\epsilon_m}{1-\epsilon_E} \leq \frac{2\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\frac{1}{1-\epsilon_E}.\end{aligned} \tag{111}$$

By assumption (AD.2) and the triangle inequality, we have

$$\begin{aligned}&\left\|\frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}}\right\|_1 \leq \frac{\hat{\sigma}_f^2}{\hat{\ell}} \max_{1\leq i\leq m}\sum_{j\neq i}\left|\frac{d_{ij}}{\hat{\ell}} c'\left(\frac{d_{ij}}{\hat{\ell}}\right)\right| \leq \frac{\hat{\sigma}_f^2 B_c L_c'}{\hat{\ell}} \max_{1\leq i\leq m}\sum_{j\neq i}\min\left\{\left(\frac{d_{ij}}{\hat{\ell}}\right)^{2p'}, 1\right\} \\ &\leq \frac{\hat{\sigma}_f^2 B_c L_c'}{\hat{\ell}} \max_{1\leq i\leq m}\sum_{j\neq i}\min\left\{\left(\frac{d_i + d_j}{\hat{\ell}}\right)^{2p'}, 1\right\} \leq m\frac{\hat{\sigma}_f^2 B_c L_c'}{\hat{\ell}}\min\left\{\left(\frac{d_m}{\hat{\ell}}\right)^{2p'}, 1\right\},\end{aligned} \tag{112}$$

where $d_{ij} := \|\boldsymbol{x}_{n,i} - \boldsymbol{x}_{n,j}\|_2$ and $d_i := \|\boldsymbol{x}_{n,i} - \boldsymbol{x}_*\|_2$. After some algebra we finally obtain

$$\left|\frac{\partial \mathbb{E}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \leq \frac{1}{\hat{\ell}} \frac{3B_f B_c L_c'(1+2L_f)}{(1-\epsilon_E)^2} \min\left\{\left(\frac{d_m}{\hat{\ell}}\right)^{2p'}, 1\right\}. \tag{113}$$

Let us next move to analysing the variance derivative. Using (77) we have

$$\frac{1}{\Gamma^2}\frac{1}{2\sigma_\xi^2}\left|\frac{\partial \mathrm{Var}\left[\tilde{\mu}_{GPnn} \mid \mathcal{X}_*, \boldsymbol{X}_n\right]}{\partial \hat{\ell}}\right| \leq \left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right| + \left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^*\right|.$$

The first term of the above inequality can be bounded as

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right| \leq \left\|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2}\right\|_1 \left\|\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_1 \leq \left(\left\|(\mathbf{k}_{\mathcal{N}}^*)^T K_{\mathcal{N}}^{-2} - \hat{\sigma}_f^2 \mathbf{1}^T \left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-2}\right\|_1 + \hat{\sigma}_f^2 \left\|\mathbf{1}^T \left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-2}\right\|_1\right) \left\|\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_1.$$

By the application of Equation 22 (Lemma OA.B.3, Appendix OA.B) we obtain

$$\begin{aligned}\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right| &\leq 2\hat{\sigma}_f^2 \left\|\mathbf{1}^T \left(K_{\hat{\Theta},\mathcal{N}}^{\infty}\right)^{-2}\right\|_1 \left\|\frac{\partial \mathbf{k}_{\mathcal{N}}^*}{\partial \hat{\ell}}\right\|_1 \frac{1}{1-\epsilon_{E,2}} \\ &\leq \frac{2}{m}\left(\frac{m\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\right)^2 B_c L_c' \frac{\hat{\sigma}_f^2}{\hat{\ell}} \frac{1}{1-\epsilon_{E,2}} \min\left\{\left(\frac{d_m}{\hat{\ell}}\right)^{2p'}, 1\right\}\end{aligned}$$

In a similar way, we get

$$\left|\mathbf{k}_{\mathcal{N}}^{*T} K_{\mathcal{N}}^{-2} \frac{\partial K_{\mathcal{N}}}{\partial \hat{\ell}} K_{\mathcal{N}}^{-1} \mathbf{k}_{\mathcal{N}}^*\right| \leq \frac{2\hat{\sigma}_f^2 B_c L_c'}{\hat{\ell}} \left(\frac{m\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\right)^3 \frac{1}{1-\epsilon_E} \frac{1}{1-\epsilon_{E,2}} \min\left\{\left(\frac{d_m}{\hat{\ell}}\right)^{2p'}, 1\right\}.$$

Thus,

$$\left|\frac{\partial \mathrm{Var}_{\boldsymbol{y}|X_n}\left[\tilde{\mu}_{\mathcal{N}}^*\right]}{\partial \hat{\ell}}\right| \leq 8\frac{\sigma_\xi^2}{\hat{\ell}} \hat{\sigma}_f^2 B_c L_c' \frac{1}{1-\epsilon_E} \frac{1}{1-\epsilon_{E,2}} \min\left\{\left(\frac{d_m}{\hat{\ell}}\right)^{2p'}, 1\right\}. \tag{114}$$

The final result (103) follows from combining the inequalities (114), (113) and (108). ■

Lemma OA.G.5 (Uniform convergence of MSE derivatives.) *Let $X = (\boldsymbol{x}_1, \boldsymbol{x}_2, \dots)$ be an infinite sequence of i.i.d. points sampled from $P_{\mathcal{X}}$ and denote by X_n its truncation to the first n points. Assume (AC.1-3), (AC.5), (AR.6) and (AR.4). Then, for almost every sampling sequence X and test point $\boldsymbol{x}_* \in \mathrm{Supp}(P_{\mathcal{X}})$ and any compact subset S of the hyper-parameters $\Theta = \left(\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\ell}, \hat{\boldsymbol{b}}\right) \in S \subset \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0} \times \mathbb{R}_{>0} \times \mathbb{R}^{d_T}$ we have that*

$$\left|\frac{\partial MSE(\boldsymbol{x}_*, X_n)}{\partial(\hat{\sigma}_\xi^2)}\right| \xrightarrow{n\to\infty} 0, \quad \left|\frac{\partial MSE(\boldsymbol{x}_*, X_n)}{\partial(\hat{\sigma}_f^2)}\right| \xrightarrow{n\to\infty} 0, \tag{115}$$

$$\left\|\nabla_{\hat{\boldsymbol{b}}} MSE_{NNGP}(\boldsymbol{x}_*, \boldsymbol{X}_n)\right\|_2 \xrightarrow{n\to\infty} 0 \tag{116}$$

and this convergence is uniform as a function of $\Theta \in S$. If additionally (AR.2) and (AD.1-2) hold, we have that

$$\left|\frac{\partial MSE(\boldsymbol{x}_*, X)}{\partial \hat{\ell}}\right| \xrightarrow{n\to\infty} 0 \tag{117}$$

uniformly on S.

Proof (For $GPnn$ – the $NNGP$ counterpart follows straightforwardly using the same techniques.) We follow the same strategy as in the proof of Theorem 15. Let us consider the derivative with respect to $\partial(\hat{\sigma}_\xi^2)$. The proofs for the remaining derivatives are fully analogous. Using Equation (100) we will construct an upper bound

$$\left|\frac{\partial MSE(\boldsymbol{x}_*, X_n)}{\partial(\hat{\sigma}_\xi^2)}\right| \leq h_\Theta(\boldsymbol{x}_*, X_n),$$

where $h_\Theta(\boldsymbol{x}_*, X_n)$ is a continuous function of the hyper-parameters $\Theta = (\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\ell})$ and is monotonically decreasing with n. As in the proof of Theorem 15 Dini's theorem combined with the sandwich theorem for uniform convergence readily implies that $\partial MSE(\boldsymbol{x}_*, X_n)/\partial(\hat{\sigma}_\xi^2)$ tends to zero uniformly on S.

To construct $h_\Theta(\boldsymbol{x}_*, X_n)$, we replace ϵ_E and $\epsilon_{E,2}$ in the RHS of Equation (100) with their suitable upper bounds that are monotonically decreasing with n. Namely,

$$\begin{aligned} h_\Theta(\boldsymbol{x}_*, X_n) =& 48 B_f^2 L_f^2 \frac{1}{m\hat{\sigma}_f^2} \frac{1}{(1-\tilde{\epsilon}_E)^3} \min\left\{d_m^{2q}, 1\right\} \\ &+ \frac{1}{m\hat{\sigma}_f^2}\left(24\sigma_\xi^2 \frac{1}{1-\tilde{\epsilon}_{E,2}} \frac{1}{1-\tilde{\epsilon}_E} + \frac{2B_f^2}{(1-\tilde{\epsilon}_E)^3}(78L_f+5)\right)\epsilon_m \end{aligned}$$

We have

$$\epsilon_m(\boldsymbol{x}_*, X_n) = \rho_c^2\left(\frac{d_m(\boldsymbol{x}_*, X_n)}{\hat{\ell}}\right), \quad d_m(\boldsymbol{x}_*, X_n) = \|\boldsymbol{x}_* - \boldsymbol{x}_{n,m}(\boldsymbol{x}_*)\|_2.$$

As in the proof of Theorem 15, $\epsilon_E(\boldsymbol{x}_*, X_n)$ is upper bounded by the ϵ_E calculated for the nearest neighbour configuration where the nearest neighbours are grouped on the antipodal points of the Euclidean ball of the radius $d_m(\boldsymbol{x}_*, X_n)$, i.e.

$$\boldsymbol{x}_{n,1} = \cdots = \boldsymbol{x}_{n,m-1} = 2\boldsymbol{x}_* - \boldsymbol{x}_{n,m}. \tag{118}$$

In other words,

$$\epsilon_E(\boldsymbol{x}_*, X_n) \leq \tilde{\epsilon}_E(\boldsymbol{x}_*, X_n) := (m-1)\,\rho_c^2\left(\frac{2d_m(\boldsymbol{x}_*, X_n)}{\hat{\ell}}\right).$$

It remains to construct suitable $\tilde{\epsilon}_{E,2}$. Recall the definition of $\epsilon_{E,2}$

$$\epsilon_{E,2} = \left(\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2}\right)^2 \max_{1\leq i\leq m} \sum_{j=1}^m \left(2\frac{\hat{\sigma}_\xi^2}{\hat{\sigma}_f^2}\epsilon_{ij} + \sum_{k=1}^m (\epsilon_{ik} + \epsilon_{jk} - \epsilon_{ik}\epsilon_{jk})\right). \tag{119}$$

Note that each $\epsilon_{ij}(\boldsymbol{x}_*, X_n)$ for $i \neq j$ is upped-bounded by ϵ_{ij} calculated for the configuration of nearest-neighbours described by Equation (118). Namely,

$$\epsilon_{ij} \leq \overline{\epsilon} := \rho_c^2 \left(\frac{2d_m(\boldsymbol{x}_*, X_n)}{\hat{\ell}} \right).$$

What is more, if $\epsilon'_{ij} \geq \epsilon_{ij}$ for all i, j, then

$$\epsilon'_{ik} + \epsilon'_{jk} - \epsilon'_{ik}\epsilon'_{jk} \geq \epsilon_{ik} + \epsilon_{jk} - \epsilon_{ik}\epsilon_{jk}.$$

Thus, we can upper-bound the RHS of (119) by replacing every ϵ_{ij} with $\overline{\epsilon}$ and setting $i \equiv m$. This leads to

$$\epsilon_{E,2} \leq \tilde{\epsilon}_{E,2} := \left(\frac{\hat{\sigma}_f^2}{\hat{\sigma}_\xi^2 + m\hat{\sigma}_f^2} \right)^2 (m-1)\overline{\epsilon} \left(2\frac{\hat{\sigma}_\xi^2}{\hat{\sigma}_f^2} + m + 2 - \overline{\epsilon} \right).$$

Since ϵ_m, $\tilde{\epsilon}_E$ and $\tilde{\epsilon}_{E,2}$ are continuous functions of Θ that are monotonically decreasing with n, so is $h_\Theta(\boldsymbol{x}_*, X_n)$.

The remaining MSE derivatives are treated in a fully analogous way. ■

Theorem 17 (Convergence Rates of Derivatives). *Let n be the size of the training set which is i.i.d. sampled from the distribution $P_\mathcal{X}$ and let the test point be also sampled from $P_\mathcal{X}$. Let m be the (fixed) number of nearest-neighbours used in GPnn/NNGP. Assume (AC.5) and (AR.1-6). In GPnn define $\alpha_\phi := \min\{p, q\}$ for each $\phi \in \{\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\boldsymbol{b}}\}$. In NNGP define $\overline{q} := \min\{q_1, \ldots, q_{d_T}\}$ and $\alpha_\phi := \min\{p, q_0, \overline{q}\}$ when $\phi \in \{\hat{\sigma}_\xi^2, \hat{\sigma}_f^2\}$ and $\alpha_{\hat{\boldsymbol{b}}} := \min\{2p, \overline{q}\}$. Then, if $d_\mathcal{X} > 4(\alpha_\phi + p)$, for each $\phi \in \{\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\boldsymbol{b}}\}$ we have*

$$\mathbb{E}\left[D_\phi(\mathcal{X}_*, \boldsymbol{X}_n)\right] \leq A_1^{(\phi)} \left(\frac{m}{n}\right)^{2\alpha_\phi/d_\mathcal{X}} + A_2^{(\phi)}\, m \left(\frac{m}{n}\right)^{2(\alpha_\phi+p)/d_\mathcal{X}}, \tag{120}$$

where $0 < A_i^{(\phi)} < \infty$ depend on p, $\{q_i\}$, $d_\mathcal{X}$, d_T, B_f, B_T, L_f, L_c, $L_{\tilde{c}}$, σ_ξ and the GPnn/NNGP hyper-paramaters. Taking $m_n = n^{\frac{2p}{2p+d_\mathcal{X}}}$ the derivatives tend to zero at the same rates as the (minimax-optimal) risk rate from Stone's theorem, i.e.,

$$\mathbb{E}\left[D_\phi(\mathcal{X}_*, \boldsymbol{X}_n)\right] \leq \left(A_1^{(\phi)} + A_2^{(\phi)}\right) n^{-\frac{2\alpha_\phi}{2p+d_\mathcal{X}}}. \tag{121}$$

Proof (Sketch.) Recall the shorthand $D_\phi(\mathcal{X}_*, \boldsymbol{X}_n)$ from (12). We start from the bias–variance derivative identity (75)–(77): for $\phi \in \{\hat{\sigma}_\xi^2, \hat{\sigma}_f^2, \hat{\boldsymbol{b}}\}$, $\partial_\phi MSE$ is a sum of a bias $\times$ $\partial_\phi \mathbb{E}[\tilde{\mu} \mid \mathcal{X}_*, \boldsymbol{X}_n]$ term and a $\partial_\phi \mathrm{Var}[\tilde{\mu} \mid \mathcal{X}_*, \boldsymbol{X}_n]$ term. Inequalities derived in the proof of Lemma OA.G.1 give uniform control of the derivative building blocks in terms of upper bounds proportional to m, and Lemma OA.G.4 yields a deterministic upper bound for D_ϕ in terms of d_m and nearest-neighbour kernel-metric distances (with Lemma OA.B.4 replacing $\epsilon_E, \epsilon_{E,2}$ by ϵ_m).

To bound $\mathbb{E}[|D_\phi|]$, use the same good/bad region decomposition as in the risk-rate proof of Theorem 13: for $0 < R \le 1$ define the bad event $\Omega_{m,n}(R) := \{(\boldsymbol{x}_*, X_n) : d_m(\boldsymbol{x}_*, X_n) \ge R\}$ and apply Lemma OA.E.1 (Cauchy–Schwarz splitting) to obtain

$$\mathbb{E}[|D_\phi|] \;\le\; \mathbb{E}[|D_\phi| \mid d_m < R] \;+\; \sqrt{\mathbb{E}[D_\phi^2]}\,\sqrt{P[\Omega_{m,n}(R)]}.$$

On $\{d_m < R\}$, plug in the deterministic bound from Lemma OA.G.4 and use the NN-distance/ϵ moment rates (as in Theorem 13) to get the first term $A_1^{(\phi)}(m/n)^{2\alpha/d_{\mathcal{X}}}$. Next, control $P[\Omega_{m,n}(R)]$ via Lemma OA.C.8 and bound $\mathbb{E}[D_\phi^2]$ by a uniform upper bound proportional to m (from Lemma OA.G.1), yielding the second term $A_2^{(\phi)}\, m\,(m/n)^{2(\alpha+p)/d_{\mathcal{X}}}$. Combining the two contributions proves the claim (and choosing m_n gives the stated rate). ■

Theorem 20. *Let n be the size of the training set which is i.i.d. sampled from the distribution $P_{\mathcal{X}}$ and let the test point be also sampled from $P_{\mathcal{X}}$. Let m be the (fixed) number of nearest-neighbours used in GPnn/NNGP. Assume (AC.5), (AR.1-6) and (AD.1- 2). Then, if $d_{\mathcal{X}} > 4(p' + 2p)$, we have*

$$\mathbb{E}\left[D_{\hat{\ell}}(\mathcal{X}_*, \boldsymbol{X}_n)\right] \le \frac{\max\{\hat{\ell}^{-2p'}, 1\}}{\hat{\ell}} A_1 \left(\frac{m}{n}\right)^{2p'/d_{\mathcal{X}}} + \frac{1}{\hat{\ell}} A_2\, m^2 \left(\frac{m}{n}\right)^{2(p'+2p)/d_{\mathcal{X}}}, \tag{122}$$

where $0 < A_1, A_2 < \infty$ depend on p, $\{q_i\}$, $d_{\mathcal{X}}$, d_T, B_f, B_T, B_c, L_f, L_c, L'_c, $L_{\tilde{c}}$, σ_ξ, $\hat{\sigma}_\xi$, $\hat{\sigma}_f$ (but not on $\hat{\ell}$). Taking $m_n = n^{\frac{2p}{2p+d_{\mathcal{X}}}}$ the derivatives tend to zero at the following rate.

$$\mathbb{E}\left[D_{\hat{\ell}}(\mathcal{X}_*, \boldsymbol{X}_n)\right] \le \frac{1}{\hat{\ell}} \left(\max\left\{\hat{\ell}^{-2p'}, 1\right\} A_1 + A_2\right) n^{-\frac{2p'}{2p+d_{\mathcal{X}}}}. \tag{123}$$

Proof (Sketch.) Use (75)–(77) with $\phi = \hat{\ell}$ and the definition $D_{\hat{\ell}} = |\partial_{\hat{\ell}} MSE|$ in (12). Bounding $\partial_{\hat{\ell}}\mathbb{E}[\tilde{\mu} \mid \mathcal{X}_*, \boldsymbol{X}_n]$ and $\partial_{\hat{\ell}}\mathrm{Var}[\tilde{\mu} \mid \mathcal{X}_*, \boldsymbol{X}_n]$ involves bounding $\partial_{\hat{\ell}}\mathbf{k}^*_{\mathcal{N}}$ and $\partial_{\hat{\ell}} K_{\mathcal{N}}$. This is handled by the kernel bounds in Lemma OA.G.4 and in the proof of Lemma OA.G.3, which extract the explicit $\hat{\ell}$-prefactors and yield i) deterministic upper bound for $|D_{\hat{\ell}}|$ (103) in terms of d_m and nearest-neighbour kernel-metric distances (with Lemma OA.B.4 replacing $\epsilon_E, \epsilon_{E,2}$ by ϵ_m) and ii) uniform upper bound on $|D_{\hat{\ell}}|$ (99) proportional to m^2 .

This allows us to bound $\mathbb{E}[|D_{\hat{\ell}}|]$ using the same good/bad event split based on $\Omega_{m,n}(R) := \{(\boldsymbol{x}_*, X_n) : d_m \ge R\}$ as in Theorem 13, i.e.,

$$\mathbb{E}[|D_{\hat{\ell}}|] \;\le\; \mathbb{E}[|D_{\hat{\ell}}| \mid d_m < R] \;+\; \sqrt{\mathbb{E}[D_{\hat{\ell}}^2]}\,\sqrt{P[\Omega_{m,n}(R)]}.$$

Then, apply the deterministic bound (103) in terms of moments of euclidean and kernel-metric NN-distances to obtain the term $\frac{\max\{\hat{\ell}^{-2p'},1\}}{\hat{\ell}} A_1 (m/n)^{2p'/d_{\mathcal{X}}}$ in (122). On $\Omega_{m,n}(R)$ control the tail probability $P[\Omega_{m,n}(R)]$ as in Theorem 13 and use the uniform bound (99), which yields the second term $\frac{1}{\hat{\ell}} A_2\, m^2 (m/n)^{2(p'+2p)/d_{\mathcal{X}}}$ and completes the proof sketch. ■

Appendix OA.H. Upper bounds on kernel functions and their derivatives

In this section we show that some popular kernel choices (exponential, squared exponential and Matérn) satisfy the assumption (A2) and the related assumption needed for calculating the convergence rates of MSE-derivatives in Section OA.G.2. Namely, we show that there exist positive constants $L_c, \widetilde{L}_c$ and $p, \tilde{p} \in]0,1]$ such that for all $r \geq 0$

$$c(r) \geq 1 - L_c\, r^{2p}, \quad \left|rc'(r)\right| \leq \widetilde{L}_c\, r^{2\tilde{p}}$$

where $c(r)$ is the normalised kernel function, $c(0) = 1$. Recall the relevant kernel function definitions

$$c_{\text{Exp}}(r) := e^{-r}, \quad c_{\text{SE}}(r) := e^{-r^2/2}, \quad c_{M,\nu}(r) := \frac{2}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}}{2} r\right)^{\nu} \mathcal{K}_\nu(\sqrt{2\nu}\, r),$$

where $\Gamma(\cdot)$ is the Euler Gamma function and $\mathcal{K}_\nu(\cdot)$ is the modified Bessel function of the second kind. The goal of this Appendix is to prove that the following inequalities hold for any $r \geq 0$.

$$c_{\text{Exp}}(r) \geq 1 - r, \quad c_{\text{SE}}(r) \geq 1 - \frac{r^2}{2}, \tag{124}$$

$$\left|r\, c'_{\text{Exp}}(r)\right| \leq r, \quad \left|r\, c'_{\text{SE}}(r)\right| \leq r^2, \tag{125}$$

$$c_{M,\nu}(r) \geq 1 - \frac{\nu}{\nu - 1}\frac{r^2}{2}, \quad \nu > 1, \tag{126}$$

$$c_{M,\nu}(r) \geq 1 - \nu^{\nu}\, \Gamma(1-\nu)\, \mathcal{I}_\nu\,(1)\; r^{2\nu}, \quad 0 < \nu < 1, \tag{127}$$

$$c_{M,1}(r) \geq 1 - L_\epsilon\, r^{2-\epsilon},\; L_\epsilon = \max_{0 \leq r \leq 1} \frac{1 - c_{M,1}(r)}{r^{2-\epsilon}} < \infty \quad \text{for any} \quad 0 < \epsilon < 2,\; 0 \leq r \leq 1, \tag{128}$$

$$\left|r\, c'_{M,\nu}(r)\right| \leq \nu \frac{\nu+1}{\nu-1}\,\frac{r^2}{2}, \quad \nu > 1, \tag{129}$$

$$\left|r\, c'_{M,\nu}(r)\right| \leq \Gamma(1-\nu)\, \nu^{\nu} \left(\frac{1}{\Gamma(\nu)\, 2^{\nu}} + \nu\, \mathcal{I}_\nu\,(1)\right) r^{2\nu}, \quad 0 < \nu < 1, \tag{130}$$

where $\mathcal{I}_\nu(\cdot)$ denotes the modified Bessel function of the first kind.

Bounds for the Kernels. The desired lower bounds for the exponential and squared exponential kernel functions (124) are readily derived from the standard inequality $e^u \geq 1+u$ which holds for any $u \in \mathbb{R}$. In a similar way, we use the fact that $e^{-u} \leq 1$ for $u \geq 0$ and get the bounds (125). Thus, we have that $p = \tilde{p} = 1$ for c_{Exp} and $p = \tilde{p} = 2$ for c_{SE}.

In order to obtain the lower bounds (126) and (127) for the Matérn family, we need to consider two different cases. The first case concerns kernels with $\nu > 1$ and the second case concerns kernels with $0 < \nu < 1$. When $\nu > 1$ we use the following integral representation of the Matérn kernel (Tronarp et al., 2018)

$$c_{M,\nu}(r) = \frac{\nu^{\nu}}{\Gamma(\nu)} \int_0^{\infty} s^{\nu-1} e^{-\nu s} e^{-\frac{r^2}{2s}}.$$

Using the fact that $e^{-\frac{r^2}{2s}} \geq 1 - \frac{r^2}{2s}$ in the above integral we obtain

$$c_{M,\nu}(r) \geq \frac{\nu^{\nu}}{\Gamma(\nu)} \int_0^{\infty} s^{\nu-1} e^{-\nu s} \left(1 - \frac{r^2}{2s}\right) = 1 - \frac{\nu}{\nu-1}\frac{r^2}{2}, \quad \nu > 1.$$

When $0 < \nu < 1$ we apply a different strategy that uses the following series expansion of $\mathcal{K}_\nu$ that converges for any $z \in \mathbb{C}$ and $\nu \notin \mathbb{Z}$.

$$\mathcal{K}_\nu(z) = \frac{\Gamma(\nu)\Gamma(1-\nu)}{2}\left(\sum_{k=0}^{\infty}\frac{1}{\Gamma(k-\nu+1)k!}\left(\frac{z}{2}\right)^{2k-\nu} - \sum_{k=0}^{\infty}\frac{1}{\Gamma(k+\nu+1)k!}\left(\frac{z}{2}\right)^{2k+\nu}\right).$$

This implies that

$$\begin{aligned} c_{M,\nu}(r) = 1 + \sum_{k=1}^{\infty}\frac{\Gamma(1-\nu)}{\Gamma(k-\nu+1)k!}\left(\frac{\sqrt{2\nu}}{2}r\right)^{2k} \\ -\Gamma(1-\nu)\left(\frac{\sqrt{2\nu}}{2}r\right)^{2\nu}\sum_{k=0}^{\infty}\frac{1}{\Gamma(k+\nu+1)k!}\left(\frac{\sqrt{2\nu}}{2}r\right)^{2k} \end{aligned} \tag{131}$$

In order to find the desired upper bound for $0 < \nu < 1$ we simply bound the first sum by 0

$$\sum_{k=1}^{\infty}\frac{\Gamma(1-\nu)}{\Gamma(k-\nu+1)k!}\left(\frac{\sqrt{2\nu}}{2}r\right)^{2k} \geq 0. \tag{132}$$

Next, using the series expansion of the modified Bessel function of the first kind $\mathcal{I}_\nu(z)$ we get that

$$\sum_{k=0}^{\infty}\frac{1}{\Gamma(k+\nu+1)k!}\left(\frac{\sqrt{2\nu}}{2}r\right)^{2k} = \left(\frac{\sqrt{2\nu}}{2}r\right)^{-\nu}\mathcal{I}_\nu\left(r\sqrt{2\nu}\right).$$

The RHS of the above equation is an increasing function of r for $r > 0$. In particular, it implies that for any $r < 1/\sqrt{2\nu}$

$$\sum_{k=0}^{\infty}\frac{1}{\Gamma(k+\nu+1)k!}\left(\frac{\sqrt{2\nu}}{2}r\right)^{2k} \leq 2^{\nu}\,\mathcal{I}_\nu(1). \tag{133}$$

Plugging the bounds (132) and (133) into the expansion (131) we get the upper bound (127) for any $r < 1/\sqrt{2\nu}$. However, the bound (127) also holds for $r \geq 1/\sqrt{2\nu}$. To see this, it suffices to note that the bound is a decreasing function of r and evaluate the bound at $r = 1/\sqrt{2\nu}$ to see that its value is negative at this point, i.e.

$$2^{-\nu}\,\Gamma(1-\nu)\,\mathcal{I}_\nu(1) > 1 \quad \text{for all} \quad 0 < \nu < 1.$$

Thus, for $r \geq 1/\sqrt{2\nu}$ the value of the bound stays negative, and hence it is strictly smaller than $c_{M,\nu}(r)$ which is positive for any $r \geq 0$.

For the normalised Matérn kernel with smoothness parameter $\nu = 1$, $c(r) = \sqrt{2}\,r\,K_1\big(\sqrt{2}\,r\big)$, the small-$r$ expansion gives

$$1 - c(r) = r^2\left(\log\frac{\sqrt{2}}{r} - \gamma + \frac{1}{2}\right) + O\left(r^4\log\frac{1}{r}\right), \qquad r \to 0,$$

with γ the Euler–Mascheroni constant. In particular,

$$1 - c(r) = r^2 \log \frac{1}{r} + O(r^2) + O\left(r^4 \log \frac{1}{r}\right), \qquad r \to 0.$$

Fix any $\epsilon \in (0, 2)$ and define, for $r \in (0, 1]$,

$$g_\epsilon(r) := \frac{1 - c(r)}{r^{2-\epsilon}}.$$

Then, as $r \to 0$,

$$g_\epsilon(r) = r^\epsilon \log \frac{1}{r} + O(r^\epsilon) + O\left(r^{2+\epsilon} \log \frac{1}{r}\right) \to 0.$$

Hence g_ϵ extends continuously to $[0, 1]$ by setting $g_\epsilon(0) := 0$. Since g_ϵ is continuous on the compact interval $[0, 1]$, it attains a finite maximum there. Therefore, for every $\epsilon \in (0, 2)$, there exists

$$L_\epsilon := \max_{0 \le r \le 1} \frac{1 - c(r)}{r^{2-\epsilon}} < \infty$$

such that

$$c(r) \ge 1 - L_\epsilon r^{2-\epsilon} \qquad \text{for all } r \in [0, 1].$$

Bounds for the Kernel Derivatives In order to derive bounds (129) and (130) involving derivatives of the Matérn kernel function, we use the following recursive formula for the derivative of the relevant Bessel function.

$$\mathcal{K}'_\nu(u) = -\frac{1}{2} \left(\mathcal{K}_{\nu-1}(u) + \mathcal{K}_{\nu+1}(u)\right).$$

Using the above formula together with the identity $\mathcal{K}_{-\nu}(u) = \mathcal{K}_\nu(u)$, one can verify by a straightforward calculation that the following expressions hold.

$$\left|r\, c'_{M,\nu}(r)\right| = \frac{\nu}{\nu - 1} \frac{r^2}{2}\, c_{M,\nu-1}\left(r\sqrt{\frac{\nu}{\nu - 1}}\right) + \nu \left(c_{M,\nu+1}\left(r\sqrt{\frac{\nu}{\nu + 1}}\right) - c_{M,\nu}\left(r\right)\right), \quad \nu > 1,$$

$$\begin{aligned}\left|r\, c'_{M,\nu}(r)\right| = \frac{\Gamma(1-\nu)}{\Gamma(\nu)} \left(\frac{\nu}{2}\right)^\nu r^{2\nu}\, c_{M,1-\nu}\left(r\sqrt{\frac{\nu}{1 - \nu}}\right) \\ + \nu \left(c_{M,\nu+1}\left(r\sqrt{\frac{\nu}{\nu + 1}}\right) - c_{M,\nu}\left(r\right)\right), \quad 0 < \nu < 1.\end{aligned}$$

Finally, we obtain the bounds (129) and (130) by plugging into the above expressions the following inequalities:

$$\begin{aligned}
&c_{M,\nu-1}\left(r\sqrt{\frac{\nu}{\nu - 1}}\right) \le 1, \quad c_{M,1-\nu}\left(r\sqrt{\frac{\nu}{1 - \nu}}\right) \le 1, \\
&c_{M,\nu+1}\left(r\sqrt{\frac{\nu}{\nu + 1}}\right) - c_{M,\nu}\left(r\right) \le 1 - \left(1 - \frac{\nu}{\nu - 1}\frac{r^2}{2}\right) = \frac{\nu}{\nu - 1}\frac{r^2}{2}, \quad \nu > 1 \\
&c_{M,\nu+1}\left(r\sqrt{\frac{\nu}{\nu + 1}}\right) - c_{M,\nu}\left(r\right) \le \nu^\nu\, \Gamma(1-\nu)\, \mathcal{I}_\nu\left(1\right) r^{2\nu}, \quad 0 < \nu < 1,
\end{aligned}$$

where in the last two inequalities we have applied the bounds (126) and (127) to bound $c_{M,\nu}(r)$ from below.

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